

The Mental Representation of Negative Integer Magnitudes Over Development and Across
Contexts

By

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Table of Contents

<i>Acknowledgements</i>	<i>ii</i>
<i>Abstract</i>	<i>v</i>
<i>Chapter 1: General Introduction</i>	<i>1</i>
<i>Chapter 2: General Method</i>	<i>30</i>
<i>Chapter 3: Study 1 with Adults</i>	<i>37</i>
RQ1: What types of representations (extension, reflection, components) do participants employ when making magnitude comparisons that involve negative integers?	39
RQ2: Does participants' use of different representations for negative numbers depend on task features (blocked/intermixed stimuli and simultaneous/sequential presentation)?.....	48
RQ3: Does participants' use of these representations depend on their executive functions?	56
<i>Chapter 4: Study 2 with Children</i>	<i>66</i>
RQ1: What types of representations (extension, reflection, components) do participants employ when making magnitude comparisons that involve negative integers?	68
RQ2: Does participants' use of different representations for negative numbers depend on task features (blocked/intermixed stimuli and simultaneous/sequential presentation)?.....	79
RQ3: Does participants' use of these representations depend on their executive functions?	88
<i>Chapter 5: General Discussion</i>	<i>97</i>
The Representational Dynamics in Integer Cognition Framework.....	109
<i>References</i>	<i>118</i>
<i>Appendix 1</i>	<i>127</i>
<i>Appendix 2</i>	<i>135</i>
<i>Appendix 3</i>	<i>136</i>
<i>Appendix 4</i>	<i>139</i>
<i>Appendix 5</i>	<i>142</i>
<i>Appendix 6</i>	<i>143</i>
<i>Appendix 7</i>	<i>144</i>
<i>Appendix 8</i>	<i>147</i>
<i>Appendix 9</i>	<i>150</i>
<i>Appendix 10</i>	<i>151</i>

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Abstract

Humans routinely engage with highly abstract mathematical concepts, such as negative numbers, across various domains. Understanding the cognitive representations people use to process negative number magnitudes is crucial for both cognitive scientists and educators. Furthermore, how people use these representations may vary depending on context, experience, and cognitive skills, such as executive function skills. This dissertation addresses these issues by examining behavioral signatures of different representations in participants' reaction times as they compare magnitudes of number pairs, focusing on how responses vary depending on the numerical distance between numbers in a pair and the predicate used in the comparison question (greater or lesser). In Study 1, adults ($N = 54$) completed a symbolic magnitude comparison task involving positive, negative, and mixed positive-negative pairs presented either simultaneously or sequentially, and either in blocks or intermixed. Group-level results showed no distance effects on mixed pairs, suggesting participants often used transformation rules based on positive numbers to represent negative numbers. However, a semantic congruence effect indicated that adults meaningfully applied magnitude semantics to negative numbers. Individual-level analyses uncovered substantial variability in the representations used by participants. In Study 2, I tested children in 6th and 7th grade in the US ($N = 54$) using the same design. I found similar patterns as for adults, indicating that variability in representation use is present even in a younger sample. In both studies, there were no significant relations between executive function skills and representations. Integrating findings from both studies, I propose the Representational Dynamics in Integer Cognition Framework. This framework characterizes how individuals dynamically shift among different representational strategies for integers based on specific task demands and individual cognitive skills.

Chapter 1: General Introduction

Mathematical knowledge is essential for success in modern society, yet many children begin to struggle with mathematics early in their education, often failing to develop a solid understanding of foundational concepts. These early struggles can lead to persistent achievement gaps, particularly for students from low-income backgrounds (Goldhaber et al., 2023). A major challenge in addressing these gaps is the limited scientific understanding of how the human cognitive system processes abstract mathematical concepts. Without a clear grasp of the cognitive mechanisms that support mathematical thinking, including how abstract concepts are mentally represented and used in problem solving, educational interventions often fail to address the root causes of students' difficulties.

A key example of an abstract mathematical concept is the concept of negative integers. Negative numbers are frequently encountered in everyday life, such as when describing temperature changes or financial debt. Yet, many people find negative numbers unintuitive and confusing. Unlike whole numbers, which map naturally onto concrete objects (for example, “two clementines”), negative numbers do not correspond to physical quantities in a straightforward way. It is not possible to see or touch “negative two clementines,” which makes the concept more abstract and challenging to grasp.

Over the last two decades, cognitive scientists have begun to shed light on how people understand negative numbers (Mende et al., 2018; Prather & Boroditsky, 2003; Varma et al., 2007), but there is currently no widely accepted theoretical account of how they are mentally represented. Existing empirical findings are often inconsistent, suggesting that multiple representations may be used depending on age, context, and individual differences.

The goal of this dissertation is to contribute to our understanding of how people represent and process negative number magnitudes. This work focuses on two groups: adults, who have mature mathematical knowledge, and late elementary school children, whose understanding of negative numbers is still developing. The dissertation also investigates how the context in which negative numbers are presented influences the mental representations people use.

In addition, this research explores whether individual differences in executive function skills—that is, domain-general skills involved in regulating cognitive processing—may relate to how people understand negative number magnitudes and how flexible their representations are across contexts. By addressing these questions, this dissertation aims to fill gaps in the current literature and support the future development of more effective approaches to teaching and learning negative numbers.

Previous Research and Current Knowledge

Understanding Numerical Magnitude

A large body of evidence suggests that people represent numbers using a *mental number line*, an abstract cognitive representation in which numbers are mapped onto an imagined spatial continuum (Dehaene et al., 1993; Dehaene & Changeux, 1993; Hubbard et al., 2005). This representation inherently reflects *magnitude*, with larger numbers perceived as having greater distances on the mental number line and smaller numbers less distance on the mental number line. The ability to understand magnitude, both approximately and precisely, is foundational to how people interpret size, quantity, and scale across many domains including number, space, and time. For instance, people intuitively grasp the difference between small and large numbers, short and long distances, and brief and extended durations. Critically, the capacity to perceive and compare numerical magnitudes supports arithmetic skills and broader mathematical

reasoning (Booth & Siegler, 2006; Fazio et al., 2014; Halberda et al., 2008; Siegler & Ramani, 2009).

Understanding the development of the mental number line is critical for understanding adult performance as well for understanding challenges in mathematical cognition. The Integrated Theory of Numerical Development (Siegler & Braithwaite, 2017; Siegler, 2016) proposes a specific progression of steps with which numerical magnitude understanding is developed and refined. This dissertation builds on two key ideas from that theory. First, people mentally represent numerical magnitudes using a flexible internal structure known as the mental number line. Early in development, this mental number line is used to represent small whole numbers. Over time, it is extended rightward to include larger whole numbers and leftward to include negative numbers. Second, knowledge of numerical magnitudes, both for whole numbers and integers, is strongly correlated with, and predictive of, success in learning arithmetic and more advanced mathematical concepts (Fazio et al., 2014; Schneider et al., 2017; Vest & Alibali, 2024).

A central claim of the Integrated Theory of Numerical Development is that the mental number line is dynamic and inclusive, capable of representing all rational numerical magnitudes. While magnitude is often associated with positive, countable quantities, some types of numbers, such as negative numbers, are less tangible and therefore more challenging to represent mentally. This makes the study of how negative numbers are mentally represented a key window into broader questions about numerical development.

Negative numbers introduce a new layer of complexity to our understanding of numerical magnitude. They represent quantities that are conceptually opposite to positive numbers, and they challenge many of the intuitive rules children acquire early in development (i.e., old rules

“expire”; Karp et al., 2014). For example, learners must revise the idea that addition always increases quantity, as in the case of $2 + (-1) = 1$. These shifts require rethinking deeply held assumptions about numbers and operations, making negative numbers particularly difficult to integrate into existing mental models.

A comparable disruption occurs with fractions, which also violate intuitive rules about number size and operation. Many learners demonstrate a “whole number bias,” such that they overextend their knowledge of whole numbers to fraction tasks (Ni & Zhou, 2005). This bias can interfere with magnitude judgments, leading individuals to rely on superficial features, such as comparing numerators or denominators, rather than evaluating the magnitudes of the fractions themselves (Meert et al., 2009). Fractions therefore illustrate how expanding the number system can demand shifts in how numerical meaning is constructed and interpreted. In this dissertation, I investigate how learning about negative numbers elicits related shifts.

These kinds of conceptual disruptions highlight a deeper theoretical puzzle in numerical cognition: How are the magnitudes of negative numbers mentally represented? Negative numbers introduce a paradox in which numerically larger absolute values correspond to conceptually smaller quantities (e.g., $-5 < -3$). This inversion complicates their placement on the mental number line and raises questions about whether negative numbers are processed as integrated magnitudes or as qualitatively different from positive numbers. Although the mental number line is a foundational framework for understanding numerical cognition, how it accommodates negative numbers remains an open and important question.

Do Negative Numbers Have Magnitude?

What does it mean for negative numbers to have magnitude? To answer this question, we must consider how people adapt and extend their existing cognitive frameworks to include

negative values. In the domain of logic, for example, toddlers must learn to interpret negation (“NOT P”; Nordmeyer & Frank, 2013), which requires overriding a default tendency toward affirmative reasoning. Similarly, understanding negative numbers involves transforming an early mental model built around natural numbers to incorporate signed values. This shift often requires revising familiar rules and reinterpreting well-practiced operations—for example, recognizing that two numbers can sum to a smaller result or that subtraction can increase quantity (Bofferding, 2014; Varma et al., 2019). Through such experiences, the mental number line may gradually extend to represent this broader number system, which entails new and different rules.

Broadly, three major theoretical accounts offer different predictions about how negative numbers are represented and processed: the number-line *extension* account, the number-line *reflection* account, and the *components* account. Each account proposes a distinct way in which negative numbers are mentally conceptualized and integrated into numerical reasoning. Here, I briefly introduce the core idea in each of the accounts; below, I describe the nature of the evidence that has been brought to bear on each account and review previous findings relevant to each of the accounts.

First, the number-line *extension* account (hereafter referred to as the *extension* representation) suggests that the mental representation of positive numbers extends leftward to include negative numbers on the mental number line (Fischer, 2003; Fischer & Rottmann, 2005). This hypothesis proposes that people *integrate* absolute magnitude with polarity information, thereby forming an understanding of negative number magnitude as part of the mental number line. The extension hypothesis emphasizes the continuity of numerical representation across positive and negative numbers, highlighting the interconnectedness of numerical concepts within the left-to-right mental number line framework.

Second, the number-line *reflection* account (hereafter referred to as the *reflection* representation) proposes that people understand negative numbers by mentally representing them as mirror images of positive numbers reflected across zero on the mental number line (Varma & Schwartz, 2011). In this framework, zero serves as the central point of symmetry, and each negative number is understood as occupying a position that is spatially opposite to its positive counterpart. For example, -3 is conceived as a location directly across from 3 on the number line, not as the result of applying a piecewise algorithm that depends on the magnitude of 3. Instead, the negative number is represented holistically, as a unified spatial construct. The Bundling Hypothesis (Blair et al., 2013) provides a conceptual basis for this reflection-based understanding. It proposes that children combine perceptual competencies such as subitizing, sequencing, and magnitude comparison to construct an understanding of natural numbers, and that they extend this system by recruiting additional perceptual mechanisms for symmetry when learning about negative numbers. In this view, the reflection representation emerges as a coherent spatial mapping grounded in perceptual symmetry, rather than as a mere extension of natural number magnitude or through a rule-based transformation of natural number magnitude.

Third, the *components* representation suggests that people do not represent negative numbers as having their own positions on the mental number line. Instead, they interpret negative numbers by relying on the magnitude of their positive counterparts and separately considering the negative sign (Nuerk et al., 2004; Tzelgov & Ganor-Stern, 2009). According to this view, people do not understand negative numbers as having a distinct magnitude of their own. Rather, people perceive them as positive magnitudes combined with a symbolic rule that signals an opposite direction (or *polarity*) or a lesser status. As a result, people who rely on a components representation may reason that all positive numbers are larger than all negative

numbers and may estimate the size of a negative number by referencing the corresponding positive magnitude, rather than by positioning the negative number directly on an integrated mental number line.

In the following sections, I describe how magnitude understanding is classically measured. Then, I outline behavioral signatures during symbolic magnitude comparison that reflect the mental number line for whole numbers. Subsequently, I discuss the potential extension of these findings to negative numbers, and I review previous research that supports or contradicts each of the three aforementioned theoretical accounts.

Measuring Magnitude: Symbolic Magnitude Comparison

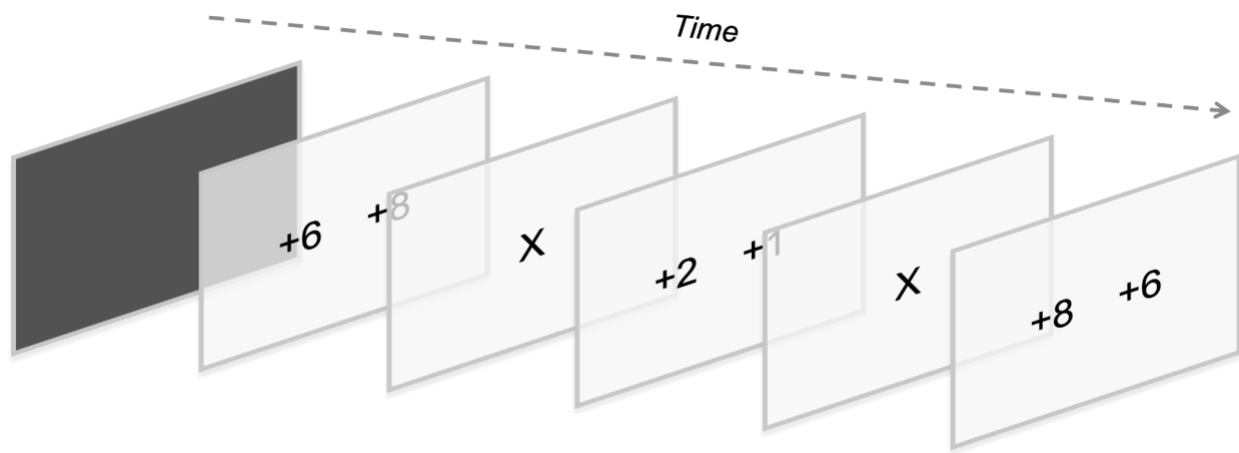
Evidence from tasks that involve symbolic magnitude comparison can help disentangle the mental representations people use for negative integers, in the same way tasks have been used to study mental representations of positive integers and positive fractions (e.g., Moyer & Landauer, 1967; Dehaene et al., 1993). In magnitude comparison tasks, people judge which number within a pair is greater or lesser. Typically, the numbers are presented simultaneously on a computer screen, and participants use a keyboard to indicate their choice of the greater or lesser number, depending on the task instructions (Figure 1). Both accuracy and reaction times (RT) are recorded. Certain RT and error-rate patterns are taken to derive from and therefore to reflect properties of people's magnitude representations for the numbers in the pair being compared.

Critically, features of the magnitude comparison task can be varied. For instance, numbers can be displayed simultaneously or sequentially. In a simultaneous condition, people see the two numbers at the same time (e.g., 1 and 4; see Figure 1), but in a sequential condition people see one digit (e.g., 1) on its own and subsequently see the second digit on its own (e.g., 4; see Figure 3 later in the Method). In a sequential-pairs condition, number pairs are presented

sequentially, and participants are required to indicate whether the second number was larger or smaller than the first (Ganor-Stern et al., 2010). Sequential presentation requires people to hold information about the first number (e.g., polarity, magnitude) in their memory until the second number is presented.

Figure 1

Depiction of Symbolic Magnitude Comparison Task in Which Participants Compare Two Numbers to Decide Which Is Larger or Smaller



Critically, patterns of reaction times, termed “behavioral signatures”, during magnitude comparison tasks can reveal aspects of the underlying representation of numbers (e.g., the distances between numbers on the mental number line).

Behavioral Signatures During Symbolic Magnitude Comparison

When comparing pairs of *positive* numbers, people tend to show a *distance effect*, which is the phenomenon that people are faster and more accurate when the gap between the two numbers is larger (Moyer & Landauer, 1967). For example, people are slower and less accurate when comparing 4 and 6 than when comparing 4 and 9. This distance effect is considered a hallmark of the way people represent numerical magnitudes mentally and suggests that people

process numerical information on a mental number line, where numbers are spaced out, and larger gaps between numbers make comparisons easier and faster.

When comparing pairs of *positive* numbers, people also display the *semantic congruence effect*, which is the phenomenon that people associate small values with the semantic meaning of “lesser” and large values with the semantic meaning of “greater” (Banks et al., 1976). In tasks in which participants compare pairs of positive numbers, they tend to respond more quickly to choose the *lesser* of two numbers when both numbers are small and more quickly to choose the *greater* of two numbers when both numbers are large. This semantic congruence effect is taken as evidence that the right and left poles of the mental number line for natural numbers are associated with greater and lesser predicates, respectively.

People’s reaction time (RT) patterns in comparisons that involve negative numbers may reveal the ways in which they represent negative numbers. In the following section, I first describe whether and how people who represent negative numbers in different ways—either in terms of their components, on a leftward extension of the mental number line, or on a reflection of the positive number line—would show a *distance effect* when comparing negative numbers (both in negative-negative comparisons and mixed positive-negative comparisons). Next, I consider how individuals who represent negative numbers in different ways would display a *semantic congruence effect* when making comparisons involving negative numbers. For brevity, I refer to the different ways of representing negative numbers as the components representation, the extension representation, and the reflection representation.

Distance Effect Predictions from the Extension, Reflection, and Components Accounts

First, consider the *distance effect*. The standard distance effect is manifested in RTs that are shorter when the distances between the numbers being compared are greater.

Extension. According to the *extension* account, the mental number line follows the same organization as the conventional number line in mathematics. Under this hypothesis, all comparisons, including positive-positive, negative-negative, and mixed positive-negative comparisons, should demonstrate standard distance effects, with shorter RTs for greater distances between the numbers being compared.

RT patterns on positive-negative comparisons are particularly crucial, because the different theoretical accounts yield different predictions for these types of comparisons. Indeed, some studies using symbolic magnitude comparison with integers have reported *standard* distance effects for mixed comparisons, in line with an extension account. For example, in two experiments with adults, Ganor-Stern and colleagues presented numbers sequentially (e.g., -3 on one slide, then 1 on the next; Ganor-Stern et al., 2010). In both experiments, adults demonstrated a standard distance effect, indicative of an extension representation.

Reflection. The *reflection* account (also referred to as *analog-x* in Varma & Schwartz, 2011) offers a distinct account of how people understand negative integers, differing from both the extension and components accounts. Like the extension account, the reflection account assumes that people represent negative numbers in an integrated way rather than constructing meaning from separate components. However, it uniquely incorporates mirror-like symmetry processing into this representation.

Symmetry processing refers to the cognitive ability to recognize when elements are balanced or mirrored across a central axis or point. For example, two sides of a shape may be visually identical. This ability is foundational to perception, spatial reasoning, and categorization (Treder, 2010; Tyler, 2002). In psychological research, sensitivity to symmetry appears early in life. Infants prefer symmetrical over asymmetrical stimuli (Bornstein et al., 1981), and adults

consistently use symmetry when recognizing objects and making aesthetic judgments (Jacobsen & Höfel, 2002). Beyond perception, symmetry supports higher-level cognitive processes such as analogical reasoning and mathematical problem-solving (Knauff, 2009).

In formal mathematics education, symmetry becomes especially relevant when students learn about negative numbers and the additive inverse property, which states that for every positive number x , there exists a corresponding negative number $-x$ such that $x + (-x) = 0$. According to the reflection account, over time, learners build an intuitive understanding of this relationship and develop representations of negative numbers that incorporate this symmetry.

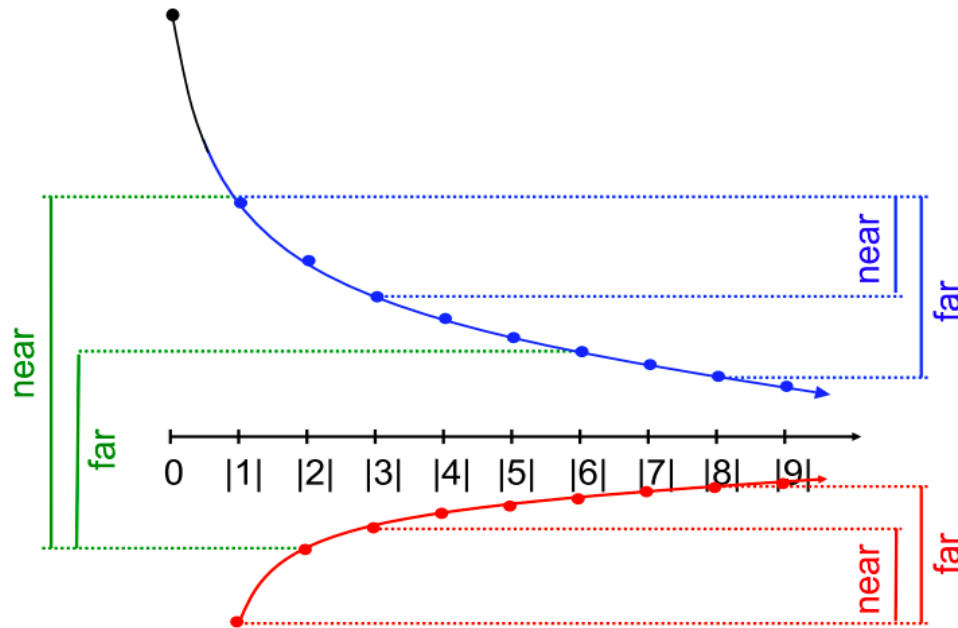
The reflection account proposes that mature representations of negative numbers involve both magnitude processing and symmetry processing. Instead of simply extending the mental number line to the left, as in the extension account, this account suggests that negative numbers are understood through a mental reflection of the number line for positives. Each negative number $(-x)$ is perceived as a mirror image of its positive counterpart (x) , reinforcing the concept of inverse relationships. This symmetrical structure captures the unique cognitive demands of negative number understanding by integrating familiar magnitude representations with spatial symmetry (see Figure 2, which is drawn from Figure 14.1 in Varma et al., 2019).

If people have a reflection representation, they should exhibit a specific pattern of distance effects during symbolic magnitude comparison. For negative-negative pair comparisons, people should show a *standard* distance effect, because the negative number mental number line follows the same organization as the positive number line, yielding the standard distance effect (similar to extension).

For mixed positive-negative comparisons, however, a reflection representation predicts an *inverse* distance effect—such that people take *longer* to respond to pairs that are further apart.

Figure 2

Model of the Number-line Reflection Representation from Varma et al. (2019), with the Curves Representing the Psychophysical Scaling of Magnitude



Note. The reference axis at the center helps in locating the positive and negative number mental number lines: the positive numbers above and the negative numbers below, which mirror each other. Standard distance effects are observable in negative-negative comparisons (marked in red) and positive-positive comparisons (marked in blue), and inverse distance effects are observable in the mixed positive-negative comparisons (marked in green).

This phenomenon occurs because of how our minds perceive and process numerical magnitudes in the context of integers, as explained in Varma et al. (2019) and shown in Figure 2. This pattern arises due to the psychophysical scaling of magnitude representations of the mental number line for positive numbers and the corresponding reflection for negative numbers. As

shown in Figure 2, the blue curve represents the natural number mental number line, which is scaled logarithmically. The red curve shows the corresponding mental number line for negative integers, modeled as a reflection of the positive curve but with greater compression. The green lines indicate the perceived psychophysical distances between positive and negative integers across the zero boundary (which Varma and Schwartz modeled from Banks & Hill, 1974; Curtis & Fox, 1969, Dehaene & Mehler, 1992; Moyer & Landauer, 1967; Shepard et al., 1975).

Essentially, when mixed pairs are close together (e.g., -1 and 2), they represent larger differences on the mental number line because they are psychophysically further apart (as indicated by the distance between the two curves), making them easier to distinguish and leading to quicker responses. Conversely, when pairs are farther apart (e.g., -1 and 7), they represent smaller differences on the mental number line, making them harder to distinguish and resulting in slower responses—a phenomenon known as an *inverse* distance effect.

Another potential reason for the inverse distance effect for mixed pairs is that comparisons are easier when numbers are close to zero—a culturally important numerical boundary. This has been observed in other studies examining boundary effects in categorical perception (Harnad, 1990) and numerical cognition (Brysbaert, 1995; Holyoak, 1978; Verguts & De Moor, 2005).

As noted above, RT patterns on positive-negative comparisons are particularly crucial, because the different theoretical accounts yield different predictions for these types of comparisons. In line with the reflection account, several studies have reported *inverse* distance effects for mixed comparisons. For example, in studies that involved simultaneous presentation of mixed pairs, Varma and Schwartz (2011) reported inverse distance effects for samples of graduate students (Experiment 1) and community college students (Experiment 2), though not

for accelerated 6th-grade students. In another study that involved simultaneous presentation of mixed pairs, Vest and Alibali (2021) reported an inverse distance effect in a sample of undergraduates; however, a parallel study with 5th- to 7th- grade students (Vest & Alibali, 2024) did not reveal an inverse distance effect. Finally, one study of adults used a symbolic magnitude classification task, in which participants compared given values to either 5 or -5, and also reported an inverse distance effect for mixed pairs (Krajsci & Igács, 2010). Thus, several studies support the idea that adults rely on a reflection representation. However, it is worth noting that Krajsci and Igács (2010) dismiss their findings as due to an idiosyncratic feature of their experimental task. They note that, in the “-5” condition, both negative numbers between -4 and -1 and positive numbers between 1 and 4 are greater than -5. Consequently, these numbers are easier to process compared to those with absolute values greater than 5 (i.e., 6 through 9 are greater than -5, but -6 through -9 are less than -5). This could explain why they found an inverse distance effect, i.e., that target numerals with an absolute value greater than 5 were processed more slowly.

Components. According to the components account, people process magnitude and polarity separately. If people hold a *components* representation, their comparisons of two negative numbers (e.g., determining which number is greater between -4 and -9) should show a standard distance effect. This is because people may initially consider absolute magnitudes on the natural number line and separately process the negative signs—thus, their RTs should manifest the standard distance effect associated with positive number magnitudes. On the other hand, people who use a components representation should *not* display a distance effect on comparisons with mixed positive-negative number pairs (e.g., -2 and 4). This is because people

Table 1

Summary Table of Studies Supporting Each Account including Distance Effects and Semantic Congruence Effects in RT

Study	Age Group	Country	N	Presentation	Order	Pair Type	Effect
Distance Effect							
Ganor-Stern, Pinhas, Kallai, & Tzelgov (2010) Exp 1	Adults	Israel	N = 34	Sequential	Intermixed	Mixed	Standard
Ganor-Stern, Pinhas, Kallai, & Tzelgov (2010) Exp 1	Adults	Israel	N = 34	Simultaneous	Intermixed	Mixed	None
Ganor-Stern, Pinhas, Kallai, & Tzelgov (2010) Exp 2	Adults	Israel	N = 20	Sequential	Intermixed	Mixed	Standard
Ganor-Stern, Pinhas, Kallai, & Tzelgov (2010) Exp 2	Adults	Israel	N = 20	Simultaneous	Intermixed	Mixed	None
Varma, Schwartz, Lindgren, & Go (2007)	Adults	USA	N = 21	Simultaneous	No Info	Mixed	Inverse
Tzelgov, Ganor-Stern, & Maymon-Schreiber (2009) Exp 1	Adults	Israel	N = 20	Simultaneous	Intermixed	Mixed	None
Varma & Schwartz (2011) Exp 1	Graduate	USA	N = 21	Simultaneous	Intermixed	Mixed	Inverse
Varma & Schwartz (2011) Exp 2	Community College	USA	N = 55	Simultaneous	Intermixed	Mixed	Inverse

Vest & Alibali (2021)	Undergraduate	USA	$N = 55$	Simultaneous	Intermixed	Mixed	Inverse
Ganor-Stern (2012) Experiment 1	Adults	Israel	$N = 22$	Simultaneous	Intermixed	Mixed	None
Krajsci & Igács (2010)	Adults	Hungary	$N = 16$	Simultaneous	Intermixed	Mixed	Inverse
Vest & Alibali (2024)	5th-7th Grade	USA	$N = 54$	Simultaneous	Intermixed	Mixed	None
Varma & Schwartz (2011)	Accelerated 6th	USA	$N = 36$	Simultaneous	Intermixed	Mixed	None
	Grade						
Semantic Congruence Effect							
Shaki & Petrusic (2005)	Adults	Canada	$N = 22$	Simultaneous	Blocked	Negative	Standard
Shaki & Petrusic (2005)	Adults	Canada	$N = 22$	Simultaneous	Intermixed	Negative	Standard
Ganor-Stern et al. (2010) Exp 1	Adults	Israel	$N = 34$	Sequential	Intermixed	Negative	Standard
Varma & Schwartz (2011) Exp 1	Graduate	USA	$N = 21$	Simultaneous	Intermixed	Negative	Standard
Varma & Schwartz (2011) Exp 2	Community						
	College	USA	$N = 55$	Simultaneous	Intermixed	Negative	Standard
Ganor-Stern & Tzelgov (2008) Exp 1	Adults	Israel	$N = 27$	Simultaneous	Intermixed	Negative	None
Tzelgov et al. (2009) Exp 1	Adults	Israel	$N = 20$	Simultaneous	Intermixed	Negative	None
Ganor-Stern (2012) Exp 1	Adults	Israel	$N = 22$	Simultaneous	Intermixed	Negative	Standard
Varma & Schwartz (2011)	Accelerated 6th	USA	$N = 36$	Simultaneous	Intermixed	Negative	None
	Grade						

may note the difference in signs and, based on this component, employ a simple rule: all positive numbers are greater than all negative numbers. This rule allows for fast judgments that do not require processing of the magnitudes of the numbers, as they are based solely on the polarity component.

Again, RT patterns on positive-negative comparisons are particularly crucial for distinguishing theoretical accounts. Several studies have yielded evidence in support of a components representation for positive-negative pairs. In the earliest such study, Tzelgov et al. (2009) presented college students of engineering or mathematics with pairs of numbers in the range of 9 to -9, excluding 0. Critically, participants demonstrated a distance effect for positive-positive pairs and negative-negative pairs, but not for mixed positive-negative pairs, suggesting that they judged mixed positive-negative pairs solely by polarity. Other studies have yielded similar results with both adults (Ganor-Stern, 2012; Ganor-Stern et al., 2010) and with children in the United States, including those in accelerated 6th-grade math (Varma & Schwartz, 2011) and in 5th through 7th grade more generally (Vest & Alibali, 2024).

Taken together, empirical findings about how people represent negative numbers are inconsistent (see Table 1 for a summary of studies). Across studies, there are findings that provide support for each of the three accounts: extension, reflection, and components. Standard distance effects for mixed pairs support the extension account. Inverse distance effects support the reflection account. No distance effects for mixed pairs support the components account. One possible explanation for the mixed findings outlined previously is that people may use different representations depending on context, task demands, or developmental stage. Rather than all people using one fixed type of representation, multiple forms of representation may coexist,

across individuals or even within individuals, and these different representations may be flexibly applied. I will return to this broad issue later. Next, I consider the semantic congruence effect.

Semantic Congruence Effect Predictions from the Extension, Reflection, and Components Accounts

Next, consider the *semantic congruence effect*. As a reminder, the semantic congruence effect is the phenomenon that people associate small values with the semantic meaning of “lesser” and large values with the semantic meaning of “greater” (Banks et al., 1976).

Extension and Reflection Accounts. The *extension* account predicts a *standard* semantic congruence effect for both positive-positive and negative-negative number pairs during symbolic magnitude comparison. This effect reflects a compatibility between the size of the numbers and the direction of the comparison. Specifically, Shaki and Petrusic (2005) argue that negative numbers are associated with “lesser” and positive numbers with “greater.” Under this account, selecting the larger of two large positive numbers (e.g., 9 vs. 7) should be faster than selecting the smaller of two large positive numbers, and selecting the smaller of two large negative numbers (e.g., -9 vs. -7) should be faster than selecting the larger of those same numbers. In other words, performance is facilitated when the magnitude of the numbers and the comparison predicate (greater or lesser) are semantically congruent.

The extension account also predicts a *fully graded* standard semantic congruence effect. According to this stronger version of the hypothesis, the compatibility between numerical magnitude and comparison direction should increase systematically with distance from zero. That is, the SCE should be strongest for comparisons involving the most extreme values (e.g., -9 vs. -7, or 9 vs. 7, in a set of stimuli that range from -9 to 9), and the effect should decrease monotonically as the numerical values approach zero (e.g., -3 vs. -1, or 1 vs. 3). This graded

pattern would reflect an orderly and continuous mental representation of numerical magnitude, with zero serving as the conceptual boundary between negative and positive values. Such a result would support the idea that people mentally extend the number line in a symmetrical and graded fashion across the zero boundary.

The *reflection* account does not specifically predict semantic congruence effects. However, Varma and Schwartz (2011, Appendix A.3) make an auxiliary assumption that positive numbers are semantically associated with “greater” and negative numbers with “lesser”—similar to that of Shaki and Petrusic (2005).¹

Several studies have examined semantic congruence effects in symbolic magnitude comparison tasks involving negative-negative number pairs, typically spanning numerical ranges such as -1 to -9. Most of these studies provide support for the predictions of the extension and reflection accounts. In one early investigation, Shaki and Petrusic (2005) found standard semantic congruence effects in adults under both blocked and intermixed conditions using simultaneous presentation. Similar effects were reported by Ganor-Stern et al. (2010, Experiment 1) in a sequential, intermixed task and by Varma and Schwartz (2011) in two studies involving simultaneous, intermixed presentation, one with graduate students (Experiment 1) and one with community college students (Experiment 2). A later study by Ganor-Stern (2012) likewise reported a standard semantic congruence effect under simultaneous and intermixed conditions. These results are consistent with both the extension and reflection accounts.

Components. In contrast, the components account predicts that comparisons of negative

¹Importantly, in their study, they adopted the convention of mapping size based on absolute value, such that -9 is a large negative number (and thus more strongly associated with “lesser”) while -1 is a small negative number (less strongly associated with “lesser”). Thus, what they call an inverse semantic congruence effect actually corresponds to what I call a standard semantic congruence effect. This difference arises because I do not define “large” and “small” negative numbers based on their absolute values. I treat -9 as a small (more negative) number and -1 as a large (less negative) number. Using this convention, a finding of a standard semantic congruence effect for negative-negative pairs would be consistent with both the reflection and extension accounts.

numbers should exhibit either an *inverse* semantic congruence effect or *no* semantic congruence effect. I describe each of these possibilities in turn.

First, people who use a components representation may display an *inverse* semantic congruence effect if they focus solely on the absolute values of negative numbers. For example, when comparing *small* negative numbers (e.g., -8 and -9), they should respond faster when asked to choose the greater number rather than the lesser. Conversely, when comparing *large* negative numbers (e.g., -1 and -2), they should respond faster when asked to choose the lesser number rather than the greater. This is due to the semantic associations of big and small with the absolute magnitudes.

Shaki and Petrusic (2005) also suggested that with a components representation, people would be expected to demonstrate an *inverse* semantic congruence effect for negative-negative pairs. This is because, when comparing *small* negative numbers (e.g., -8 and -9), participants should respond faster in identifying the larger one (-8) compared to when comparing *large* negative numbers (e.g., -1 and -2). However, the researchers did not find supporting evidence for this pattern in response times; instead, they found a standard semantic congruence effect.

Alternatively, people who use a components representation may display *no* semantic congruence effect if they activate magnitude and polarity components separately and these counterbalance one another. According to Ganor-Stern and Tzelgov (2008), in this scenario, the sign information might dominate, overshadowing the absolute magnitude information, thereby eliminating the semantic congruence effect altogether. Indeed, in a symbolic magnitude comparison study, Ganor-Stern and Tzelgov (2008) discovered a semantic congruence effect for two-digit positive numbers but not for two-digit negative numbers. This outcome aligns with a components representation of negative numbers. In this representation, the minus sign is

combined with the absolute value of the number. Since the minus sign is typically represented as small and the absolute value as large, these opposing effects offset each other, resulting in the elimination of the semantic congruence effect for negative numbers. This evidence was corroborated in another study with single-digit positive and negative numbers (Tzelgov et al., 2009). The results of these studies suggest that negative numbers are not understood as “small” or “large”.

In line with this idea, two studies, Ganor-Stern and Tzelgov (2008, Experiment 1) and Tzelgov et al. (2009, Experiment 1), have reported no semantic congruence effects for negative numbers. Notably, the null finding from Ganor-Stern and Tzelgov (2008) included a wider numerical range extending from -99 to 99, which may have encouraged more componential thinking rather than integrated magnitude processing.

Taken together, the majority of studies support the presence of semantic congruence effects for negative number comparisons, consistent with the extension account and also consistent with the reflection account; however, some studies also report no semantic congruence effect, consistent with a components account. The variability across studies suggests that the presence and magnitude of the effect may depend on contextual differences or individual variation in representation use.

These behavioral patterns are what distinguishes the three accounts (see Table 2 for a summary of predicted effects by representation).

Research Gaps and New Directions

Effects of Development and Context on Negative Number Magnitude Context Representations

From a developmental perspective, Varma and Schwartz (2011) have argued that adults use a *reflection* representation, whereas children, due to their limited exposure to negative

numbers, have not yet developed an integrated magnitude representation for negative numbers. Consequently, Varma and Schwartz (2011) suggested that children tend to reason using a *components* representation (i.e., in terms of natural number magnitude combined with a polarity rule).

Table 2

Hypothesized Distance and Semantic Congruence Effects for Each Representation for Each Comparison Type

Effect	Comparison Type	Extension	Reflection	Components
Distance Effect	Negative-Negative	Standard	Standard	Standard
	Positive-Negative	Standard	Inverse	None
Semantic Congruence Effect	Negative-Negative	Standard	Standard	Inverse or None

Note. A standard distance effect means that people judge pairs of numbers more quickly when the numbers are farther apart, while an inverse distance effect means that people are faster at judging pairs that are closer together. Similarly, a standard semantic congruence effect means people are quicker to choose the greater number when comparing two large numbers and the lesser number when comparing two small numbers. An inverse semantic congruence effect, on the other hand, means people are faster at choosing the greater number from two small numbers and the lesser number from two large numbers.

If people shift from a components to an extension or reflection representation over development, it seems likely that adults may have both types of representations at their disposal.

In other words, adults may typically use extension or reflection representations, but they may still be able to represent numbers in terms of their components when it might be advantageous to do so. If this is the case, adults may adaptively use different representations, depending on the task. This idea aligns with a large body of work on strategy choice in mathematics (e.g., Siegler, 1991, 1998) which suggests that people often have multiple strategies available and use them adaptively depending on the situation.

Indeed, some studies suggest that adults employ different mental representations of negative numbers based on the specific characteristics of the task they are performing. For example, in a study by Shaki and Petrusic (2005), adults' RTs were examined during tasks involving the comparison of symbolic magnitudes between positive and negative number pairs. In one condition, participants were asked to judge all positive number pairs first, followed by all negative number pairs (blocked condition). In another condition, positive and negative number pairs were mixed together (intermixed condition). To determine which mental representation participants employed, Shaki and Petrusic (2005) used a different RT pattern, the Spatial-Numerical Association of Response Codes (SNARC) effect. The SNARC effect represents a systematic association between numerical magnitude and spatial position. It is based on the idea that people tend to associate smaller numbers with the left side of space and larger numbers with the right side of space (Dehaene et al., 1993). In the blocked condition, Shaki and Petrusic (2005) observed a *reverse* SNARC effect for negative numbers, indicating a representation based on separate components for negative numbers. Conversely, in the intermixed condition, they observed a *standard* SNARC effect, suggesting an integrated, magnitude-based representation. This finding suggests that people use an extension or reflection representation of negative

numbers when they are encountered in a context that includes both positive and negative numbers.

In a related study, Ganor-Stern et al. (2010) asked adults to compare magnitudes of positive number pairs, negative number pairs, and mixed positive-negative pairs (e.g., determining which number is greater between 2 and -5). The number pairs were presented in two different formats: simultaneously (both numbers displayed at the same time) or sequentially (one number displayed first followed by the other). When numbers were presented simultaneously, participants could employ a strategy of comparing the polarity signs first. If the signs differed, they could then disregard the magnitude of the digits. In contrast, when numbers were presented sequentially, participants needed to retain both the polarity and magnitude of the numbers in memory simultaneously. The results indicated that adults used different mental representations depending on the timing condition. In the simultaneous condition, where participants could focus on polarity signs first, no distance effect was observed for mixed positive-negative pairs, suggesting a components representation. However, in the sequential condition, where participants had to hold both polarity and number magnitude in memory, a distance effect emerged, suggesting an integrated representation. Thus, adults may adaptively draw on different representations, depending on the demands of the specific task.

In this dissertation, I systematically varied contextual factors and examined whether participants draw on different representations in different contexts. I also examine these patterns in both adults and children, to examine whether participants with different levels of experience and cognitive skills show different patterns of variability across contexts. These aspects represent one novel contribution of the current research—with adults and children examined under contexts never before tested.

Given that I expect to find variability over development and across contexts, it is natural to consider individual factors that may also contribute to shaping patterns of performance and variability. A large body of past research has implicated executive functions (henceforth, EFs) as critical for math performance (e.g., Bull & Scerif, 2001; Cragg & Gilmore, 2014). In this work we explore the possibility that individual differences in EFs are associated with different patterns of performance in magnitude comparisons that involve negative integers, or in explaining contextual variability in these patterns.

Executive Function and Numerical Representations

Executive functions (EFs) are a set of domain-general cognitive skills including working memory, inhibitory control, and cognitive flexibility. These skills play a central role in supporting goal-directed thinking and problem-solving across a wide range of tasks (Miyake et al., 2000). In the context of mathematics, EFs are critical for tasks that involve managing multiple pieces of information, suppressing misleading intuitions, and flexibly adapting to shifting problem structures (Bull & Scerif, 2001; Cragg & Gilmore, 2014). These demands may be especially pronounced when reasoning about negative integers, which are abstract and conceptually counterintuitive for many learners (Bofferding, 2014; Vlassis, 2004). Unlike natural numbers, negative integers do not correspond to concrete quantities in early experience and often require learners to override prior understandings of numbers as always increasing or as measures of “how many.”

As I have argued above, to reason effectively about negative numbers, people may adopt different mental representations. An extension representation involves placing negative numbers on an extended number line left of zero and drawing on spatial-numerical mappings (Fischer & Shaki, 2014). A reflection representation involves mapping negative numbers as symmetrical

counterparts to positive numbers, requiring conceptual integration and symmetry-based reasoning (Varma & Schwartz, 2011). A components representation treats negative numbers as composed of a magnitude and a polarity sign, requiring the application of transformation rules (Nuerk et al., 2004; Tzelgov et al., 2009). These representations differ in their cognitive demands. For example, a components representation may place heavier demands on working memory, as it requires the maintenance and coordination of two distinct components (magnitude and sign) and the application of transformation rules during problem solving. In contrast, a reflection representation may rely more heavily on inhibitory control, as it requires suppressing default left-to-right spatial mappings and overriding entrenched whole number intuitions. The extension representation may require fewer EF resources, as it builds more directly on well-established number line schemas and spatial intuitions.

Prior research supports the idea that individual differences in EF skills relate to how children and adults engage with mathematical content, particularly when tasks involve conflicting representations or require abstract reasoning. For example, working memory and inhibitory control have been linked to performance on symbolic number comparison and arithmetic tasks (Gilmore et al., 2010; Van der Ven et al., 2012), and children with stronger EF skills often show better conceptual understanding of number structure and operations (Fuhs & McNeil, 2013). Within the domain of negative numbers, inhibitory control has been implicated in the ability to suppress positive-number biases (e.g., Coulanges et al., 2021) while cognitive flexibility may be particularly important when students are asked to switch between tasks or representations (see Yeniad et al., 2013, for a meta-analysis).

The present study explores whether executive functions are related to the type of mental representation participants use when reasoning about negative integers. Specifically, I examine

whether individuals with stronger working memory are more likely to use a components representation, whether those with stronger inhibitory control are more likely to use a reflection representation, and whether cognitive flexibility predicts whether participants switch between representations across different task features (e.g., intermixed vs. blocked presentation order). By exploring whether domain-general cognitive abilities are associated with specific representations, this research will contribute to our understanding of how people construct and apply abstract numerical concepts, and how individual differences in EF shape mathematical thinking.

Current Studies

In this dissertation, I analyzed patterns of reaction times during symbolic magnitude comparison tasks involving both positive and negative integers. To investigate how people mentally represent negative numbers, I examined distance effects (how the difference between numbers affects performance) and semantic congruence effects (whether numerical sizes align with the predicate used in instructions), both at the group level and for individual participants. Because previous research suggests that how stimuli are presented can influence the types of representations people use (e.g., Ganor-Stern et al., 2010), I included magnitude comparison tasks in which number pairs were presented in blocks (e.g., all positive-positive pairs first, followed by negative-negative pairs, and finally mixed positive-negative pairs) and tasks in which they were randomly intermixed. Additionally, I varied the presentation format by presenting number pairs either simultaneously (e.g., both 2 and -4 displayed together on the same screen) or sequentially (e.g., 2 shown first, followed by -4). These two factors (order and presentation) were crossed to generate four unique symbolic magnitude comparison conditions.

The primary goal of this research is to examine the cognitive representations of negative numbers that people employ and under what conditions they employ them. Specifically, in two

studies, one with adults and one with children, I addressed three research questions. First, what types of representations (i.e., extension, reflection, or components) do people employ when making magnitude comparisons that involve negative integers? I hypothesize that (1a) both children and adults will show evidence for using each type of representation of negative numbers at the individual level. Previous studies provide evidence that during symbolic magnitude comparison with positive and negative integers, children and adults utilize different mental representations (Varma & Schwartz, 2012; Vest & Alibali, 2024). I further hypothesize that, when comparing the magnitudes of negative numbers or pairs that include both a positive and a negative integer, (1b) most children will rely on a components representation of negative numbers, in which they break the numbers down into parts. In contrast, I predict that (1c) most adults will use a more integrated magnitude representation of negative numbers—i.e., extension or reflection representations—in which they view the numbers as integrated units rather than in terms of their components.

Second, does participants' use of these representations depend on task features (blocked/intermixed stimuli and simultaneous/sequential presentation)? Given past research, I hypothesize that children and adults will employ different representations depending on the order in which they encounter the magnitude comparisons (i.e., blocked versus intermixed) and timing of presentation of pairs (i.e., simultaneously or sequentially).

For adults, I hypothesize that (2a) in the blocked-simultaneous condition, adults will rely on a components representation of negative numbers, as shown in a prior study (Ganor-Stern, 2005). However, in all other conditions, I hypothesize that (2b) adults will use an extension or reflection representation of negative numbers. This is because, with time and experience, adults likely develop well-integrated representations of negative numbers, possibly organized along a

mental number line. Additionally, previous neuroscience research has shown that magnitudes become automatically activated even in routine, well-practiced tasks (Delazer et al., 2005), supporting the idea that adults may use a more automatic and integrated approach. These hypotheses are organized in Table 3.

Table 3

Table with the Experimental Conditions and Hypothesized Representations for Each Sample

Age Group	Presentation	Order	Representation
Adults	Simultaneous	Blocked	Components
		Intermixed	Extension/Reflection
	Sequential	Blocked	Extension/Reflection
		Intermixed	Extension/Reflection
Children	Simultaneous	Blocked	Components
		Intermixed	Components
	Sequential	Blocked	Components
		Intermixed	Extension/Reflection

For children, I hypothesize that (2c) in both blocked conditions and the intermixed-simultaneous condition, children will rely on a components representation of negative numbers. This prediction is based on previous studies showing that adults tend to use a components-based approach when comparisons are blocked. Given that children have less experience with negative numbers, I hypothesize that they are likely to use a components approach.

However, I further predict that (2d) in the intermixed-sequential condition, children may show evidence of using an integrated representation of negative numbers that incorporates both

polarity and magnitude. The sequential nature of the task will require them to hold both magnitude and polarity in memory, while the intermixed format will limit their ability to rely on simple heuristics, encouraging a more integrated representation (see Table 3).

Third, does participants' use of these representations depend on their executive function skills? I hypothesize that (3a) participants with stronger working memory will be more likely to rely on a components representation of negative numbers, given that this representation involves maintaining and coordinating two distinct pieces of information: numerical magnitude and polarity. Second, I hypothesize that (3b) participants with stronger inhibitory control will be more likely to adopt a reflection representation when processing negative integers. This is because the reflection representation may require suppressing more intuitive spatial mappings tied to natural numbers and overriding default biases. Last, I hypothesize that (3c) cognitive flexibility will predict whether participants change their mental representations between order conditions (intermixed vs. grouped). Specifically, I expected that participants with higher cognitive flexibility will be more likely to shift their representation when the order condition changes.

Chapter 2: General Method

To investigate how people mentally represent negative integers and how these representations depend on context, I conducted a series of experimental tasks with both children and adults. The study was designed to test competing theoretical accounts of negative number representation: the components account, the extension account, and the reflection account. These accounts were evaluated by examining RT patterns during symbolic magnitude comparison. In this section, I describe the general procedure and materials used across studies to address the

research questions. In the method sections of Study 1 and Study 2, I provide additional details about the participants and age-specific design decisions relevant to each sample.

Design and Procedure

Each participant completed a session lasting approximately 45 minutes. Adults participated in an unmoderated online setting, while children completed the study via Zoom with an experimenter present to assist with instructions and technical issues. Both age groups completed similar tasks with minor age-appropriate modifications described below.

Children also completed two computerized magnitude comparison tasks. As with adults, stimulus order was manipulated within subjects and presentation format was assigned between subjects (Table 4). However, unlike adults, the comparison prompt was manipulated between subjects: each child was randomly assigned to either the “greater” or “lesser” prompt condition. After each task, children also answered the same strategy reflection question.

Table 4

Table with the Symbolic Magnitude Comparison Conditions

Presentation of Pairs	Order of Comparison Type
Simultaneous	Blocked
	Intermixed
Sequential	Blocked
	Intermixed

Note. “Presentation of Pairs” is a between-subjects factor and “Order of Comparison Type” is a within-subjects factor.

All stimuli were presented on a white background using black text in a sans-serif font.

Number pairs consisted of integers and were formatted to include a “+” sign for positive values (e.g., “+4”, “-3”). In simultaneous trials, the two numbers were presented side-by-side in large font (approximately 80 px), centered horizontally and spaced apart by 50% of the viewport width to ensure clear visual separation. In sequential trials, each number was presented individually in the center of the screen for 1000 ms, with no fixation cross or inter-stimulus interval between the two presentations. Participants then saw a response screen with instructions. Instruction text was displayed in grey, centered near the bottom of the screen with a 200 px top margin.

On each trial, participants judged which of the two numbers was greater or lesser, depending on the prompt. Responses were made using the “A” key to indicate the first number and the “L” key to indicate the second. Before the symbolic magnitude comparison task, participants viewed task-specific instructions and were encouraged to take breaks between blocks. After each condition, they provided open-ended responses about their strategies.

Following the comparison tasks, participants completed a battery of executive function assessments: the Backward Corsi Block Task (to measure working memory), the Flanker Task (to measure inhibitory control), and the Wisconsin Card Sorting Task (to measure cognitive flexibility); see below for details of these tasks and see Appendix 2 for figure samples. These tasks were selected because they assess core executive functions using non-numerical stimuli, minimizing overlap with numerical processing. To support engagement and reduce fatigue, participants were offered breaks between tasks.

Stimuli and Measures

Magnitude Understanding

The magnitude comparison tasks were presented using jsPsych hosted on GitHub. Participants completed symbolic magnitude comparisons involving integer values ranging from -

9 to 9. All possible number pairs within this range were included, except those containing the culturally significant boundary numbers 0 and 5. These were excluded to avoid potential conceptual and perceptual biases, consistent with prior studies that omit such values due to their special status in both mathematical conventions and cognitive processing (e.g., Lyons & Beilock, 2011; Varma & Schwartz, 2011). This resulted in three categories of comparisons: positive-positive (e.g., 2 vs. 4), negative-negative (e.g., -2 vs. -4), and mixed positive-negative (e.g., -2 vs. 4). Additionally, pairs with equal magnitudes but opposite signs (e.g., -1 vs. 1) were excluded to discourage reliance on strategies that ignored numerical magnitude entirely.

The numerical distance between the two numbers in each pair ranged from 1 to 8 for positive-positive and negative-negative comparisons, and from 3 to 8 for mixed positive-negative comparisons. For mixed pairs, distances of 1 and 2 were excluded: a distance of 1 would involve 0, and a distance of 2 would include the equal-magnitude pair -1 vs. 1, both of which violated my inclusion criteria. This resulted in 74 unique number pairs. Each pair was presented in both left-right and right-left order (e.g., 1 vs. 9 and 9 vs. 1), yielding 148 trials per comparison prompt (i.e., “choose the greater number” or “choose the lesser number”).

To avoid confounding numerical distance with overall number size, stimuli were constructed to balance size across distance levels. For positive-positive and negative-negative comparisons, near-distance stimuli paired two small integers or two large integers, while far-distance stimuli paired a small integer with a large integer. For negative numbers, those closer to zero (e.g., -1) were treated as large, and those further from zero (e.g., -9) were treated as small. For mixed comparisons, near-distance stimuli paired a large negative integer with a small positive integer (e.g., -1 vs. 2), while far-distance stimuli paired a small negative integer with a large positive integer (e.g., -9 vs. 7), or vice versa (e.g., -7 vs. 9). This construction scheme

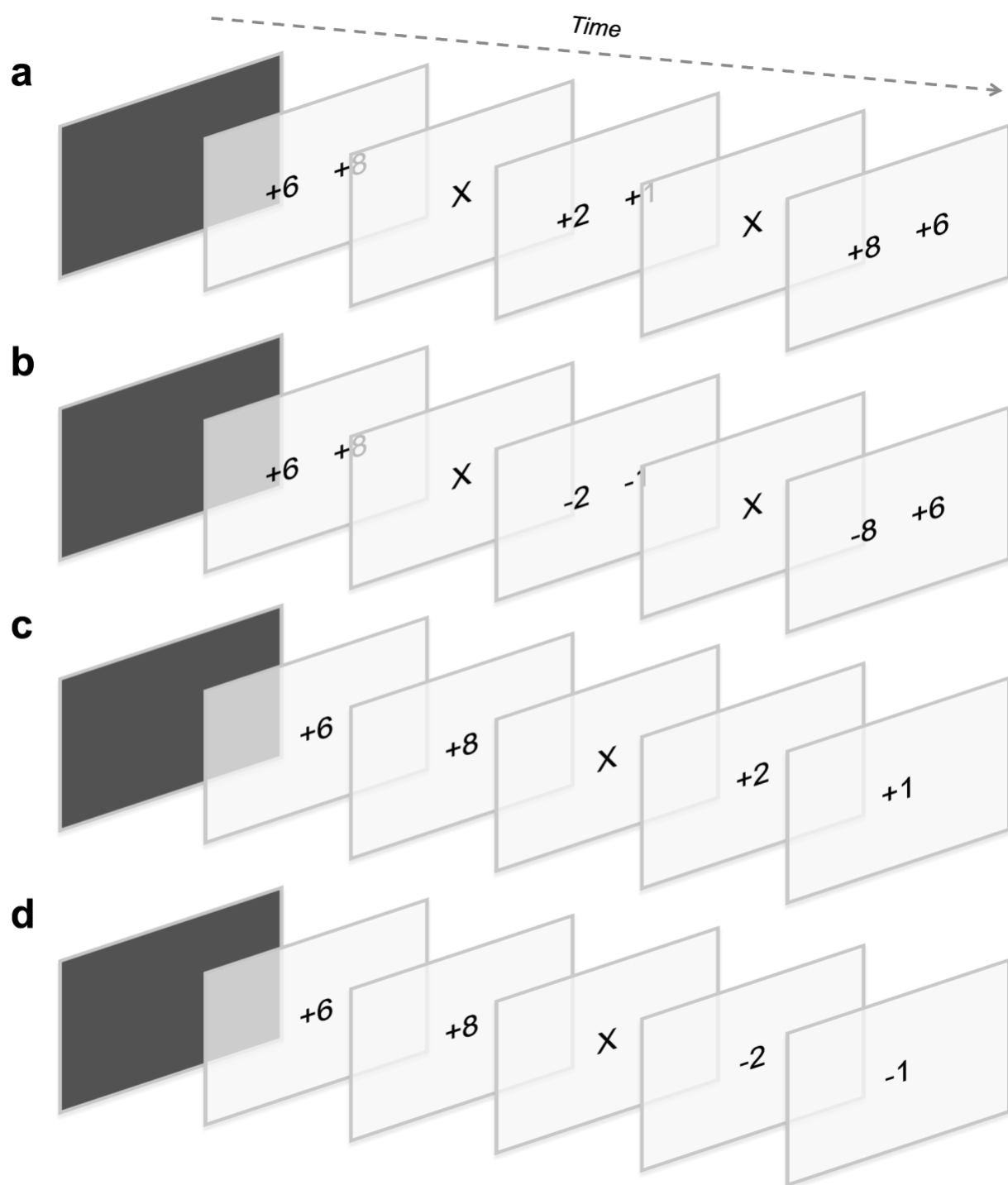
equated the numerical distance of the near and far levels across comparison types and ensured that distance and size were not confounded, particularly in the mixed condition.

For adults, the comparison prompt (predicate) was manipulated within subjects. This meant that each adult completed 148 trials for the “greater” prompt and 148 trials for the “lesser” prompt within each condition; see Appendix 1 for a list of the stimuli. Since adults experienced two stimulus order conditions (blocked and intermixed, a within-subjects factor), they completed 296 trials per condition, for a total of 592 trials across the entire task.

For children, the comparison prompt was manipulated between subjects. Each child was randomly assigned to either the “greater” or “lesser” prompt and thus completed 148 trials per condition. With two stimulus order conditions (blocked and intermixed), children completed 296 trials in total.

On each trial, participants indicated their choice by pressing either the A key (left) or the L key (right) using their left and right index fingers, and reaction time (RT) and accuracy were recorded. To maintain visual consistency and control for perceptual complexity, all positive numbers were displayed with an explicit plus sign (e.g., “+4”), following the recommendation of Krajsci and Igács (2010).

The four conditions varied in how the number pairs were presented. In the simultaneous-blocked condition, two digits were displayed together on the screen at the same time, and comparisons were organized by type, with all positive-positive pairs shown first, followed by negative-negative pairs, and then mixed pairs. In the simultaneous-intermixed condition, the two digits were shown together on the screen, but the comparison types were randomly intermixed, with no grouping by type. The sequential-blocked condition presented the digits one at a time on separate screens, and like the simultaneous-blocked condition, comparisons were grouped by

Figure 3*Demonstration of Each Symbolic Magnitude Comparison Condition*

type. Finally, in the sequential-intermixed condition, the digits were shown one at a time on separate screens, but the comparison types were randomly intermixed rather than blocked. This setup introduces different presentation formats and numerical contexts while ensuring a range of comparison types across conditions (Figure 3).

Working Memory

To assess working memory, I used the digital version of the backward Corsi Block test using PsyToolkit (Stoet, 2017). Each participant was shown nine randomly arranged blocks on a computer screen. Then, the participant was asked to click the same blocks, but in an order that was the reverse of the order in which they were originally presented. Across items, the number of blocks increased until the participant was no longer able to click the blocks in the correct, reverse sequence. Participants' scores were based on how many blocks they were able to tap in the correct order, otherwise known as the participant's "spatial span" score (Arce & McMullen, 2021).

Inhibitory Control

To assess inhibitory control, I used the classic Eriksen Flanker Task (Eriksen & Eriksen, 1974), implemented in PsyToolkit. On each trial, participants viewed five letters presented in a horizontal row. Participants were instructed to respond based only on the identity of the central letter, while ignoring the surrounding "flanker" letters.

The letters "X" and "C" required a response using the left arrow key, while "V" and "B" required the right arrow key. On congruent trials, the flanker letters matched the target (e.g., "CCCCC"), supporting the correct response. On incongruent trials, the flankers conflicted with the target letter's response (e.g., "XXCXX"), creating response interference. This interference

places demands on participants' inhibitory control, as they must suppress the tendency to respond based on the irrelevant flankers.

RTs were recorded for all trials. Inhibitory control was quantified as the mean reaction times on correct incongruent trials. Larger scores reflect greater difficulty inhibiting interference from irrelevant information, and thus weaker inhibitory control.

Cognitive Flexibility

To assess cognitive flexibility, I used a digital version of the Wisconsin Card Sorting Task (Berg, 1948). Participants were shown digital cards, each displaying a different shape type, color, and number of shapes. For example, one card might show two pink circles, while another might show three orange rectangles. Four reference cards were displayed on the screen at the same time. Participants were instructed to match each new card to one of the four reference cards based on an unknown sorting rule.

After each match, participants received feedback indicating whether their choice was correct or incorrect. The sorting rule changed periodically during the task without informing participants. They had to use the feedback to infer the new rule and adjust their matching strategy accordingly.

To quantify cognitive flexibility, I focused on participants' performance after these rule changes. Specifically, I measured the mean RT on correct trials following a rule switch. Outlier trimming was applied at plus or minus 3 standard deviations for each participant. Faster RTs on these post-switch correct trials generally reflect higher cognitive flexibility, indicating more efficient adjustment to new rules.

Chapter 3: Study 1 with Adults

Method

Participants

Participants were 60 undergraduate students enrolled in a course in the Department of Psychology at the University of Wisconsin-Madison. I recruited these students from the departmental subject pool. To ensure data quality, I excluded six participants who had an overall error rate exceeding 45%, resulting in a final sample of 54 participants. Participants' average age was 19 years and 7 months ($SD = 9$ months), with ages ranging from 18 years and 3 months to 46 years and 4 months, and there were 43 women and 11 men. Thirty-eight were White (of these, 3 indicating Hispanic ethnicity), 8 Asian, 2 Native American (of these, 2 indicating Hispanic ethnicity), 4 some other race, and 2 bi- or multi-racial.

All participants completed the study asynchronously and in an unmoderated online context. The sessions lasted approximately 45 minutes. Participants received extra credit in their Introductory Psychology Course for completing the experiment.

Design and Procedure

I used the methods described in the general procedure section. Predicate (i.e., comparison task, “choose greater” or “choose lesser”) was varied within-subjects for adults.

Results

The final sample of participants contributed a total of 31,968 trials. Reaction time (RT) analyses were limited to trials with correct responses; on this basis, 3,749 incorrect trials (11.7%) were removed. I then applied condition-specific RT cutoffs to exclude implausibly fast or slow responses: for sequential trials, I retained only those with RTs between 50 and 3,000 milliseconds; for simultaneous trials, I retained those with RTs between 200 and 3,000 milliseconds. These ranges were selected based on visual inspection of RT histograms, which revealed natural drop-offs suggesting the presence of anticipatory responses (very fast) and

attentional lapses (very slow) within each condition; on this basis, 1,240 trials (4.4%) were removed. Such global exclusion criteria are widely used to eliminate anticipatory responses and attentional lapses (Luce, 1986; Ratcliff, 1993). Following this, I trimmed within-subject outliers by removing RTs more than three standard deviations above or below the mean for each participant * comparison type * condition cell, which excluded an additional 579 trials (2.1%). These cleaning procedures ensured that the final dataset included only valid and representative responses for RT analyses, yielding a total of 26,400 trials.

I examined participants' numerical representations by analyzing distance effects and semantic congruence effects across conditions. For each comparison type—positive, negative, and mixed—I evaluated RT, focusing first on group-level patterns within the three comparison types, and then on individual-level variability, specifically for negative and mixed comparisons.

RQ1: What types of representations (extension, reflection, components) do participants employ when making magnitude comparisons that involve negative integers?

To address this question, I examined participants' performance for the three types of comparisons: positive-positive, negative-negative, and mixed positive-negative. I focused on two key behavioral signatures—numerical distance effects and semantic congruence effects—to infer the types of mental representations participants employed: extension, reflection, or components. These effects were analyzed both at the group level, to capture general trends across participants, and at the individual level, to identify meaningful variability in representations. By combining these analyses, I aimed to characterize how participants represent and reason about negative numbers overall, and to explore whether different comparison types elicit different mental representations.

Distance Effects

As described in the introduction, the distance effect refers to the robust finding that people compare number pairs more quickly when the numerical distance between them is larger (Moyer & Landauer, 1967). For example, distinguishing between 4 and 9 is typically easier than between 4 and 6. This effect is thought to reflect the organization of numerical magnitudes along a mental number line.

Group-Level Analysis. To assess distance effects at the group level, I fit linear mixed-effects models (LMEs) predicting RTs. Each model included numerical distance (1-8) as a continuous predictor, as well as order (blocked vs. intermixed), presentation format (simultaneous vs. sequential), and their three-way interaction, along with all lower-order terms. Predicate was included as a covariate. I also included random intercepts and random slopes for within-subject predictors to account for individual variability. Separate models were fit for positive, negative, and mixed comparisons (see Appendix 3 for full model outputs). In the following text, I report F-values from Type III ANOVAs to provide omnibus tests of the significance of each fixed effect and interaction; these were calculated using the *lmerTest* package, which applies Satterthwaite's approximation for degrees of freedom.

As expected, a standard distance effect was observed in *positive* comparisons: participants responded more slowly to pairs that were closer in magnitude, $b = -8.45$, $F(1, 53.18) = 15.13$, $p < .001$. Note that because the beta reflects change in RT, a negative beta indicates faster responses as numerical distance increases. A similar standard distance effect emerged in *negative* comparisons, $b = -12.72$, $F(1, 202.65) = 51.55$, $p < .001$. I did not make specific predictions for negative comparisons, as a standard distance effect could arise from either a components representation that transforms negatives into positive magnitudes, or from a

Table 5

Distance Effects Reported in Milliseconds

	Overall ¹	Far	Near	$t(53)^2$	p^2	Direction
Positive	742.06 (4.34)	734.43 (7.01)	770.62 (5.50)	-3.89	< .001	Standard
Negative	803.39 (4.60)	770.01 (7.46)	822.23 (5.82)	-7.18	< .001	Standard
Mixed	742.06 (5.41)	735.07 (7.14)	750.84 (8.28)	-0.71	.479	None
Mixed Simultaneous-Intermixed	1098.45 (8.70)	1092.09 (11.34)	1106.37 (13.51)	-0.67 ^a	.502 ^a	None
Mixed Simultaneous-Blocked	733.46 (5.04)	723.63 (6.44)	745.91 (7.97)	-0.50 ^a	.621 ^a	None
Mixed Sequential-Intermixed	505.31 (11.30)	499.17 (15.00)	512.93 (17.18)	0.16 ^a	.873 ^a	None
Mixed Sequential-Blocked	507.68 (11.78)	504.95 (15.61)	511.13 (17.97)	-0.40 ^a	.692 ^a	None

¹Measures are reported as $M (SE)$.

² t -values and p -values are reported to show the size and direction of the main or ^asimple effect of distance (1-8), estimated from a linear mixed-effects model including condition and predicate; p -values were obtained using Satterthwaite's method via the *lmerTest* package.

representation that integrates negative numbers onto the mental number line (extension or reflection). Thus, distance effects alone are not diagnostic for negative comparisons.

In contrast, mixed comparisons (those involving both a positive and a negative number) did not show a significant distance effect at the group level, $b = -1.82$, $F(1, 6320.5) = 0.50$, $p = .479$; see Table 5 for RT means and Appendix 5 for error rates. This finding initially suggests that participants might be relying on a components representation when comparing mixed-sign pairs—that is, rather than processing magnitude, they use a quick heuristic or rule such as “all positive numbers are greater than negatives.” This finding is inconsistent with findings in past work that revealed inverse distance effects for mixed pairs (Varma & Schwartz, 2011, Vest & Alibali, 2021). However, the lack of a significant distance effect at the group level could also reflect an averaging across participants employing different representations. Because these representations produce opposing distance effects (no effect for components, standard effect for extension, inverse effect for reflection), they may cancel each other out when combined in aggregate data. I turn next to assessing individual-level variability in how participants represent negative numbers.

Individual-Level Analysis. To investigate individual variability, I examined each participant’s distance effect on response time (RT) separately for each comparison type. For each participant, I computed a slope quantifying the effect of numerical distance on RT, controlling for predicate type. Following prior work (Vest et al., 2025), I categorized these slopes as reflecting a standard effect (slope < -0.25 SD), an inverse effect (slope $> +0.25$ SD), or no effect (slope between -0.25 and $+0.25$ SD). Importantly, these thresholds were defined relative to zero, using the sample standard deviation of slope values to set cutoffs at $+0.25$ SD or -0.25 SD.

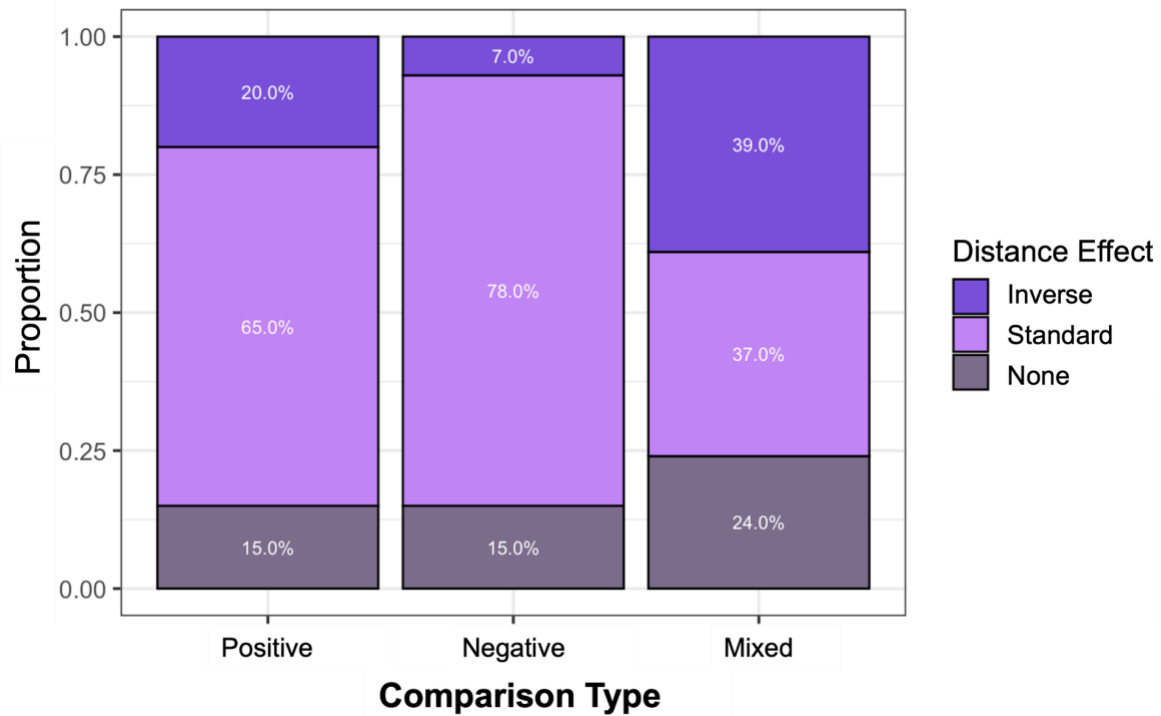
As seen in Figure 4, for *positive* comparisons, 65% of participants exhibited a standard distance effect (35 out of 54), while 20% showed an inverse effect (11 out of 54) and 15% showed no effect (8 out of 54)—closely mirroring the group-level results. Similarly, in *negative* comparisons, 78% of participants (42 out of 54) demonstrated a standard distance effect, suggesting that they represented negative numbers either on an integrated number line or by transforming them into positive equivalents. Only 7% showed an inverse effect (4 out of 54) and 15% showed no effect (8 out of 54).

In contrast, *mixed* comparisons revealed substantial variability. I had hypothesized that adults would predominantly exhibit an inverse distance effect for mixed comparisons, consistent with prior findings suggesting that adults often conceptualize negative integers using a reflection representation (Varma & Schwartz, 2011; Vest & Alibali, 2021). Contrary to this hypothesis, the inverse pattern did not emerge in the individual-level data. While 39% of participants (21 out of 54) showed an inverse distance effect, 37% also showed a standard effect (20 out of 54) and 24% showed no effect (13 out of 54; Figure 4). This variation underscores that different individuals may employ different representations when comparing mixed-sign pairs.

Specifically, those showing no distance effect likely rely on a components representation, treating positive and negative numbers as categorically distinct and applying rule-based representations rather than magnitude comparisons. Participants who display a standard distance effect appear to use an extension representation, mentally placing negatives on a continuous extension of the number line to the left of zero and comparing magnitudes linearly. Conversely, participants with an inverse distance effect demonstrate a reflection representation, mentally flipping negative numbers about zero, which causes faster responses for numbers closer in absolute value despite their opposite signs.

Figure 4

Proportions of Each Distance Effect as a Function of Comparison Type



Together, the range of effects underscores the importance of individual-level analyses and highlights the diverse representations people use when comparing across the zero boundary. These results support prior findings of heterogeneous patterns in mixed comparisons (e.g., Ganor-Stern et al., 2010; Krajsci & Igács, 2007).

Semantic Congruence Effects

To further explore how participants represented magnitude, I examined semantic congruence effects (SCE). These effects capture the tendency to respond more quickly when the magnitude of a number pair aligns with the comparative instruction. For example, participants are typically faster to choose the “lesser” of two small numbers or the “greater” of two large numbers (Banks et al., 1976), reflecting alignment between semantic and numerical content. Note that semantic congruence cannot be meaningfully evaluated for mixed comparisons, as

these pairs inherently involve one small and one large number and therefore do not offer within-pair congruence or incongruence. Therefore, my analysis of the SCE focuses on positive and negative comparisons only.

Group-Level Analysis. I again used LMEMs with RTs to examine semantic congruence effects. The models included predicate (greater vs. lesser), size (small vs. large), order, presentation format, and their four-way interaction, along with all main effects. I also included random intercepts and random slopes for within-subject predictors to account for individual variability. Separate models were conducted for each comparison type. My primary focus was the interaction between predicate and size, which indexes semantic congruence (see Appendix 4 for full model output).

In *positive* comparisons, participants showed a significant semantic congruence effect, $b = 72.93$, $F(1, 2420.4) = 9.62$, $p = .002$; they responded more quickly when the comparative predicate matched the numerical size of the pair (e.g., choosing the “lesser” of two small numbers; Banks et al., 1976). It is worth noting that this effect was statistically reliable only in the simultaneous conditions; effects of condition will be considered in detail in the next section.

For negative comparisons, both extension and reflection representations predict a standard SCE, due to preserved mappings between numerical magnitude and semantic direction (e.g., “greater” aligns with less negative values). In contrast, a componential representation may yield an inverse SCE, reflecting reliance on the magnitude of the corresponding positive number, or could yield no SCE, if polarity and magnitude components counterbalance each other.

At the group level, a standard SCE emerged for negative comparisons: participants responded faster when choosing the greater of two large negative numbers and the lesser of two

Table 6

Semantic Congruence Effects Reported in Milliseconds

	Overall	Greater		Lesser		$t(53)^2$	p^2	Direction
		Small	Large	Small	Large			
Positive	820.26 (6.15)	744.16 (12.20)	739.00 (12.06)	792.81 (14.44)	822.07 (14.99)	3.10	.002	Standard ³
Negative	773.09 (6.71)	860.44 (11.01)	851.30 (14.00)	768.64 (11.07)	799.52 (13.94)	2.03	.042	Standard
Neg Sim-Intermix	1168.77 (9.73)	1190.84 (16.52)	1178.69 (21.24)	1151.70 (19.56)	1143.54 (21.56)	0.20	.840	None
Neg Sim-Blocked	904.47 (7.06)	952.83 (13.39)	915.29 (17.09)	852.96 (11.36)	896.12 (15.39)	2.23	.026	Standard
Neg Seq-Intermix	583.22 (13.20)	624.97 (23.29)	636.82 (30.82)	517.13 (22.63)	551.59 (31.63)	0.60	.547	None
Neg Seq-Blocked	498.79 (11.12)	538.78 (21.71)	542.19 (27.62)	443.57 (16.98)	479.28 (24.33)	1.07	.283	None

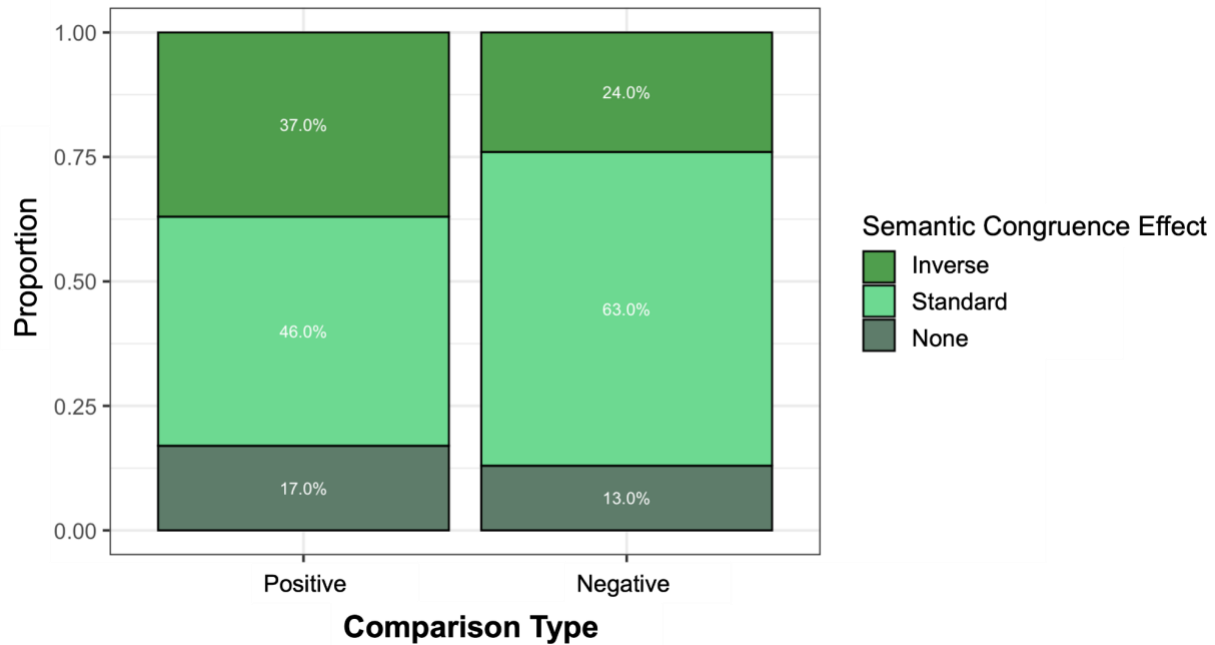
¹Measures are reported in $M (SE)$.

² t -values and p -values are reported for the main or ^asimple two-way interaction between predicate (greater/lesser) and size (small/large), estimated from a linear mixed-effects model with condition as a factor.

³Reflects the standard semantic congruence effect observed specifically in the simultaneous conditions.

Figure 5

Proportions of Each Semantic Congruence Effect as a Function of Comparison Type



small negative numbers. This pattern was statistically significant, as reflected in a predicate * size interaction, $b = 39.78$, $F(1, 5137.6) = 4.12$, $p = .042$. These findings suggest that, on average, participants processed negative magnitudes in a manner consistent with either an extension or reflection representation (see Table 6).

Individual-Level Analysis. To complement the group-level findings, I also examined semantic congruence effects at the individual level. This approach allows us to capture variation in representation use that might be obscured in aggregate analyses. For each participant and comparison type, I computed a semantic congruence slope, defined as the interaction between predicate and size in the participant's own data. Participants were categorized based on their slope values using a threshold of +0.25 or -0.25 standard deviations from zero. Slopes greater than +0.25 SD were classified as showing a standard effect, slopes less than -0.25 SD as showing

an inverse effect, and slopes between -0.25 SD and +0.25 SD as showing no effect. These thresholds were calculated using the sample standard deviation of slope values.

The results aligned with my hypotheses regarding representation types. For positive comparisons, the most common pattern was the standard semantic congruence effect, seen in 46% of participants (25 out of 54), consistent with a spatial magnitude representation. In contrast, 37% (20 out of 54) showed an inverse effect, which may reflect alternative or less typical processing strategies, and 17% (9 out of 54) showed no clear semantic congruence effect.

For negative comparisons, a majority of participants (63%, 34 out of 54) demonstrated the standard semantic congruence effect, suggesting the use of extension or reflection representations in which magnitude and semantic labels (greater/lesser) align. Meanwhile, 24% (13 out of 54) exhibited an inverse effect, consistent with a componential representation that dissociates magnitude from semantic congruence. The remaining 13% (7 out of 54) showed no reliable effect, also consistent with a componential representation in which polarity and magnitude components counterbalance one another (Figure 5).

These patterns provide evidence that most participants represent positive and negative numbers with integrated magnitude-semantic mappings, but a meaningful subset relies on componential approaches, especially for negative numbers.

RQ2: Does participants' use of different representations for negative numbers depend on task features (blocked/intermixed stimuli and simultaneous/sequential presentation)?

I next zeroed in on participants' representations of negative numbers by examining whether their use of components, extension, or reflection-based representations depended on task features. To address this question, I examined whether the effects of distance and semantic congruence varied across task conditions. These task conditions were defined by the presentation

structure—simultaneous or sequential—and by whether comparison types were blocked or intermixed (see Figure 3 for demonstrations). For the distance effect, I focus on mixed positive-negative pairs, as the absence of a distance effect for such pairs is a signature of a components representation of negative numbers, whereas the presence of a distance effect is diagnostic of an integrated number line representation. For the semantic congruence effect, I focus on negative-negative pairs, as the presence an *inverse* semantic congruence effect or the absence of a semantic congruence effect is a signature of a components representation of negative numbers, whereas the presence of a *standard* semantic congruence effect is a signature of an extension or reflection representation of negative numbers.

I expected adults to primarily use extension or reflection representations in negative and mixed comparisons as indicated by their distance and semantic congruence effects. However, in the simultaneous-blocked condition, I anticipated that participants might be more likely to rely on a components representation, quickly making judgments based on the sign alone without fully engaging magnitude comparisons.

Below, I report both group-level and individual-level analyses to capture patterns in both aggregate trends and individual differences.

Distance Effects

As a reminder, the distance effect reflects whether participants' response times varied depending on the numerical distance between values.

Group-Level Analysis. At the group level, I conducted linear mixed-effects models (LMEMs) with RTs as the dependent variable. Key predictors included distance (treated as a continuous variable from 1 to 8 for positive-positive and negative-negative pairs, and 3 to 8 for mixed pairs), order (blocked vs. intermixed), presentation (simultaneous vs. sequential), and their

three-way interaction. All lower-order effects were included, and predicate was included as a covariate. I also included random intercepts and random slopes for within-subject predictors to account for individual variability.

To explore the simple effect of distance within each condition, I re-coded the order and presentation variables to form four task conditions: simultaneous-intermixed, simultaneous-blocked, sequential-intermixed, and sequential-blocked. No statistically significant simple effects of distance were observed in any of the four conditions (see Table 6), indicating that, at the group level, task structure did not elicit a consistent or dominant representation strategy for mixed comparisons. This is also compatible with the idea that, overall, participants used the components representation for mixed positive-negative comparisons.

Individual-Level Analysis. Given that group-level trends may mask substantial individual variability, I next examined distance effects at the participant level within each condition, using the approach described above. Based on theoretical considerations, I hypothesized that participants in the simultaneous-blocked condition would predominantly rely on a components representation, which should manifest as no distance effect. In contrast, I expected that participants in the intermixed conditions (both simultaneous and sequential) would be more likely to use extension or reflection representations, reflected by standard or inverse distance effects, respectively.

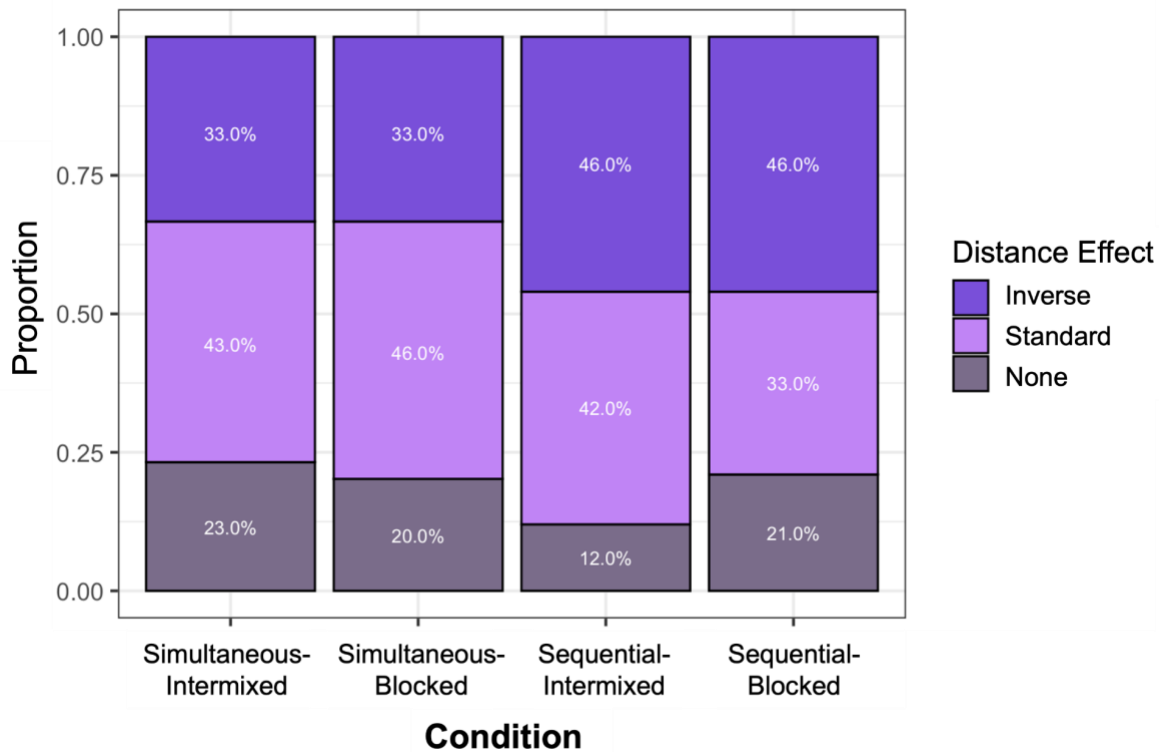
As depicted in Figure 6 and summarized in Table 2, participants exhibited distinct patterns of distance effects corresponding to different mental representations of negative numbers. Specifically, a standard distance effect indicates an extension representation, whereby negative integers are mentally positioned to the left of zero on an extended mental number line. An inverse distance effect is indicative of a reflection representation, in which negative numbers

are conceptualized as reflections about zero. The absence of a distance effect corresponds to a components representation, where magnitude processing relies on separate positive and negative polarity components rather than an integrated spatial representation.

Contrary to my initial hypothesis, participants in the simultaneous-blocked condition were not more likely to exhibit the components representation. Only 20% of participants (6 out of 30) demonstrated no distance effect, while the majority displayed distance effect patterns consistent with extension (47%, 14 out of 30) or reflection (33%, 10 out of 30) representations. Similarly, in the sequential-blocked condition, only 21% (5 out of 24) showed no distance effect, whereas 33% (8 out of 24) and 46% (11 out of 24) exhibited standard and inverse distance effects, respectively.

Figure 6

Distance Effects for Mixed Comparisons in Each Condition



In the simultaneous-intermixed condition, as anticipated, a larger proportion of participants exhibited distance effects aligned with extension (43%, 13 out of 30) or reflection (33%, 10 out of 30) representations, with fewer participants showing no effect (23%, 7 out of 30). The sequential-intermixed condition demonstrated a similar distribution, with 42% extension (10 out of 30), 46% reflection (11 out of 30), and 12% no effect (3 out of 30).

Collectively, these findings indicate that across all task conditions, only a minority of participants predominantly employed components representations. Instead, most participants displayed distance effect patterns consistent with integrated mental representations—either extension or reflection—regardless of presentation mode or task blocking. These results suggest that blocked presentation alone may be insufficient to elicit widespread use of components representations, highlighting lower context-sensitivity in mental representations of negative numbers than previously hypothesized.

Semantic Congruence Effects

Next, I analyzed semantic congruence effects, which measure how participants respond when the magnitude and the semantic predicate (greater or lesser) are either aligned or misaligned. Here, the focus is on *negative* comparisons, as they are particularly informative for distinguishing between different representations of negative number magnitude. I hypothesized that presentation format and stimulus structure would influence the mental representations participants use for negative number comparisons. Specifically, in blocked conditions, I expected participants to rely on components representations during simultaneous presentation, and extension or reflection representations during sequential presentation. In contrast, in intermixed conditions, I anticipated a stronger reliance on extension or reflection representations regardless of presentation format.

Group-Level Analysis. First, I employed LMEMs with RT as the dependent variable. Predictors included predicate (greater vs. lesser), size (small vs. large negative numbers), order, presentation, and their four-way interaction, along with all main effects. I also included random intercepts and random slopes for within-subject predictors to account for individual variability. To examine condition-specific effects, I re-coded the condition variables to create the four task contexts. SCEs were used as behavioral markers: an inverse SCE indicating a componential representation, and a standard SCE or no NCE indicating either an extension or reflection representation.

A significant *standard* semantic congruence effect was observed in one of the four conditions. In the simultaneous-blocked condition, participants were faster when selecting the greater of two large negative numbers (e.g., -1 vs. -2) and the lesser of two small negative numbers (e.g., -8 vs. -9), $b = 0.01$, $F(1, 5116.8) = 4.00$, $p = .046$; see Table 6 above. This finding is consistent with participants relying on either an extension or reflection representation in this condition—contrary to my hypothesis, which predicted greater use of a components representation under simultaneous-blocked presentation. No other condition showed a statistically significant effect at the group level.

Individual-Level Analysis. Individual-level analyses revealed meaningful variability in semantic congruence effects across task conditions (see Figure 7), offering insight into the types of representations participants used when comparing negative numbers.

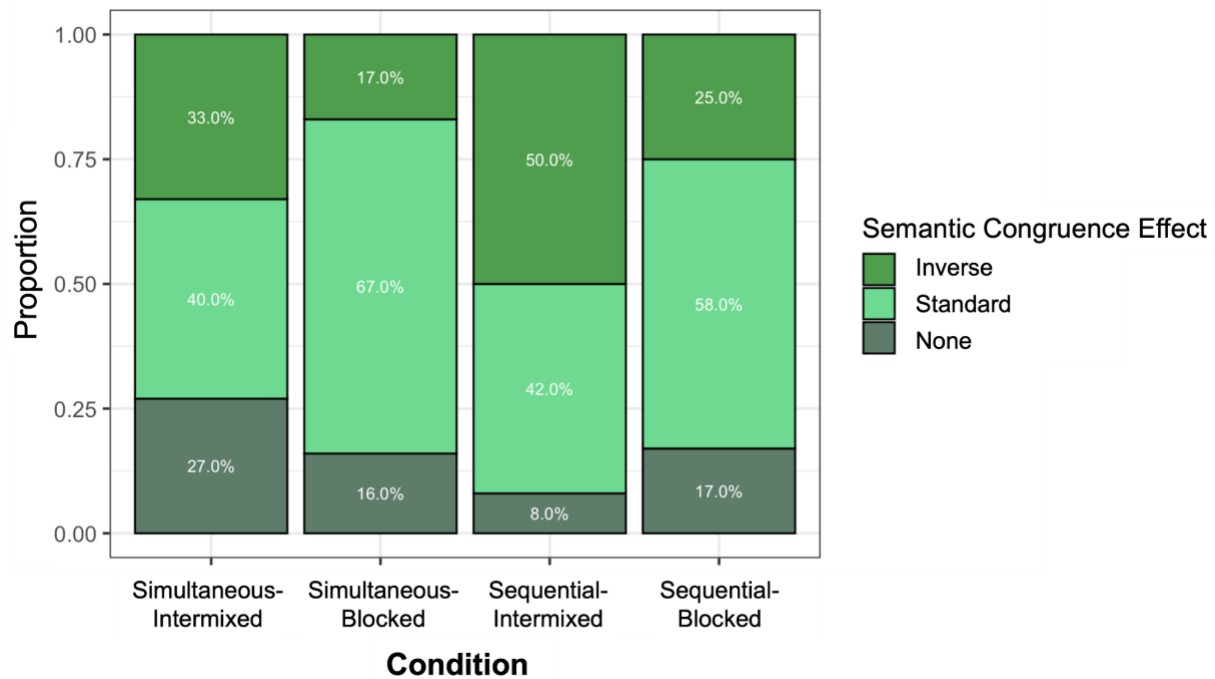
In the simultaneous-blocked condition, most participants (20 out of 30; 67%) showed a standard SCE, consistent with an extension or reflection representation. Fewer participants showed an inverse SCE (5 out of 30; 17%) or no effect (5 out of 30; 17%). A similar pattern was

observed in the sequential-blocked condition: 14 out of 24 (58%) showed a standard SCE, 6 out of 24 (25%) an inverse SCE, and 4 out of 24 (17%) no effect.

By contrast, intermixed conditions revealed greater variability in representations. In the simultaneous-intermixed condition, only 40% (12 out of 30) showed a standard SCE, while 33% (10 out of 30) showed an inverse SCE and 27% (8 out of 30) showed no effect. The sequential-intermixed condition displayed the highest proportion of inverse SCEs (12 out of 24; 50%), with 10 participants (42%) showing a standard SCE and just 2 (8%) showing no effect.

Figure 7

Semantic Congruence Effects for Negative Comparisons in Each Condition



Taken together, these findings suggest that standard SCEs—indicative of integrated number line representations (extension or reflection)—were most common overall. However, components representations—indicated by inverse or null SCEs—were especially prevalent in intermixed conditions, with inverse SCEs prevalent under sequential presentation, and null SCEs

prevalent under simultaneous presentation. These results align with the prediction that stimulus structure and presentation format shape the representations participants adopt when reasoning about negative numbers.

Summary of RQ1 and RQ2

To address the first research question—about what types of representations adults use when comparing negative numbers—my results show that most adults access and use magnitude information, as reflected by both distance effects and semantic congruence effects. At the group level, there were no consistent distance effects for mixed positive-negative pairs, which could suggest a components representation that does not rely on magnitude. However, individual-level analyses revealed that many adults do exhibit clear distance effects in mixed pairs, indicating they access the magnitudes of negative numbers rather than relying solely on sign-based rules. This finding is supported by the semantic congruence effects observed in negative-negative pairs, for which most adults demonstrated standard semantic congruence effects consistent with extension or reflection representations. Together, these behavioral markers confirm that adults frequently use integrated mental representations of negative numbers, engaging with their magnitudes both when negative numbers are compared directly and when mixed with positive numbers.

Regarding the second research question—whether task features influence the representations adults use—the results suggest that the presentation format (simultaneous vs. sequential) and stimulus order (blocked vs. intermixed) do not strongly determine the type of mental representation employed. Across all four task conditions, individual-level analyses in distance and semantic congruence effects persisted, with most participants showing behavioral patterns indicative of extension or reflection representations, regardless of task structure.

Although blocked-simultaneous conditions were hypothesized to promote rule-based, componential representations, most adults exhibited distance and semantic congruence effects consistent with extension or reflection representations in this context as well. Thus, adults appear to access negative number magnitudes robustly across different presentation modes and stimulus orders, indicating that their mental representations of negative numbers are relatively stable and less context-dependent than previously expected.

RQ3: Does participants' use of these representations depend on their executive functions?

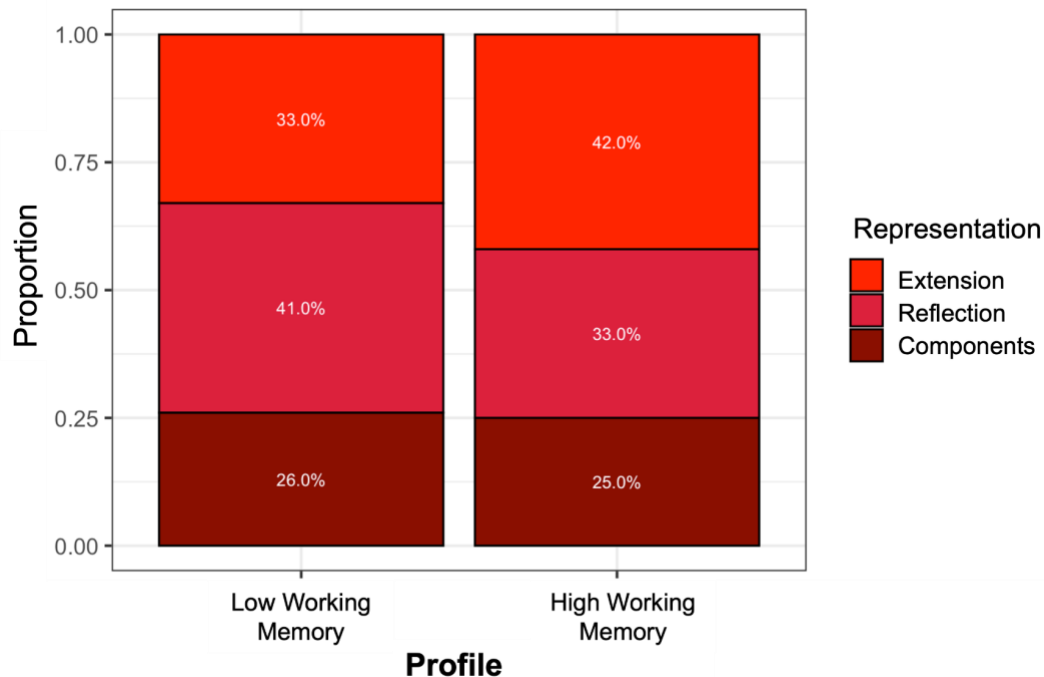
To investigate the relation between executive functions and negative number representations, I analyzed participants' EF scores in relation to the type of representation they displayed, based on the individual analysis of their data for the distance effect for mixed pairs. I focused on distance effects because my analytic procedure for such effects was identical for the studies with adults and children. Specifically, I examined whether participants with stronger EF skills, including greater working memory and stronger inhibitory control, were more likely to adopt certain representation strategies. Additionally, I tested whether cognitive flexibility predicted whether participants changed their representations between order conditions (intermixed vs. blocked). This analysis provides insight into how domain-general cognitive abilities may support different ways of thinking about negative numbers and adapt to varying task demands.

First, I hypothesized that adults with stronger working memory would be more likely to rely on a components representation of negative numbers, given that this approach involves maintaining and coordinating two distinct pieces of information: numerical magnitude and polarity.

To examine whether working memory was related to the type of representation participants used, I conducted a Pearson's chi-square test of independence. This test assesses whether there is a statistically significant relationship between two categorical variables—in this case, working memory level (low vs. high) and representation (extension, reflection, or components). Participants were divided into low and high working memory groups using a median split, and their representation use was categorized based on their distance effect (i.e., RT patterns) for mixed pairs. The distribution of representations across working memory groups is shown below in Figure 8.

Figure 8

Proportion of Participants in Low and High Working Memory Profiles Exhibiting Each Representation



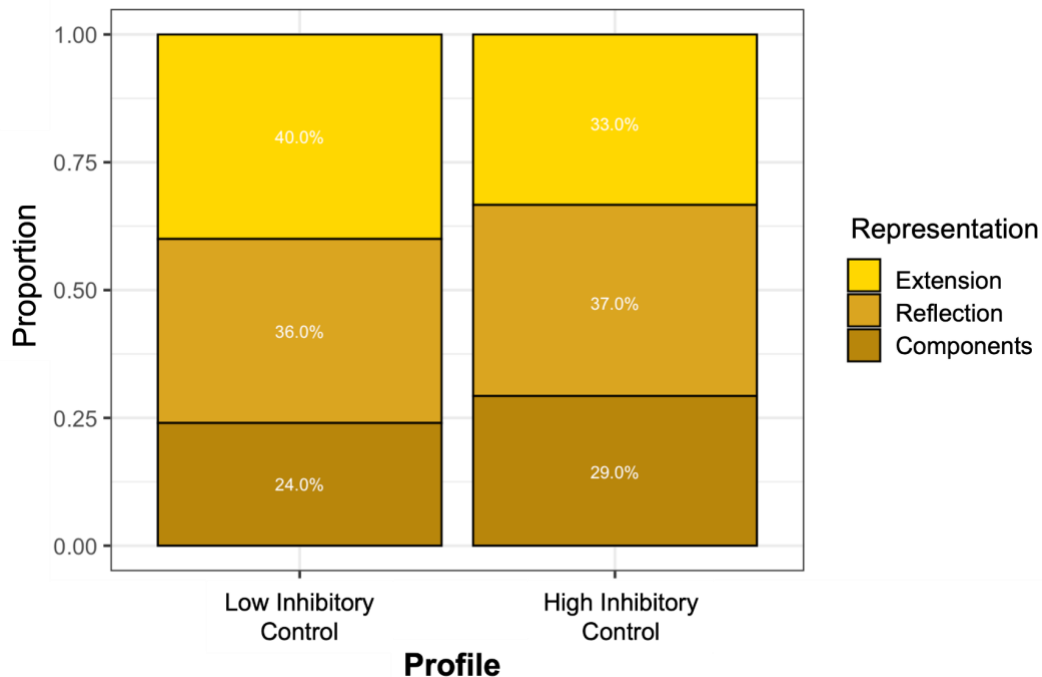
Contrary to the hypothesis, similar proportions of participants in both working memory groups (low: 7 out of 27, high: 6 out of 24) displayed the components representation, and the

distribution of participants across representations did not differ for the two working memory groups, $\chi^2(2, N = 51) = 0.43, p = .807$.

Second, I hypothesized that adults with stronger inhibitory control would be more likely to adopt a reflection representation when processing negative integers. This is because the reflection representation may require suppressing more intuitive spatial mappings tied to natural numbers and overriding default biases. To test this, participants were divided into low and high inhibitory control groups based on a median split, and each participant's negative number representation

Figure 9

Proportion of Participants in Low and High Inhibitory Control Profiles Exhibiting Each Representation



(extension, reflection, or components) was identified based on their individual analysis of the distance effect for mixed pairs. Contrary to the hypothesis, a similar proportion of participants in

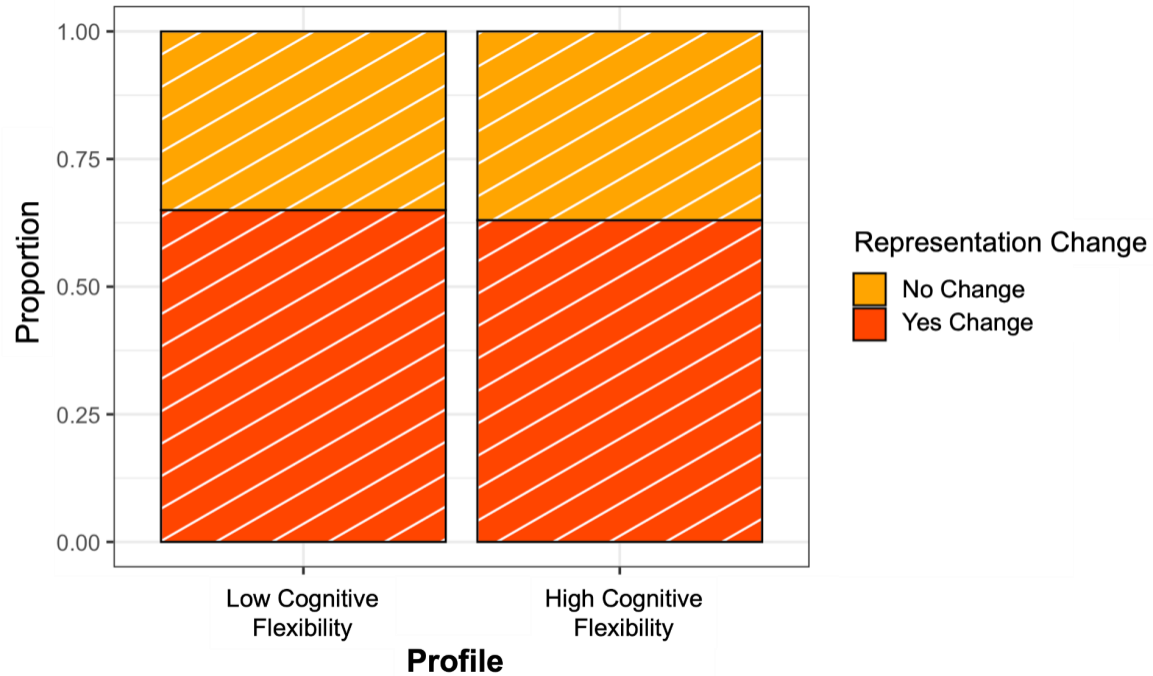
both inhibitory control groups (low: 9 out of 25, high: 9 out of 24) displayed the reflection representation, and the distribution of participants across representations did not differ for the two inhibitory control groups, $\chi^2(2, N = 49) = 0.28, p = .870$; Figure 9.

Third, I hypothesized that cognitive flexibility would predict whether participants changed their mental representations between order conditions (intermixed vs. grouped). Specifically, I expected that participants with higher cognitive flexibility would be more likely to shift their representation when the order condition changed. To address this, I categorized individual participants' distance effects on mixed pairs and identified whether participants used the same representation for both types of trials (no change) or used different representations for both types of trials (change), collapsing across sequential and simultaneous formats. I then categorized participants as low vs. high in cognitive flexibility based on a median split.

Descriptive analyses showed that among participants with low cognitive flexibility, 7 out of 20 (35%) did not change their representations between order conditions, while 13 out of 20 (65%) did change. Among participants with high cognitive flexibility, 7 out of 19 (37%) did not change, and 12 out of 19 (63%) did change representation. A chi-square test revealed no significant association between cognitive flexibility and representation change, $\chi^2(2, N = 39) = 0.28, p = .870$. Thus, there was no evidence that participants with higher cognitive flexibility were more likely to shift their representation use compared to those with lower cognitive flexibility (Figure 10).

Figure 10

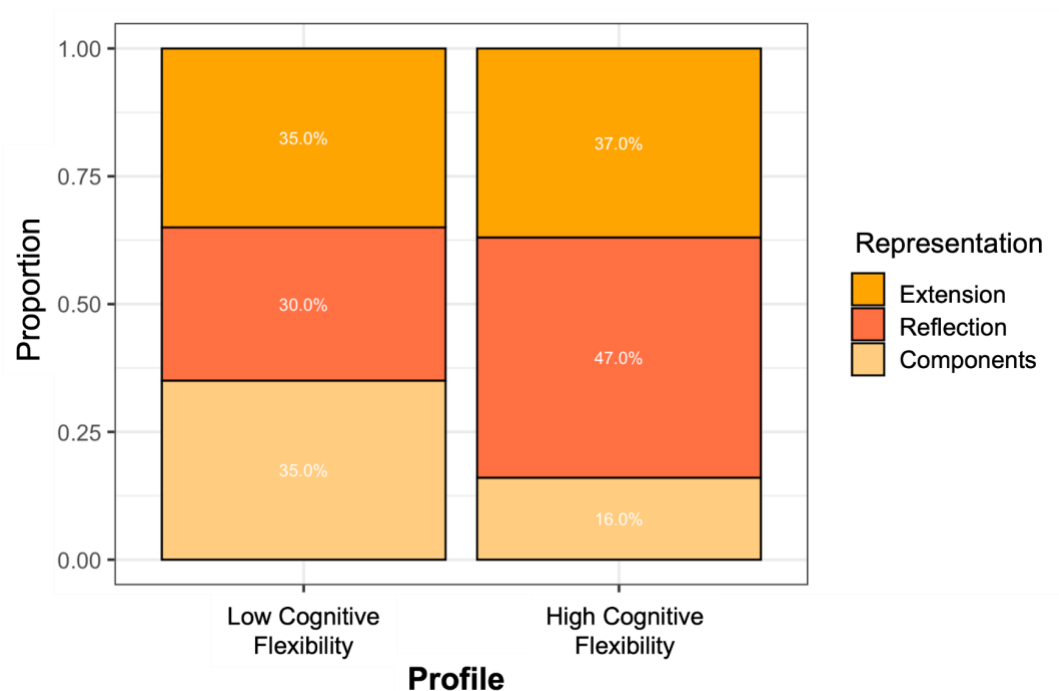
Proportion of Participants in Low and High Cognitive Flexibility Profiles Exhibiting Representation Change or Not



As an exploratory measure, I also examined whether cognitive flexibility was related to the type of representation participants used. Among participants with low cognitive flexibility, the components and extension representations were each used by 7 out of 20 participants (35%), followed closely by reflection (6 out of 20, 30%). In contrast, among participants with high cognitive flexibility, the most common representation was reflection, used by 9 out of 19 participants (47%), followed by extension (7 out of 19, 37%) and components (3 out of 19, 16%). A chi-square test comparing representation strategy by cognitive flexibility level did not reach statistical significance, $\chi^2(2, N = 39) = 2.18, p = .337$. Nevertheless, the nonsignificant trend suggested that participants with high cognitive flexibility may have been more likely to use the reflection representation, whereas those with low cognitive flexibility showed a more even distribution across all three representations (Figure 11).

Figure 11

Proportion of Participants in Low and High Cognitive Flexibility Profiles Exhibiting Each Representation



Discussion

This study addressed three primary research questions. First, I asked what mental representations adults used when comparing pairs of negative and mixed-signed integers. Second, I examined whether these representations varied depending on order and presentation format, specifically, whether the comparison types (positive, negative, or mixed) were blocked or intermixed and whether stimuli were presented simultaneously or sequentially. Last, I tested whether different representations were more likely to be employed by participants with distinct profiles of executive function skills (e.g., low vs. high working memory).

At the group level, adults showed no reliable distance effect when comparing mixed-sign pairs such as -3 versus 5. This absence of a distance effect suggests that participants may not

have been relying on a consistent, integrated mental number line that includes both positive and negative values. Instead, mixed-sign comparisons likely elicited variable or competing representations that cancelled out any systematic effect at the group level. This pattern is consistent with some prior findings. For example, Tzelgov et al. (2009) found that college students showed distance effects for comparisons within the positive or negative range, but not for mixed-sign pairs, suggesting that such comparisons were judged primarily based on polarity. Similar results have been reported with adults (Ganor-Stern, 2012; Ganor-Stern et al., 2010) and with U.S. children, including those in advanced 6th-grade math (Varma & Schwartz, 2011) and in a broad sample of children in grades 5-7 (Vest & Alibali, 2024).

However, other studies *have* found distance effects in mixed comparisons at the group level. For example, in two experiments with adults, Ganor-Stern and colleagues presented numbers sequentially (e.g., -3 on one slide, then 1 on the next; Ganor-Stern et al., 2010), and in this context they reported *standard* distance effects for mixed comparisons, in line with an extension account. Other studies have reported *inverse* distance effects. For example, in studies that involved simultaneous presentation of mixed pairs, Varma and Schwartz (2011) reported inverse distance effects for samples of graduate students (Experiment 1) and community college students (Experiment 2). In another study that involved simultaneous presentation of mixed pairs, Vest and Alibali (2021) reported an inverse distance effect in a sample of undergraduates—providing evidence for a reflection representation. In the current study, an absence of distance effects results suggests that perhaps without compatible contextual cues, adults default to heterogeneous representations or representations that do not draw on a positive-negative integrated mental number line for mixed-sign magnitude comparisons.

It is increasingly recognized that group-level analyses may obscure meaningful variability at the individual level (Fisher et al., 2018). For example, Siegler's overlapping waves theory illustrates how averaging across children can conceal diverse and coexisting representations during cognitive development (Siegler, 1987; Siegler, 1996). A key contribution of the present study is its in-depth examination of individual-level representations. In contrast to the group-level results, individual-level analyses revealed that most adults showed either a standard or an inverse distance effect for mixed-sign comparisons. A standard distance effect, such that participants are faster to respond when the numerical distance is greater, suggests use of an integrated number line representation. An inverse distance effect, such that responses are faster for close pairs that cross the 0 boundary, may reflect the psychophysical representation of a reflected number line. This variability is consistent with growing evidence that adult numerical cognition is far from uniform (see Gilmore, 2023). These findings underscore the importance of conducting individual-level analyses, as group averages can obscure meaningful differences in how participants mentally represent and process numbers.

I also examined whether adults displayed a semantic congruence effect. When comparing two negative numbers, the group-level data revealed a robust, standard semantic congruence effect. Participants were faster at choosing the greater of two larger negative numbers (-1, -2) than the greater of two small negative numbers (-8, -9). This supports the idea that adults do successfully extend the mental number line to include negative values, at least in contexts in which both numbers are negative.

This group-level pattern was also reflected in the individual-level data. Most adults exhibited a standard semantic congruence effect for negative-negative pairs, suggesting a widespread tendency to use integrated number representations in this context. The consistency

across both group- and individual-level results supports the idea that, for many adults, comparing two negative numbers is a well-practiced and relatively intuitive task that reliably activates the mental number line.

I also examined whether participants' representations were influenced by the way stimuli were presented, specifically, whether the representations participants used varied depending on order and presentation format. At the group level, no reliable distance effect emerged for mixed-sign pairs in any of the order or presentation format conditions. Regardless of whether comparisons were blocked or intermixed, or whether they were presented simultaneously or sequentially, the overall pattern remained unchanged. This suggests that task format alone did not substantially alter the variability or consistency of adults' representations for mixed-sign comparisons. This result is consistent with prior findings that behaviors, even those that are maladaptive, with mixed-sign problems can be persistent (e.g., mixed-sign arithmetic, Prather, 2023; Prather & Alibali, 2008). It also points to the persistence of rule-based representation strategies, such as "negatives are always smaller," which may occur regardless of how problems are structured, although this was infrequent in this sample.

Despite the lack of group-level effects, individual-level data again revealed that a majority of participants showed either standard or inverse distance effects in mixed comparisons in all order and presentation format conditions. Some participants shifted representations depending on the condition (see Sankey diagrams in Appendix 6), with different distance effect patterns emerging in blocked versus intermixed trials or in simultaneous versus sequential presentations. These exploratory patterns suggest that presentation format may influence representation use to some extent, but not enough to shift the group-level pattern. Further

research using within-subject designs manipulating these contexts could help clarify whether particular conditions reliably elicit certain kinds of representations.

For negative-negative comparisons, a group-level standard semantic congruence effect was observed most clearly in the simultaneous-blocked condition. In this format, participants saw comparison types in one block and could view both numbers at the same time. This may have reduced cognitive load and supported relational processing, allowing participants to more easily use a spatially organized mental number line with semantic mappings to lesser and greater. My results suggest that adults are particularly likely to activate integrated representations of negative numbers when the task structure encourages consistency and relational reasoning. This highlights the context-dependence of representation use, even in well-practiced symbolic comparison tasks.

At the individual level, most participants showed a standard semantic congruence effect in the blocked conditions, but this was less common in the intermixed conditions. In the intermixed format, in which positive and negative comparisons were mixed within a block, participants may have experienced greater task-switching demands or ambiguity. This likely made it harder to maintain a consistent representation of the negative number line. The reduced prevalence of standard semantic congruence effects in the intermixed conditions highlights how even adults' use of sophisticated numerical representations can be context-sensitive and disrupted by increased cognitive demands.

I also found no associations between participants' executive function skills and the representations they displayed. These null results suggest that, within this undergraduate sample, executive function skills may not play a strong role in determining the type of mental representations adults use for negative numbers. Nevertheless, exploratory trends hinted at a

possible link between cognitive flexibility and use of the reflection representation. Participants with high flexibility were somewhat more likely to use the reflection representation, while those with lower flexibility showed a more even distribution across representations. Although these patterns were not statistically significant, they raise the possibility that more subtle or continuous measures of flexibility—or a larger sample—might reveal stronger associations.

Taken together, the findings demonstrate both variability and context sensitivity in how adults represent negative integers. While group-level analyses revealed consistent use of a mental number line for negative-negative comparisons, though not for mixed comparisons, individual-level data showed that most participants employed either extension or reflection representations, even for mixed pairs. These patterns suggest that most adults reason with an integrated mental number line, but occasionally under certain conditions they may default to componential, rule-based representations. Importantly, these representations appear to be relatively stable across tasks and do not seem to vary systematically with individual differences in executive function—at least not in simple categorical terms.

Chapter 4: Study 2 with Children

Method

Participants

Participants were 60 children in the 6th and 7th grades recruited from a list of families who had previously participated in studies in the research lab. Each participant was compensated \$15. To ensure data quality, we excluded six participants who had an overall error rate exceeding 45%, resulting in a final sample of 54 participants. Participants' average age was 12 years and 5 months ($SD = 7$ months), with ages ranging from 10 years and 9 months to 13 years and 4 months. There were 25 girls, 27 boys, and 2 who identified as non-binary. Forty-seven were

White (of these, 3 indicating Hispanic ethnicity), 2 Asian, 1 Native American, 1 Black/African American, 1 some other race, and 2 bi- or multi-racial.

All participants completed the study on Zoom with an experimenter present to read instructions and resolve technical difficulties. The sessions lasted approximately 45 minutes.

Design and Procedure

For comparison tasks, I used the methods described in the general procedure section. However, note that predicate (i.e., instructions, “choose greater or lesser”) was between subjects for children to reduce the length of the study.

Results

For the symbolic magnitude comparison task, the final sample of participants contributed a total of 15,984 comparison trials (e.g., 3 vs. 7). To ensure data quality, I used the same exclusion criteria as described for adults. Thus, reaction time (RT) analyses were limited to trials with correct responses; on this basis, 1,326 incorrect trials (8.30%) were removed. I then applied condition-specific RT cutoffs to exclude implausibly fast or slow responses: for sequential trials, I retained only those with RTs between 50 and 3,000 milliseconds; for simultaneous trials, I retained those with RTs between 200 and 3,000 milliseconds. These ranges were selected based on visual inspection of RT histograms, which revealed natural drop-offs (similar to those of the adults) suggesting the presence of anticipatory responses (very fast) and attentional lapses (very slow) within each condition; on this basis, 746 trials (5.09%) were removed. Following this, I trimmed within-subject outliers by removing RTs more than three standard deviations above or below the mean for each participant * comparison type * condition cell, which excluded an additional 211 trials (1.5%). These cleaning procedures ensured that the final dataset included only valid and representative responses for RT analyses, yielding a total of 13,701 trials.

I examined participants' numerical representations by analyzing distance effects and semantic congruence effects across conditions. For each comparison type—positive, negative, and mixed—I evaluated RT, focusing first on group-level patterns within the three comparison types (i.e., positive-positive, negative-negative, and mixed), and then on individual-level variability, specifically for negative and mixed comparisons.

RQ1: What types of representations (extension, reflection, components) do participants employ when making magnitude comparisons that involve negative integers?

To address this question, I examined participants' performance across the three types of comparisons: positive-positive, negative-negative, and mixed (positive-negative). I focused on two key behavioral signatures—numerical distance effects and semantic congruence effects—to infer the types of mental representations participants employed: extension, reflection, or components. These effects were analyzed both at the group level, to capture general trends across participants, and at the individual level, to identify meaningful variability in representations within individual participants. By combining these analyses, I aimed to characterize how participants represent and reason about negative numbers overall, and to explore whether different comparison types elicit different mental representations of negative integers.

Distance Effects

As described in the introduction, the distance effect refers to the robust finding that people compare number pairs more quickly when the numerical distance between them is larger (Moyer & Landauer, 1967). For example, distinguishing between 4 and 9 is typically easier than distinguishing between 4 and 6. This effect is thought to reflect the organization of numerical magnitudes along a mental number line.

Group-Level Analysis. To assess distance effects at the group level, I fit linear mixed-effects models (LMEMs) predicting RTs. Each model included numerical distance (1-8) as a continuous predictor, as well as order (blocked vs. intermixed), presentation format (simultaneous vs. sequential), and their three-way interaction, along with all lower-order terms. Predicate was included as a covariate. I also included random intercepts and random slopes for within-subject predictors to account for individual variability. Separate models were fit for positive, negative, and mixed comparisons (see Appendix 7 for full model outputs). In the following text, I report F-values from Type III ANOVAs to provide omnibus tests of the significance of each fixed effect and interaction; these were calculated using the *lmerTest* package, which applies Satterthwaite's approximation for degrees of freedom.

As expected, for *positive* comparisons, a standard distance effect was observed, replicating past findings of a distance effect for positive integer comparisons in children (De Smedt et al., 2009; Holloway & Ansari, 2009; Lyons et al., 2014). Participants responded more slowly to pairs that were closer in magnitude, $b = -10.70$, $F(1, 5148.7) = 15.55$, $p < .001$. Note that because the beta reflects change in RT, a negative beta indicates faster responses as numerical distance increases. A similar standard distance effect emerged in *negative* comparisons, $b = -13.35$, $F(1, 5089.2) = 20.90$, $p < .001$. I did not make specific predictions for negative comparisons, as a standard distance effect could arise from either a components representation that focuses solely on magnitudes, or from a representation that integrates negative numbers onto the mental number line (extension or reflection). Thus, for negative

Table 7*Distance Effects Reported in Milliseconds*

	Overall ¹	Far	Near	$t(42)^2$	p^2	Direction
Positive	952.70 (8.02)	928.29 (13.11)	966.36 (10.13)	-3.94	< .001	Standard
Negative	993.97 (8.53)	960.00 (13.76)	1013.17 (10.84)	-4.57	< .001	Standard
Mixed	941.59 (10.11)	943.12 (13.76)	939.67 (14.88)	1.33	.185	None
Mixed Simultaneous-Intermixed	1415.55 (15.60)	1413.51 (21.54)	1418.08 (22.56)	-0.25 ^a	.806 ^a	None
Mixed Simultaneous-Blocked	1021.93 (11.50)	1004.51 (15.34)	1043.86 (17.32)	-0.22 ^a	.826 ^a	None
Mixed Sequential-Intermixed	689.31 (22.20)	734.94 (31.67)	631.52 (30.09)	3.02 ^a	.003 ^a	Inverse
Mixed Sequential-Blocked	537.18 (17.01)	528.35 (22.95)	548.42 (15.36)	-0.32 ^a	.750 ^a	None

¹Measures are reported as $M (SE)$.

² t -values and p -values are reported to show the size and direction of the main or ^asimple effect of distance (1-8), estimated from a linear mixed-effects model including condition and predicate; p -values were obtained using Satterthwaite's method via the *lmerTest* package.

comparisons, distance effects alone are not diagnostic of mental representations of negative integers.

In contrast, mixed comparisons (those involving both a positive and a negative number) did not show a significant distance effect at the group level, $b = 6.03$, $F(1, 3274.5) = 1.76$, $p = .185$; see Table 7 for RT means and Appendix 9 for error rates. This finding initially suggests that, at the group level, participants might be relying on a components representation when comparing mixed-sign pairs—that is, rather than processing magnitude, they use a quick heuristic or rule based solely on the polarity component of the numbers, such as “all positive numbers are greater than negatives.” This finding is consistent with findings in past work with children (Varma & Schwartz, 2011; Vest & Alibali, 2021). However, the lack of a significant distance effect at the group level could also reflect averaging across participants employing different representations. Because different representations may produce opposing distance effects (no effect for components, standard effect for extension, inverse effect for reflection), they may cancel each other out when combined in aggregate data.

Individual-Level Analysis. To investigate individual variability, I examined each participant’s distance effect on response time (RT) separately for each comparison type. For each participant, I computed a slope quantifying the effect of numerical distance on RT. Following prior work (Vest et al., 2025), I categorized these slopes as reflecting a standard effect (slope $< 0 - 0.25$ SD), an inverse effect (slope $> 0 + 0.25$ SD), or no effect (slope between -0.25 and $+0.25$ SD). Importantly, these thresholds were defined relative to zero, using the sample standard deviation of slope values to set cutoffs at $+0.25$ SD or -0.25 SD.

As seen in Figure 10, for *positive* comparisons, 35 out of 54 children (65%) exhibited a standard distance effect, indicating that a majority of the children used a well-established mental

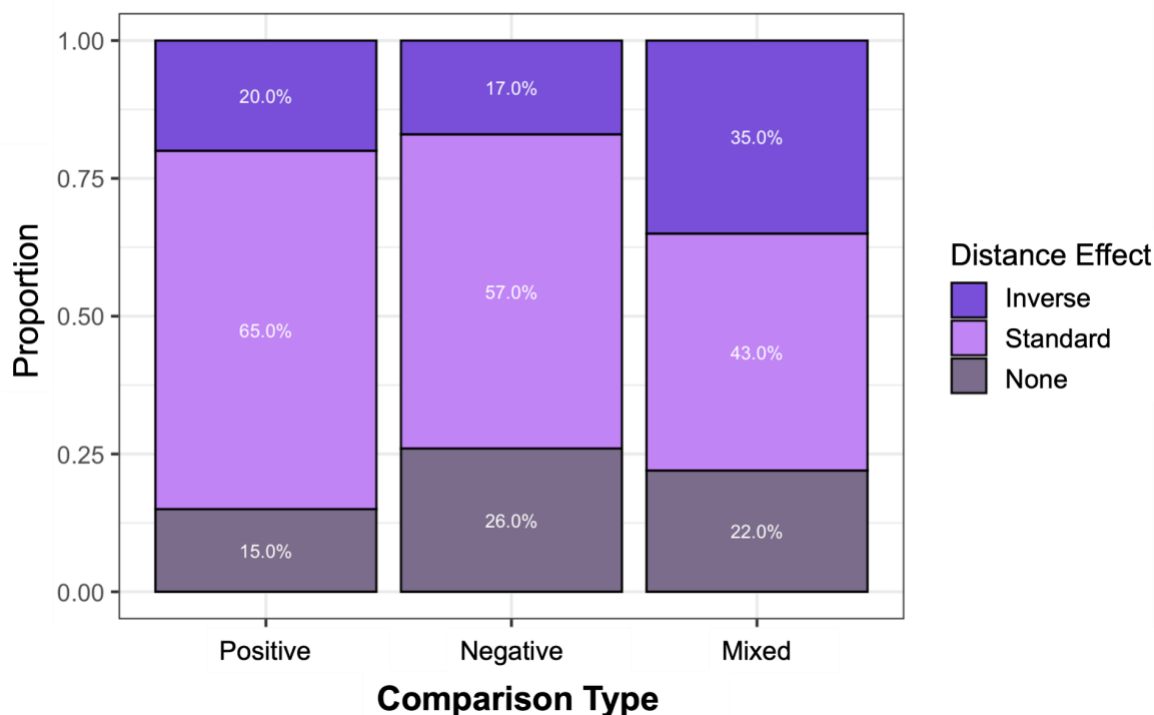
number line for positive numbers. Eleven children (20%) showed an inverse distance effect, and eight (15%) showed no DE. This distribution of individual patterns closely mirrors the group-level effects, and it suggests a moderate level of consistency across children in their representations of positive magnitudes.

For *negative* comparisons, the pattern was similar. Thirty-one out of 54 children (57%) showed a standard distance effect. Nine children (17%) showed an inverse distance effect, while 14 (26%) showed no distance effect. These findings suggest that most children activate numerical magnitudes during negative-negative comparisons. However, because standard distance effects are not diagnostic of whether children are using positive magnitudes or integrated negative magnitudes, the underlying representation remains unclear.

For *mixed* comparisons, the data revealed the greatest variability and diverged from our initial hypothesis. Based on prior work suggesting that children often lack integrated representations of negative numbers (Varma & Schwartz, 2011; Vest & Alibali, 2024), we predicted that most children would show no distance effect in mixed comparisons, consistent with a components representation. However, this was not the case. Only 12 out of 54 children (22%) showed no distance effect. In contrast, 23 children (35%) showed a standard distance effect, and 19 children (43%) showed an inverse distance effect. These findings suggest that many children may have engaged in integrated, magnitude-based comparisons across the zero boundary, using either an extension or, more frequently, a reflection representation—a mental representation that I had expected to emerge later in development.

Figure 10

Proportions of Each Distance Effect as a Function of Comparison Type



These findings underscore the importance of individual-level analyses in capturing the diversity of children’s mental representations. Children showing no distance effect likely apply componential, rule-based reasoning rather than relying on magnitude. Those who displayed a standard distance effect appear to extend their number line representations to the left to include negatives, enabling comparisons that integrate magnitude and polarity. Finally, those showing an inverse distance effect may be mentally reflecting the negative numbers about zero, leading to faster responses for numbers closer in absolute value but on opposite sides of zero. The presence of all three patterns in the sample—especially in mixed-sign comparisons—highlights that multiple representational formats may coexist in development and that children’s use of representations may vary across individuals.

Semantic Congruence Effects

To further explore how participants represented magnitude, I examined semantic congruence effects. These effects capture the tendency to respond more quickly when the

magnitude of a number pair aligns with the comparative instruction. For example, participants are typically faster to choose the “greater” of two large numbers or the “lesser” of two small numbers (Banks et al., 1976), reflecting alignment between semantic and numerical content. Note that semantic congruence cannot be meaningfully evaluated for mixed comparisons, as these pairs inherently involve one small and one large number and therefore do not offer within-pair congruence or incongruence. Therefore, my analysis of the semantic congruence effect focuses on positive and negative comparisons only.

Group-Level Analysis. I again used LMEMs with RTs to examine semantic congruence effects. The models included predicate (greater vs. lesser; between subjects in this sample with children), size (small vs. large), order (blocked vs. intermixed), presentation format (simultaneous vs. sequential), and their four-way interaction, along with all lower-order main effects. I also included random intercepts and random slopes for within-subject predictors to account for individual variability. Separate models were conducted for each comparison type. My primary focus was the interaction between *predicate* and *size*, which indexes semantic congruence (see Appendix 8 for full model output).

In *positive* comparisons, this group-level analysis revealed a standard semantic congruence effect in the predicted direction, but it did not reach statistical significance, $b = 61.92$, $F(1, 2154.51) = 2.91$, $p = .088$; participants responded more quickly when the comparative predicate matched the numerical size of the pair (e.g., choosing the “lesser” of two small numbers; Banks et al., 1976).

For *negative* comparisons, both extension and reflection representations predict a standard semantic congruence effect, due to established mappings between numerical magnitude and semantic direction (e.g., “greater” aligns with less negative values). In contrast, a

components representation may yield an inverse semantic congruence effect, reflecting reliance on the magnitude of the corresponding positive number, or no semantic congruence effect, if polarity and magnitude components counterbalance each other.

At the group level, I found a standard semantic congruence effect for negative comparisons in the predicted direction—participants responded at faster speeds when choosing the greater of two large negative numbers (-1, -2) and the lesser of two small negative numbers (-8, -9). However, this effect did not reach statistical significance, as shown by the non-significant interaction between predicate and size, $b = 61.90$, $F(1, 2843.96) = 3.02$, $p = .082$. These results suggest that, on average, participants may have processed negative numbers in a way consistent with either the extension or reflection representations (see Table 8).

Individual-Level Analysis. To complement the group-level findings, I also examined semantic congruence effects at the individual level to better capture variation in representations that may be hidden in aggregate data. In this study with children, the predicate (i.e., whether participants were instructed to choose the greater or the lesser number) was manipulated between subjects to limit task length. Therefore, I analyzed *size* effects separately within each predicate group. Traditionally, the size effect refers to faster responses for smaller numbers (Parkman, 1971; Dehaene & Mehler, 1992; Krajcsi et al., 2016). However, in this context, I interpret size effects within each predicate condition as an index of the semantic congruence effect. Because predicate was a between-subjects variable, I could not statistically test a size*predicate interaction in a single model for individual participants. Therefore, in the following analyses, I describe how response patterns change across levels of numerical size for participants within each predicate group, using a modified definition of the size effect tailored to my design.

For each participant and each comparison type, I calculated a slope representing the size effect (small coded -.5, large coded .5) based on that participant's individual data. Participants were categorized into three groups based on these slope values: those with slopes greater than +0.25 SD from zero were classified as showing a standard size effect, those with slopes less than -0.25 SD as showing an inverse size effect, and those within +0.25 SD and -0.25 SD as showing no size effect. These thresholds were determined using the sample standard deviation of slope values.

This approach allowed me to identify individual-level representations. For both positive and negative comparisons, a standard size effect is expected in the “greater” predicate group, where participants are faster to judge pairs of large numbers, and an inverse size effect is expected in the “lesser” group, where participants are faster to judge pairs of small numbers. For negative comparisons specifically, this pattern would be consistent with an extension or reflection representation of negative numbers. In contrast, for negative comparisons, a consistent *inverse* size effect across both predicates, indicating judgments based on absolute (positive) magnitude rather than integer value, would suggest a componential representation. Alternatively, a finding of no size effect would also suggest a componential representation, if magnitude and polarity counterbalance one another.

For positive comparisons, the most common pattern in the “greater” group was the inverse effect, observed in 42% of participants (14 out of 33). The standard effect was seen in 33% (11 out of 33), and no effect in 24% (8 out of 33). This suggests that although many participants relied on a typical spatial-magnitude alignment (i.e., faster responses to larger

Table 8

Semantic Congruence Effects Reported in Milliseconds

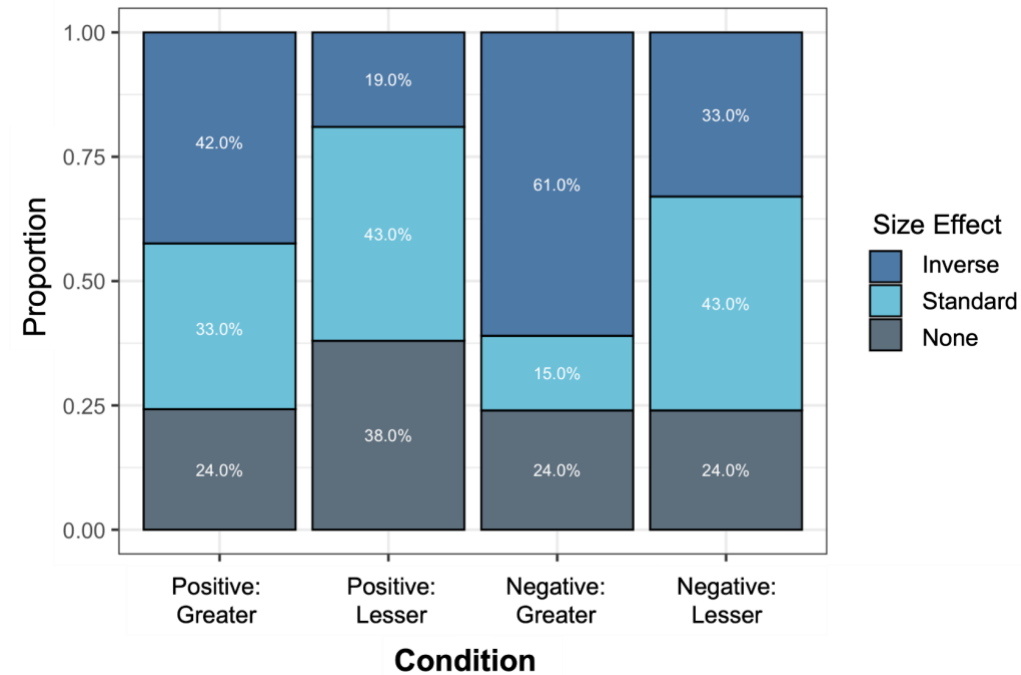
	Overall	Greater		Lesser		$t(42)^2$	p^2	Direction
		Small	Large	Small	Large			
Positive	979.53 (12.52)	974.98 (20.71)	954.26 (20.60)	984.29 (31.95)	1024.39 (32.63)	1.71	.088	(Standard)
Negative	1017.35 (11.53)	1099.35 (18.39)	1041.51 (23.21)	917.00 (24.01)	918.98 (30.17)	1.74	.082	(Standard)
Neg Sim-Intermix	1505.17 (17.09)	1523.93 (24.08)	1503.85 (34.03)	1485.50 (42.34)	1471.74 (54.51)	0.05	.961	None
Neg Sim-Blocked	1220.50 (15.08)	1272.94 (22.09)	1174.18 (28.25)	1208.46 (37.52)	1159.76 (42.64)	0.77	.444	None
Neg Seq-Intermix	694.66 (22.34)	770.92 (40.56)	689.94 (46.96)	591.66 (36.73)	705.43 (57.89)	2.59	.010	Standard
Neg Seq-Blocked	551.92 (18.47)	601.15 (35.81)	572.45 (41.10)	518.27 (32.60)	480.71 (33.33)	-0.15	.883	None

¹Measures are reported in $M (SE)$.

² t -values and p -values are reported for the main or ^asimple two-way interaction between predicate (greater/lesser) and size (small/large), estimated from a linear mixed-effects model with condition as a factor.

Figure 11

Proportions of Each Size Effect as a Function of Comparison Type and Predicate



numbers when asked “which is greater”) a greater proportion showed an inverse pattern, possibly reflecting alternative representations or weaker semantic associations (Figure 11).

In contrast, for the “lesser” predicate in positive comparisons, the most common pattern was the standard effect, found in 43% of participants (9 out of 21), followed by no effect (38%, 8 out of 21), and the inverse effect (19%, 4 out of 21). This pattern suggests that standard semantic alignment between “lesser” and small numbers was uncommon for children.

For negative comparisons, the majority of participants in the “greater” group showed an inverse effect, with 61% of participants (20 out of 33) responding faster to smaller (i.e., less negative) numbers. Only 15% (5 out of 33) showed the standard pattern, and 24% (8 out of 33) showed no effect. This supports the idea that participants often relied on absolute magnitude rather than signed value, consistent with a componential representation of negative numbers. In

the “lesser” negative group, the pattern was more evenly distributed: 43% (9 out of 21) showed the standard effect, 33% (7 out of 21) the inverse effect, and 24% (5 out of 21) showed no effect.

Together, these patterns suggest that many children struggled to integrate relational language (“greater”/“lesser”) with signed number concepts. Another possible explanation is that the “greater” predicate elicits componential thinking in many children while the “lesser” predicate elicits variable responses across children. Even when comparing positive numbers, for which standard effects might be expected to dominate, many children appeared to apply alternative representations or lacked stable mappings. For negative numbers, the prevalence of inverse effects for the “greater” predicate indicates that many children may default to reasoning based on absolute magnitude rather than the signed integer values, further supporting the presence of a componential mental representation of negative numbers.

RQ2: Does participants’ use of different representations for negative numbers depend on task features (blocked/intermixed stimuli and simultaneous/sequential presentation)?

I next zeroed in on participants’ representations of negative numbers by examining whether their use of extension, reflection, or components representations depended on task features. To address this question, I examined whether the effects of distance and semantic congruence varied across conditions. These conditions were defined by the presentation structure—simultaneous or sequential—and whether comparison types were blocked or intermixed (see Figure 3 for demonstrations). For the distance effect, I focus on mixed positive-negative pairs, as the absence of a distance effect for such pairs is a signature of a componential representation of negative numbers, whereas the presence of a distance effect is diagnostic of an integrated number line representation. For the semantic congruence effect, I focus on negative-negative pairs, as the presence an *inverse* semantic congruence effect or *no* semantic congruence

effect is a signature of a components representation of negative numbers, whereas the presence of a *standard* semantic congruence effect is a signature of an extension or reflection representation of negative numbers.

I expected children to primarily use components representations when comparing negative and mixed number pairs. However, in the sequential-intermixed condition, I anticipated that participants would be more likely to use an extension or reflection representation, since this condition requires holding magnitude information in working memory due to sequential presentation and may draw on magnitude information across different types of pairs due to intermixed order.

Below, I report both group-level and individual-level analyses to capture patterns in both aggregate trends and individual differences.

Distance Effects

As a reminder, the distance effect reflects whether participants' response times varied depending on the numerical distance between values. To examine how children's mental representations of negative numbers varied by condition, I conducted both group- and individual-level analyses of distance effects.

Group-Level Analysis. At the group level, I analyzed the data using linear mixed-effects models (LMEMs) with response times (RT) as the dependent variable. Key predictors included distance (treated as a continuous variable from 3 to 8 in mixed comparisons), order (blocked vs. intermixed), presentation (simultaneous vs. sequential), and their three-way interaction. All lower-order effects were included, and predicate was included as a covariate. I also included random intercepts and random slopes for within-subject predictors to account for individual variability.

To explore the simple effect of distance within each condition, I re-coded the order and presentation variables to form four conditions: simultaneous-intermixed, simultaneous-blocked, sequential-intermixed, and sequential-blocked.

I predicted that the simultaneous-blocked, simultaneous-intermixed, and sequential-blocked conditions would primarily elicit components representations, reflected in the absence of a distance effect. Only in the sequential-intermixed condition did I expect to see more frequent use of integrated representations, with some children potentially showing standard or inverse distance effects, consistent with extension or reflection representations.

As seen in Table 7, of these four groups, only the *sequential-intermixed* condition showed a statistically significant distance effect at the group level. Overall, and in line with my hypothesis, children in this condition exhibited a significant *inverse* distance effect, $b = 30.55$, $F(1, 174.10) = 9.09$, $p = .003$; see Table 7. While inverse distance effects have previously been reported in adults (Varma & Schwartz, 2011; Vest & Alibali, 2021), this is the first report of a significant group-level inverse effect in children. Further, the sequential-intermixed condition is the condition that I expected would be most likely to yield evidence for integrated magnitude representations, because I expected sequential presentation would require participants to hold magnitude information in working memory, and because intermixing different types of pairs could encourage the use of an integrated magnitude representation. None of the other three conditions showed significant effects of distance, suggesting that children tended to use components representations in those conditions. This is in line with my hypothesis that overall, children would tend to use components representations.

Individual-Level Analysis. Given that group-level trends may mask substantial individual variability, I next examined distance effects at the participant level within each condition, using the approach described above.

My hypotheses were guided by the idea that most children would rely on componential reasoning in the absence of strong contextual support for integrating negative numbers into a spatial mental model. Specifically, I predicted that the simultaneous-blocked, simultaneous-intermixed, and sequential-blocked conditions would primarily elicit components representations, reflected in the absence of a distance effect. Only in the sequential-intermixed condition did I expect to see more frequent use of integrated representations, with some children potentially showing standard or inverse distance effects, consistent with extension or reflection representations.

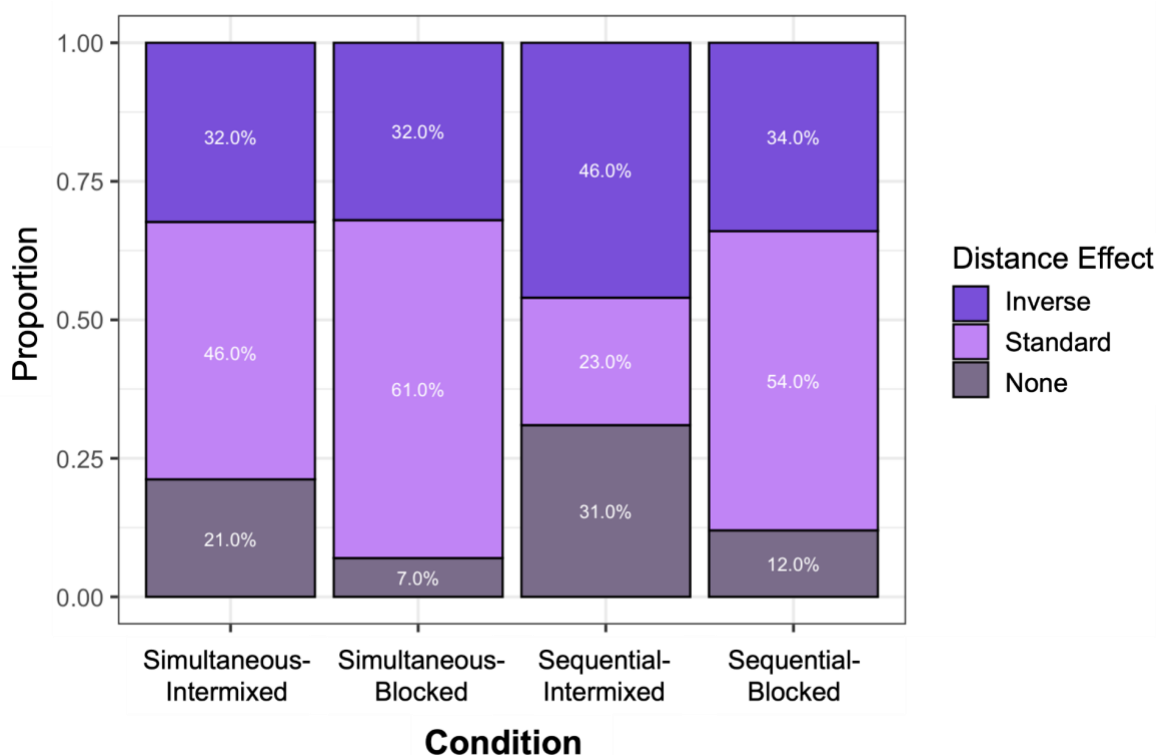
Contrary to these predictions, at the individual level, children in *all four* conditions showed substantial use of integrated representations; see Figure 12. In the simultaneous-blocked condition, only 7% of participants (2 out of 28) showed no distance effect. In contrast, 61% (17 out of 28) showed a standard distance effect and 32% (9 out of 28) showed an inverse distance effect. Similarly, in the sequential-blocked condition, just 12% of participants (3 out of 26) showed no distance effect, while 54% (14 out of 26) exhibited a standard distance effect and 34% (9 out of 26) showed an inverse distance effect.

In the simultaneous-intermixed condition, again, fewer participants relied on components representations than predicted: only 21% (6 out of 28) showed no distance effect, while 46% (13 out of 28) showed a standard distance effect and 33% (9 out of 28) showed an inverse distance effect. Only the sequential-intermixed condition came close to my original hypothesis, showing the highest proportion of children with no distance effect (31%, 8 out of 26). However, even in

this condition, 46% (12 out of 26) showed an inverse distance effect, and 23% (6 out of 26) showed a standard distance effect. Notably, this individual-level pattern aligns with the significant inverse distance effect observed at the group level.

Figure 12

Distance Effects for Mixed Comparisons in Each Condition



Taken together, these results show that in all four conditions, most children demonstrated either standard or inverse distance effects—indicative of integrated processing of magnitude and polarity. Only a minority relied on componential, rule-based representations, even in task conditions expected to elicit them. These findings challenge my initial hypothesis that three of the four conditions would be dominated by components representations. Instead, they suggest that children’s mental representations of negative numbers are more robust and less context-sensitive than hypothesized. The presence of extension and reflection representations across

conditions indicates that many children are capable of engaging with negative numbers as meaningful magnitudes, even in the absence of explicit instructional support or scaffolding—and that in group-level analyses, standard and inverse patterns may cancel one another out.

Semantic Congruence Effects

Next, I analyzed semantic congruence effects, which measure how participants respond when the magnitude and the semantic predicate (greater or lesser) are either aligned or misaligned. Here, my focus is on *negative* comparisons, as they are particularly informative for distinguishing between different representations of negative number magnitude. I predicted that presentation format and stimulus structure would influence the mental representations participants use for negative number comparisons. Specifically, in the blocked conditions, I expected participants to rely on components representations during simultaneous presentation, and extension or reflection representations during sequential presentation. In contrast, in intermixed conditions, I anticipated a stronger reliance on extension or reflection representations, regardless of presentation format.

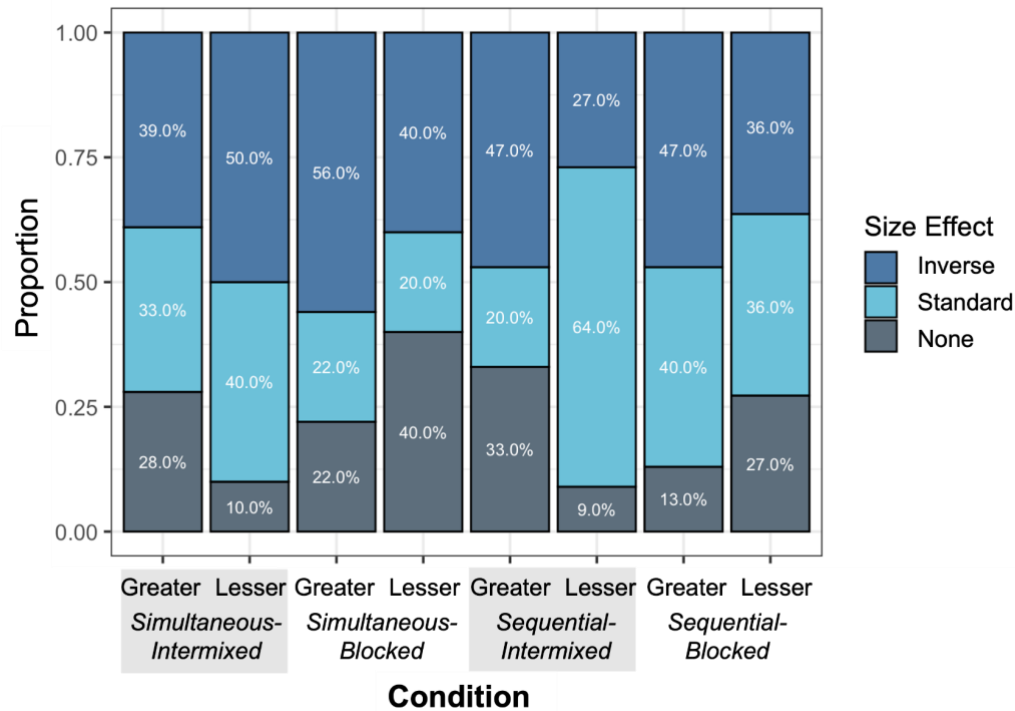
Group-Level Analysis. First, I employed LMEMs with RT as the dependent variable. Predictors included predicate (greater vs. lesser), size (small vs. large negative numbers), order, presentation, and their four-way interaction, along with all main effects. I also included random intercepts and random slopes for within-subject predictors to account for individual variability. To examine condition-specific effects, I re-coded the condition variables to create the four conditions. SCEs were used as behavioral markers, with an inverse SCE or no SCE indicating a components representation, and a standard SCE indicating either an extension or reflection representation.

At the group level, a significant standard semantic congruence effect was found only in the simultaneous-blocked condition. In this condition, participants responded faster when choosing the greater of two large negative numbers (e.g., -1 vs. -2) and the lesser of two small negative numbers (e.g., -8 vs. -9), $b = 142.58$, $F(1, 1169.63) = 3.88$, $p = .049$ (see Table 8). This result suggests that participants used an extension or reflection representation in this condition. This finding does not align with my hypothesis, which predicted more reliance on components representations in the simultaneous-blocked context. Participants did not show significant semantic congruence effects at the group level in any of the other conditions, suggesting the use of a components representation in which polarity and magnitude counterbalance one another.

Individual-Level Analysis. Recall that in this study with children, predicate (i.e., whether participants were instructed to choose the greater or the lesser number) was manipulated between subjects. As a result, I analyzed the effect of (numerical) size separately within each predicate group (see the more thorough analytic explanation provided above for RQ1). For both positive and negative comparisons, a standard size effect is expected in the “greater” predicate group, such that participants are faster to judge pairs of large numbers, and an inverse size effect is expected in the “lesser” group, such that participants are faster to judge pairs of small numbers. For negative comparisons specifically, this pattern would be consistent with an extension or reflection representation of negative numbers. In contrast, for negative comparisons, a consistent inverse size effect across both predicates, indicating judgments based on absolute (positive) magnitude rather than integer value, would suggest a components representation. Alternatively, the absence of a size effect for both predicates could also suggest a components representation, with the polarity and magnitude components counterbalancing one another.

Figure 13

Semantic Congruence Effects for Negative Comparisons in Each Condition



Due to substantial variability in the data and the limited sample sizes, I could not draw definitive inferences regarding consistent semantic-magnitude patterns (see Figure 13). Notably, across conditions, I observed more variability in the “lesser” predicate—with the proportion of participants displaying inverse patterns ranging from 27% to 50% across predicate*condition combinations and the proportion displaying standard patterns ranging from 20% to 64%. The data suggest that traces of traditional magnitude-processing patterns may persist even across contexts of varying demands, and that behavioral patterns may be especially variable under certain contexts.

Summary of RQ1 and RQ2

Thus far, these analyses have addressed two key research questions. RQ1 asked what

types of mental representations children use when comparing negative integers—specifically, whether they rely on componential representations, or instead use integrated representations such as extension (placing negatives on a leftward number line) or reflection (mirroring around zero). RQ2 asked whether children’s use of these representations depends on task features, including whether trials were blocked or intermixed and whether numbers were presented simultaneously or sequentially.

In response to RQ1, analyses of distance and semantic congruence effects revealed substantial individual variability. While some children showed no sensitivity to numerical distance, consistent with components-based reasoning, many others demonstrated either standard or inverse distance effects. This indicates that they were engaging in integrated comparisons, even across the zero boundary. Mixed comparisons revealed the greatest diversity in patterns, suggesting that multiple representational formats coexist in development and are flexibly applied.

In response to RQ2, task conditions influenced representational patterns to some extent, but not in the ways that were predicted. Although the sequential-intermixed condition yielded a significant group-level inverse distance effect, use of integrated representations was observed in all four task conditions. Components representations were less dominant than hypothesized. Only the simultaneous-blocked condition revealed a significant semantic congruence effect at the group level, and individual-level data revealed considerable variability across all contexts.

Together, these findings suggest that although task features may shape children’s reasoning in subtle ways, children are generally capable of engaging with negative numbers as meaningful magnitudes across a variety of contexts. This points to a greater degree of robustness

and flexibility in children's developing representations of negative integers than previously assumed.

RQ3: Does participants' use of these representations depend on their executive functions?

To investigate the relation between executive functions and negative number representations, I analyzed participants' EF scores in relation to the types of representation they used, as assessed in individual analyses of distance effects for mixed pairs. Specifically, I examined whether participants with stronger EF skills, including greater working memory and stronger inhibitory control, were more likely to adopt certain representations. Additionally, I tested whether cognitive flexibility predicted whether participants changed their representations between the blocked and intermixed conditions. These analyses provide insight into how domain-general cognitive abilities may support different ways of thinking about negative numbers and adaptation to varying task demands.

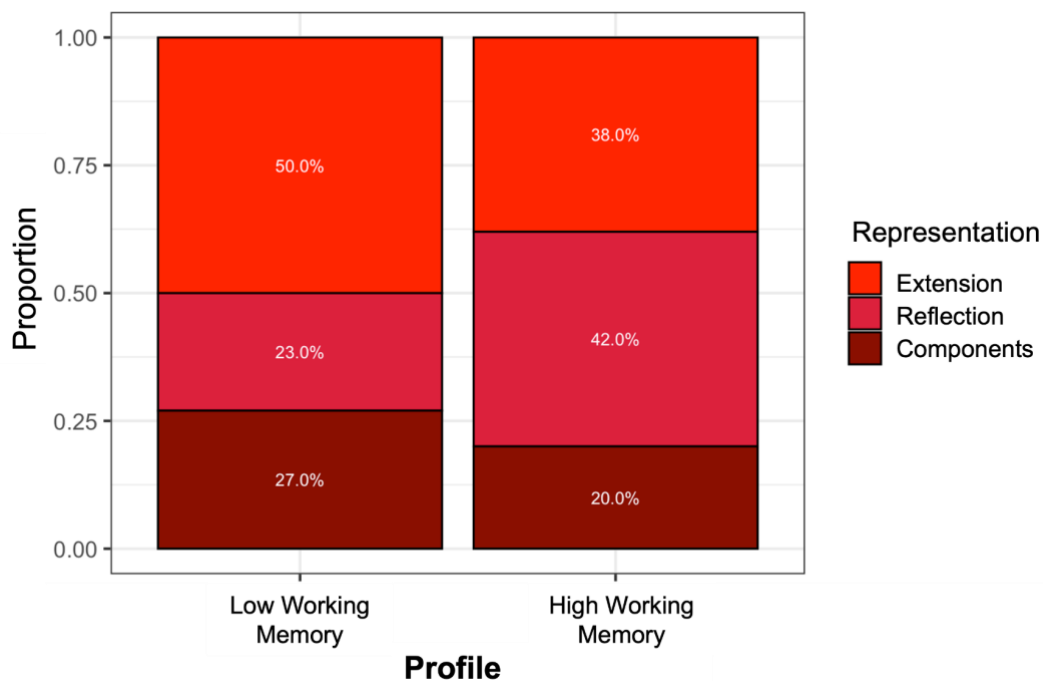
First, I hypothesized that participants with stronger working memory would be more likely to rely on a components representation of negative numbers, given that this representation involves maintaining and coordinating two distinct pieces of information: numerical magnitude and polarity. To test this, participants were divided into low and high working memory groups based on a median split, and their negative number representation (extension, reflection, or components) was identified from individual-level analyses of the distance effect for mixed pairs. Descriptively, participants with higher working memory were *less likely* to use a components representation. Only 5 of 24 high working memory participants (21%) used the components representation, compared to 7 of 27 low working memory participants (26%). High working memory participants were most likely to use a reflection representation (10 of 24; 42%), while low working memory participants were most likely to use an extension representation (13 of 27;

48%).

To examine whether working memory was related to the type of representation participants used, I conducted a Pearson's chi-square test of independence. The distribution of representations across working memory groups is shown below in Figure 14. The chi-square test revealed that the association between working memory group and representation strategy was not statistically significant, $\chi^2(2, N = 52) = 1.98, p = .371$. Thus, there was no evidence that representation use varied between participants with high versus low working memory.

Figure 14

Proportion of Participants in Low and High Working Memory Profiles Exhibiting Each Representation

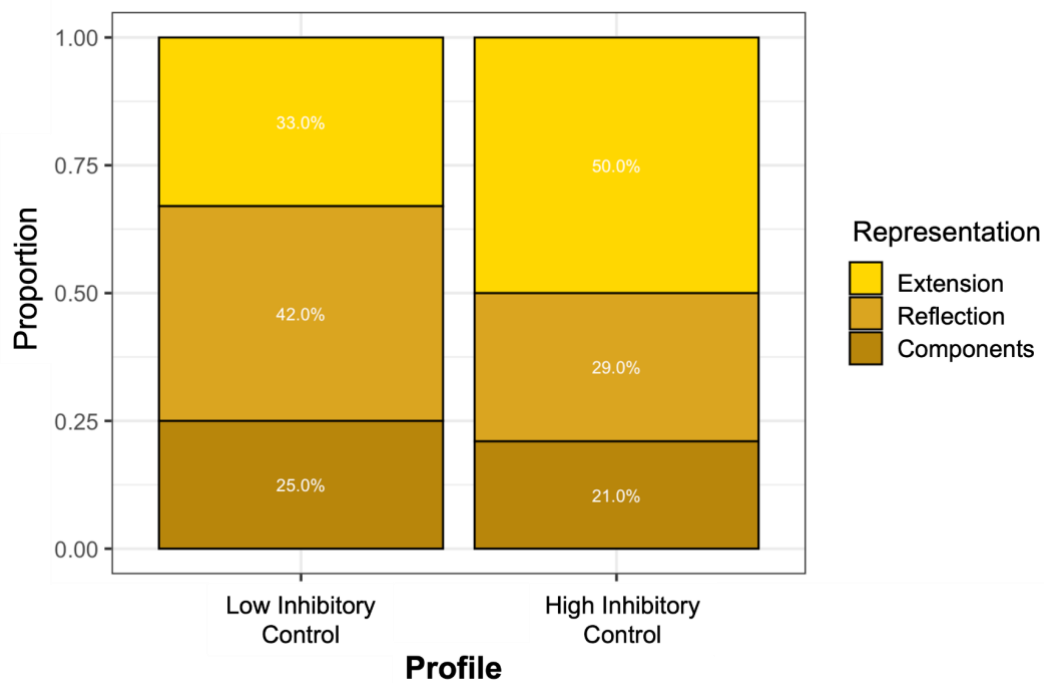


Second, I hypothesized that children with stronger inhibitory control would be more likely to adopt a reflection representation when processing negative integers. This is because the reflection representation may require suppressing more intuitive spatial mappings tied to natural

numbers. However, the observed counts and proportions did not support this prediction (Figure 15). Children with high inhibitory control were actually *less* likely to use the reflection representation (7 out of 24, 29%) than those with low inhibitory control (10 out of 24, 42%). In contrast, children with high inhibitory control more often used the extension representation (12 out of 24, 50%) compared to those with low inhibitory control (8 out of 24, 33%). Components representations were used at similar rates in both groups (high: 5 out of 24, 21%; low: 6 out of 24, 25%).

Figure 15

Proportion of Participants in Low and High Inhibitory Control Profiles Exhibiting Each Representation

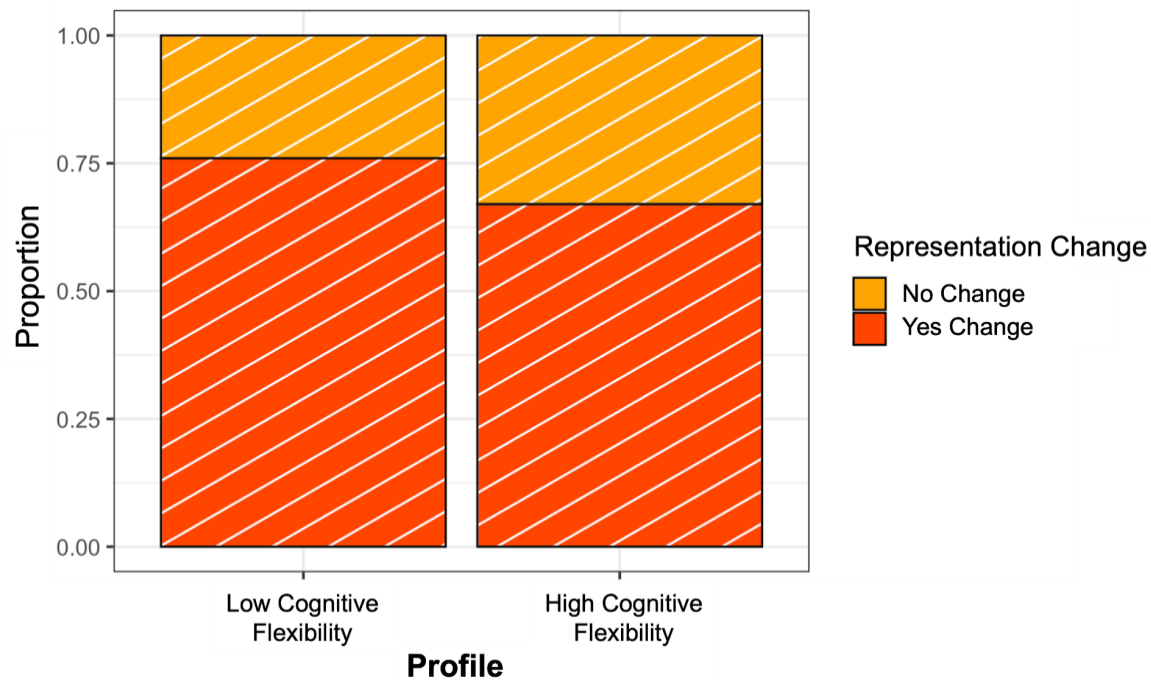


A chi-square test indicated that there was no significant association between inhibitory control and representation strategy, $\chi^2(2, N = 48) = 1.42, p = .492$. Thus, there was no evidence that the representation participants used varied based on the strength of their inhibitory control.

Third, I hypothesized that cognitive flexibility would predict whether participants change their mental representations between order conditions (intermixed vs. blocked). Specifically, I expected that participants with higher cognitive flexibility would be more likely to shift their representation across different order conditions.

Figure 16

Proportion of Participants in Low and High Cognitive Flexibility Profiles Exhibiting Representation Change or Not



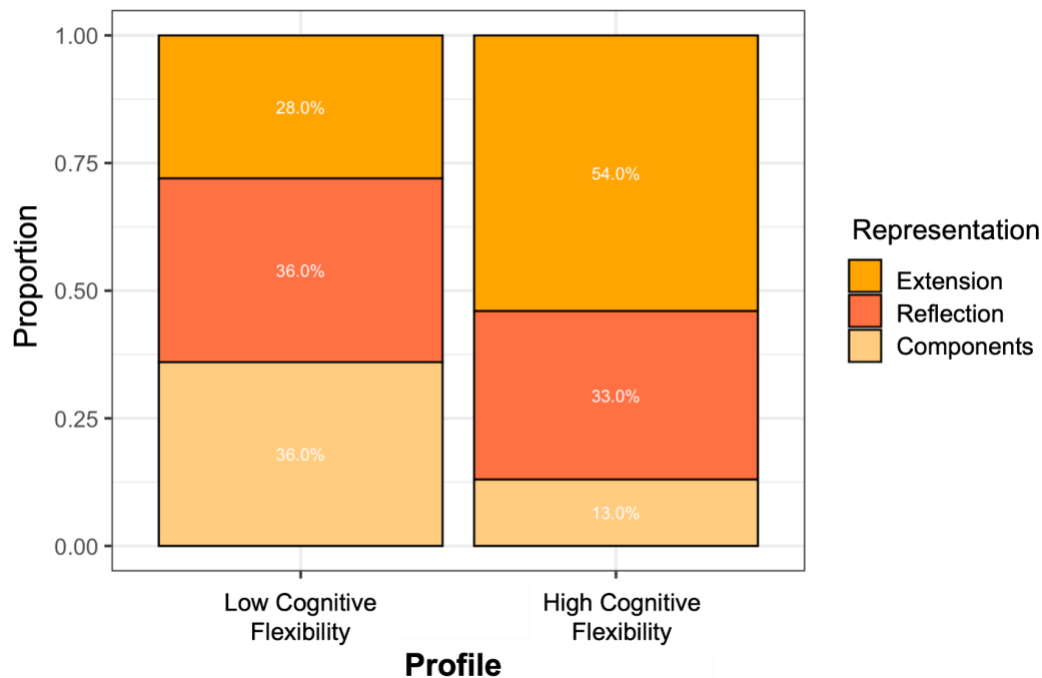
Descriptive analyses showed that among participants with low cognitive flexibility, 6 out of 25 (24%) did not change their representation between order conditions, while 19 out of 25 (76%) did change. Among participants with high cognitive flexibility, 8 out of 24 (33%) did not change, and 16 out of 24 (67%) did change representation (Figure 16). A chi-square test revealed no significant association between cognitive flexibility and representation change, $\chi^2(1, N = 49) = 0.17, p = .684$. Thus, there was no evidence that participants with higher cognitive flexibility

were more likely to shift their representation, compared to those with lower cognitive flexibility. See Sankey diagrams in Appendix 10 for representation changes.

As an exploratory analysis, I also examined whether cognitive flexibility was related to the type of representation participants used. Among participants with low cognitive flexibility, the most common representations were components (36%) and reflection (36%), each used by 9 out of 25 participants, followed by extension (7 out of 25, 28%). In contrast, among participants with high cognitive flexibility, the most common representation was extension, used by 13 out of 24 participants (54%), followed by reflection (8 out of 24, 33%), and components (3 out of 24, 13%; Figure 17).

Figure 17

Proportion of Participants in Low and High Cognitive Flexibility Profiles Exhibiting Each Representation



A chi-square test revealed that the association between cognitive flexibility and representation strategy did not reach statistical significance, $\chi^2(2, N = 49) = 4.84, p = .089$.

However, the nonsignificant trend suggests that participants' choice of representation strategy may have differed depending on their level of cognitive flexibility. Notably, the high cognitive flexibility group used the extension representation more frequently, whereas the low cognitive flexibility group showed a more balanced distribution, with comparable levels of use of the components and reflection representations.

Overall, my third research question examined whether participants' use of negative number representations was related to their executive function skills, including working memory, inhibitory control, and cognitive flexibility. Although descriptive trends suggested that participants with stronger executive functions may be more likely to use abstract representations, such as reflection, and to shift representations across task conditions, none of these relations reached statistical significance. Participants with low cognitive flexibility may be more likely to rely on the components representation, while those with high cognitive flexibility may be more likely to use integrated representations, especially the reflection representation. However, cognitive flexibility was not significantly associated with whether participants changed representations across conditions. Overall, these findings provide limited but suggestive evidence that executive functions may be associated with patterns of representation use. While the results do not offer consistent statistical support, observed patterns point to potentially meaningful relationships that warrant further investigation in future research.

Discussion

One goal of this dissertation was to investigate how children mentally represent negative integers. In this study, I examined how middle-school children engaged with symbolic magnitude comparison tasks involving positive, negative, and mixed positive-negative number pairs. RT patterns from this task revealed the kinds of mental representations children employed

for negative numbers. I systematically varied the format of the tasks such that the *order* of pairs differed (pair types intermixed vs. pair types blocked) and the presentation format of pairs differed (simultaneous vs. sequential). Based on prior work (e.g., Ganor-Stern et al., 2010; Shaki & Petrusic, 2005), I hypothesized that children would use components representations across conditions, except for in the sequential-intermixed condition—in which I expected that children would be more likely to use integrated representations. This prediction was supported at the group level: the sequential-intermixed condition yielded a statistically significant inverse distance effect, consistent with the reflection representation, and all other conditions yielded no distance effects. The null findings for the other conditions align with prior work in children (Varma & Schwartz, 2011; Vest & Alibali, 2024). The significant finding in the sequential-intermixed condition is, to my knowledge, the first report of a significant group-level inverse distance effect in children.

However, individual-level analyses painted a more complex picture. Across all four conditions, most children showed either standard or inverse distance effects, and only a minority used components representations. Surprisingly, integrated representations were common even in conditions in which I expected componential reasoning to dominate (e.g., simultaneous-blocked). For example, in the simultaneous-blocked condition, 93% of children showed a distance effect—either standard or inverse. This suggests that children’s magnitude representations of negative numbers are more robust and potentially less sensitive to variability in context than previously assumed.

These findings challenge the notion that children’s reasoning about negative numbers is entirely fragile. Instead, they suggest a transitional developmental period during which children flexibly apply different representations depending on task context. They also highlight a

mismatch between group-level and individual-level patterns: group-level null effects can obscure meaningful variation and mask the presence of reflection or extension representations in subgroups of participants (see Fisher et al., 2018, for a metascience commentary).

I also examined whether children's use of negative number representations was related to individual differences in executive function skills: working memory, inhibitory control, and cognitive flexibility. Although most of the observed associations were not statistically significant, descriptive trends offered several suggestive patterns.

First, children with stronger working memory skills were descriptively *less* likely to use components representations, contrary to the prediction that maintaining separate magnitude and polarity information would favor such a representation. Instead, children with higher working memory more often used the reflection representation. Second, children with stronger inhibitory control showed more frequent use of the extension representation and less frequent use of reflection representations—again contrary to prediction. Third, although I expected cognitive flexibility to predict representation switching across task conditions, no significant relation was found. However, exploratory analyses suggested that children with higher flexibility were more likely to use integrated representations, particularly extension.

These null results may reflect limited statistical power due to sample size and to use of a median split for categorizing children. They may also indicate that the hypothesized cognitive demands of each representation type (e.g., inhibition for reflection, working memory for components) do not map neatly onto EF tasks that use non-numerical stimuli. Nonetheless, the observed trends suggest that children with stronger executive function skills may be better positioned to engage with more abstract and integrated representations, particularly in contexts that demand flexibility and inhibition of default rules.

These findings contribute to ongoing debates in numerical cognition about how abstract number concepts such as negative integers are represented and understood. They provide compelling evidence that children's representations of negative numbers are not monolithic. Rather than progressing from components to integrated representations in a strictly linear fashion, children appear to experiment with and draw on multiple coexisting representations. The findings echo broader themes in developmental science that emphasize variability, context sensitivity, and representational flexibility (Siegler, 1996; Thelen & Smith, 1994).

Moreover, the surprising prevalence of reflection representations among children challenges assumptions that this representation is a late-developing, adult-like pattern. Given that reflection-based reasoning incorporates perceptual symmetry—a process that emerges early in development—these findings raise the possibility that reflection representations may be more intuitive or accessible to children than previously thought.

These results also speak to broader questions about the nature of the mental number line. While the extension account posits a single, continuous left-to-right mapping, and the components account assumes disjoint representations of polarity and magnitude, the reflection account suggests a more nuanced, symmetrical structure. The findings indicate that all three forms of representation may be available to children, and their use may depend on subtle features of task design and individual differences.

Overall, this study reveals that children employ a diverse array of representations when reasoning about negative numbers, including extension, reflection, and components representations. Task context plays a role in shaping which representations children use, and many children demonstrate robust and flexible use of integrated reasoning about magnitude and polarity, even without explicit scaffolding. While executive function skills may contribute to

individual differences, the evidence points most clearly to a developmental landscape characterized by variability, adaptability, and surprising sophistication. These findings contribute to our understanding of how abstract mathematical concepts are mentally represented and lay a foundation for educational practices that support deeper numerical reasoning.

Chapter 5: General Discussion

In this dissertation, I conducted two studies to investigate how adults and children mentally represent negative numbers. Drawing on prior research, I developed a symbolic magnitude comparison task that included three types of number pairs: positive-positive, negative-negative, and mixed positive-negative comparisons. By analyzing patterns in response times, specifically distance effects and semantic congruence effects, I was able to infer the types of mental representations participants used at both the group and individual levels.

To examine whether and how these representations are sensitive to context, I introduced two experimental manipulations. First, I varied the presentation format of number pairs between subjects so that participants saw numbers either simultaneously (both numbers at once) or sequentially (one number followed by the other). Second, I varied the order of pair types within subjects, presenting them either in blocked format (for example, all positive pairs followed by all negative pairs) or intermixed.

Additionally, I measured participants' executive function skills, including working memory, inhibitory control, and cognitive flexibility, to explore how these abilities might relate to different mental representations of negative numbers. Although this part of the research was largely exploratory, I hypothesized that executive function could serve as a modulating factor, influencing which representations participants rely on.

In the sections that follow, I summarize the main findings and discuss their theoretical and practical implications.

Summary of Key Findings

One of the central findings of this dissertation is the diversity of mental representations that people use when reasoning about negative numbers. Across both studies, participants drew on all three hypothesized representation types, including components, extension, and reflection representations, highlighting that there is no single, uniform way in which negative numbers are mentally represented. Instead, representations varied somewhat systematically across individuals, tasks, and age groups. Another central finding was that group-level analyses obscured this variability—highlighting the importance of individual-level analyses when examining the possible use of multiple representations.

First, I hypothesized that adults would primarily use a reflection representation of negative numbers, whereas children would rely on a less mature, components representation. To test this, I examined the distance effect specifically in mixed positive-negative comparisons, which can help distinguish between these types of representations. At the group level, neither adults nor children exhibited a distance effect during mixed positive-negative comparisons. This absence of a distance effect is consistent with a components representation, in which participants rely solely on polarity cues (e.g., “positives are always greater than negatives”) in mixed pairs, rather than processing numerical magnitudes. On the surface, these results suggest that both age groups “default” to rule-based, non-integrative strategies for mixed comparisons. Indeed, some past work suggests people do not form integrated magnitude representations of negative numbers (Fischer & Rottman, 2005; Ganor-Stern & Tzelgov, 2008). However, this group-level pattern also contrasts with other reports of inverse distance effects in adults under similar task conditions

(e.g., Varma & Schwartz, 2011; Varma et al., 2007; Vest & Alibali, 2021)—a pattern interpreted as evidence for a reflection representation.

One possible reason for this discrepancy lies in subtle differences in task design between this study and prior studies reporting inverse distance effects during mixed comparisons. For instance, most previous studies included inverse number pairs (e.g., -1 and 1), which are near in distance and symmetrical across zero. These pairs may strongly cue a reflection representation, leading to greater use of such a representation. Alternatively, they may allow participants to make judgments without engaging magnitude processing at all. In contrast to prior studies, the current study excluded exact inverse pairs to minimize heuristic responding and encourage genuine magnitude comparisons. This design choice may have reduced the likelihood of eliciting inverse distance effects at the group level, underscoring how representation patterns may be shaped by seemingly minor task features.

Critically, individual-level analyses told a different story. Although the group-level data aligned with a components account, only a minority of participants in each age group showed the pattern diagnostic of a pure components representation—no distance effect (24% of adults; 22% of children). The majority showed either standard or inverse distance effects, indicating reliance on extension and reflection representations, respectively. These contrasting individual patterns likely cancelled each other out at the group level, masking the underlying variability in representation strategies. This finding underscores the importance of examining individual data to reveal heterogeneity in cognitive representations that is obscured in aggregated results.

In addition to examining distance effects, I also measured semantic congruence effects. For this measure, comparisons between two negative numbers were of particular interest because they are useful for distinguishing between components and integrated representations. A standard

semantic congruence effect occurs when participants respond more quickly when the instruction (such as “choose greater” or “choose lesser”) matches the size of the numbers. For example, participants are typically faster to select the greater of two large numbers or the lesser of two small numbers.

Among adults, I found a standard semantic congruence effect for negative pairs. Participants were quicker to judge the greater of two large negative numbers (such as -1 vs. -2) and the lesser of two small negative numbers (such as -8 vs. -9). This pattern suggests that people associate distinct sizes with negative numbers in a way that does not map directly onto natural number magnitude, consistent with an integrated representation of polarity and magnitude—either a reflection or extension representation. If participants had been relying on natural or absolute magnitude, suggestive of a components representation, I would have expected an inverse semantic congruence effect. Individual-level analyses supported the group-level pattern. A majority of adult participants, 63%, showed the standard semantic congruence effect.

A different pattern emerged for children. At the group level, they did not show a significant semantic congruence effect for negative number pairs, although the trends were in a similar direction. This absence of a clear effect may be consistent with the use of a components representation, in which the sign and magnitude of a number are processed separately. According to Ganor-Stern and Tzelgov (2008), when negative numbers are represented this way, the sign information may dominate or interfere with the processing of magnitude, effectively counterbalancing or even overshadowing it. As a result, semantic congruence effects may be diminished or eliminated altogether. However, this null result may also be due to other factors. First, children likely have weaker semantic associations with negative numbers due to less experience. In contrast, adults encounter negative numbers more frequently in contexts such as

finance or algebra, which may strengthen those associations over time. Second, the study design for children reduced the statistical power. Because predicate instructions were assigned between subjects, the sample size was half that of the adults in each group. As a result, I analyzed the effect of size separately within each predicate condition, rather than examining a within-subjects interaction. This further reduced the number of data points contributing to the analysis. Both limited experience and reduced power may have contributed to the absence of a significant effect in children.

A primary goal of this study was to assess whether people's mental representations of negative numbers are flexible and context-dependent, or whether they are relatively stable. Specifically, I examined whether features of the task, including whether trials were blocked or intermixed and whether numbers were presented simultaneously or sequentially, influenced participants' use of components, extension, or reflection representations.

Adults showed some sensitivity to context, but this flexibility was more limited than initially expected. At the group level, there were no significant distance effects across task conditions, suggesting that no single representation dominated. However, individual-level analyses revealed that the majority of adults used integrated representations—either extension or reflection—in all four task conditions. Even in the simultaneous-blocked condition, which was predicted to encourage rule-based, componential processing, most adult participants showed distance effects in mixed positive-negative comparisons (76%) or semantic congruence effects (63%) consistent with integrated reasoning. These results suggest that adults' representations of negative numbers are relatively robust. Although some variation emerged across conditions, adults generally accessed the magnitudes of negative numbers across formats and presentation

types. This suggests that adults' magnitude representations are relatively stable and not strongly shaped by task features.

Children, in contrast, demonstrated more contextual variability. At the group level, a significant distance effect emerged in the sequential-intermixed condition, in which children showed an inverse distance effect, but not in the other three conditions. I had predicted that the sequential-intermixed condition would be most likely to elicit integrated representations, and the group-level results aligned with this prediction.

At the individual level, however, children showed substantial use of integrated, magnitude-based representations across all four task conditions. Even in contexts designed to support rule-based, componential reasoning, the majority of children exhibited either standard or inverse distance effects. This finding was unexpected. It suggests that children are not rigidly bound to polarity-based rules, and they can engage with negative numbers as meaningful magnitudes, even in different kinds of contexts.

The pattern of differences across task conditions—in distance effects, particularly—varied between adults and children. For adults, task features had minimal influence on the types of representations used. For children, task demands shaped reasoning more noticeably. In particular, the most demanding condition (sequential-intermixed) elicited the clearest group-level evidence for an integrated representation (reflection). Moreover, at the individual level, in the blocked conditions, many children exhibited standard distance effects (54% and 61% in the sequential and simultaneous conditions), whereas in the intermixed conditions these patterns were less frequent (46% and 23%). At the same time, the diversity of representations across all conditions suggests that representational flexibility for negative numbers is already present in childhood.

In sum, these findings suggest that mental representations of negative numbers are neither entirely fixed nor entirely context dependent. Adults tended to use stable, integrated representations across task conditions. Children showed more context sensitivity, with task structure influencing their use of different representations. Nonetheless, even young learners are often capable of reasoning about negative numbers as meaningful magnitudes. This suggests that the foundations for integrated number line representations are already in place by middle school, and they are applied more flexibly than previously assumed.

I also investigated whether people's use of negative number representations was related to their executive function skills, including working memory, inhibitory control, and cognitive flexibility. I hypothesized that domain-general executive skills might support the adoption or flexible use of different representations.

Among adults, I did not find support for the hypothesis that higher working memory would be associated with greater use of the components representation. Participants with both high and low working memory showed similar rates of components representation use (25% and 26%, respectively), and the overall distribution of representations did not differ between working memory groups. Similarly, no significant association was found between inhibitory control and the use of a reflection representation. Individuals with high and low inhibitory control were equally likely to use a reflection representation (37% and 36%, respectively). Finally, cognitive flexibility did not predict whether adults changed representations across task conditions, nor did it significantly predict the type of representation adults used. Nonetheless, there was a nonsignificant trend suggesting that participants with higher cognitive flexibility may have been somewhat more likely to use reflection representations (47%, vs. 30% for those with lower cognitive flexibility) and less likely to rely on components representations (16%, vs. 35% for

those with lower cognitive flexibility). These patterns suggest that although executive functions may play some role in supporting representation use or adaptability, they were not strong predictors of representation use in this sample of adults.

Among children, the results revealed a similar story. Working memory, inhibitory control, and cognitive flexibility were not significantly related to the types of representations children used. In particular, children with higher working memory were not more likely to use components representations than children with lower working memory (20% and 27%, respectively), as initially hypothesized. In fact, they were slightly more likely to use reflection representations (42% vs. 23%). Likewise, inhibitory control was not associated with greater use of the reflection representation. Cognitive flexibility also did not predict whether children shifted their representations across task contexts. However, a nonsignificant trend suggested that children with higher cognitive flexibility may have been more likely to use extension representations (54%), whereas those with lower flexibility showed more balanced use of the three strategies. This pattern hints that integrated representations, particularly extension, may become more accessible to children with stronger cognitive flexibility.

Together, these findings indicate that executive function skills were not robust predictors of how adults or children represented negative numbers. At the same time, the patterns observed in the descriptive data suggest potential relationships that may be worth exploring further.

One possibility is that executive function does contribute to representation use, but in more nuanced ways than captured in these analyses. For example, specific combinations of executive skills, rather than individual components, may jointly support the adoption or flexible use of representations. Alternatively, the relations between executive function and negative number representations may emerge more clearly under increased cognitive load, or in

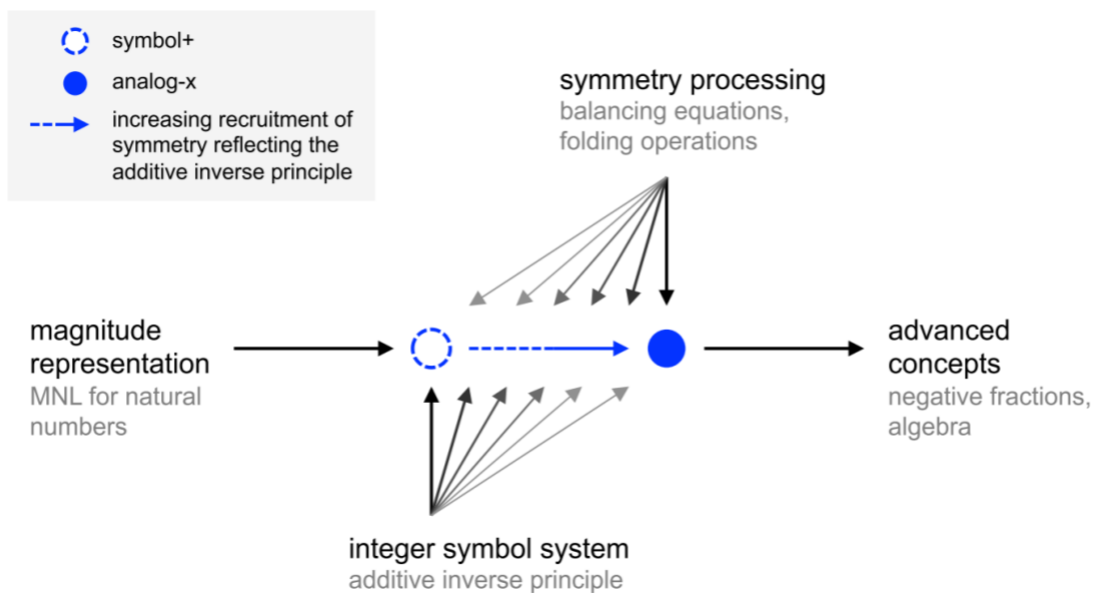
longitudinal designs that track developmental changes over time. Future research should continue to examine these possibilities, ideally with larger samples, richer measures of cognitive skills, and more sensitive analytic approaches.

Age-Related Changes in Representations

Some developmental theories propose that people's representations of negative numbers shift as they move from childhood to adulthood. One influential account is the *analog-x* hypothesis (Varma & Schwartz, 2011). According to this view, learners gradually integrate sign and magnitude into a unified, spatially organized representation—eventually forming a reflection-based mental number line.

Figure 18

Model of the Developmental Trajectory to Analog-x from Varma et al. (2019)



In this model (see Figure 18), people initially reason about integers using symbolic rules, especially the additive inverse principle (e.g., +3 and -3 sum to zero). With experience, particularly through balancing equations, they begin to engage symmetry processing. This leads

to a transformation in their mental representation of integers—from symbolic reasoning (termed *symbol+* by Varma and Schwartz, 2011) to a spatial, symmetry-based format (termed *analog-x* by Varma and Schwartz, 2011) that directly reflects the additive inverse principle. This shift in representation supports the development of more advanced mathematical thinking.

A complementary perspective comes from conceptual change theories, which view development as the restructuring of mental models in response to instruction and experience (e.g., Vosniadou, 1994; Bofferding, 2014, 2019). Within this framework, children begin with an initial model that interprets negative numbers through the lens of whole number knowledge—for example, treating -5 as larger than -3 because 5 is larger than 3. Through exposure to new number principles and repeated conflict between existing and novel ideas, children construct synthetic models that blend old and new conceptions. Eventually, some learners arrive at a formal model, in which the number line extends symmetrically around zero and negative integers are fully integrated into a magnitude-based system.

Across the two studies in this dissertation, I observed evidence of integrated processing of polarity and magnitude in many children, and a continued reliance on components representations in a subset of adults. These patterns suggest that the developmental trajectory of negative number representations is neither linear nor uniform. Instead, my findings point to a more dynamic process in which multiple representations coexist and are flexibly deployed depending on context, experience, and cognitive resources.

My findings argue against a rigid or universal sequence of representational change. While adult participants more frequently used integrated representations, many younger children also demonstrated such reasoning—sometimes even in conditions designed to favor rule-based reasoning. Conversely, a substantial minority of adults relied on components representations.

These patterns indicate that development does not involve a simple replacement of one representation with another but rather an increase in the *flexibility* and *accessibility* of multiple representations over time. In other words, changes in mental representations may not follow a fixed sequence but may instead unfold in ways that are shaped by the demands of the task and the representational possibilities available to the learner. This suggests that representational change is both probabilistic—that is, more likely under certain conditions—and context-dependent. Rather than progressing through discrete stages, people may come to flexibly draw on multiple representations as needed. Factors such as task demands, prior experience, and available cognitive resources likely shape which representation is activated in a given moment.

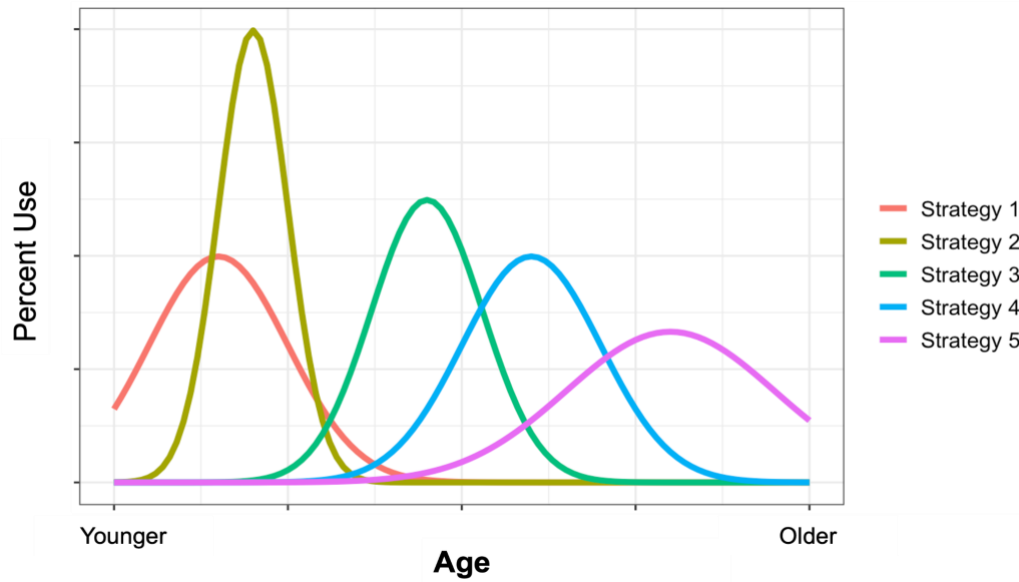
Strategy Selection and Representational Dynamics

This idea aligns with a large body of work on strategy choice in mathematics (e.g., Obersteiner et al., 2022; Siegler, 1991, 1998; Van Der Auwera et al., 2023), including the overlapping waves theory, which suggests that people often have multiple strategies available and select among them adaptively depending on development (Figure 19), individual differences, and task context. Rather than relying on a single, fixed representation, people flexibly deploy strategies that suit the demands of the task, their prior experiences, and their cognitive resources.

Some prior research shows that mental representations of negative numbers vary by context. Shaki and Petrusic (2005) found a reverse SNARC effect for negative numbers in blocked trials, suggesting a components representation, and a standard SNARC effect in intermixed trials, suggesting an integrated magnitude representation. Similarly, Ganor-Stern et al. (2010) found that simultaneous presentation encouraged reliance on polarity signs, while sequential presentation led to distance effects, reflecting an integrated approach. They proposed a hybrid-holistic representation in which integrated and rule-based representations coexist.

Figure 19

Schematic Depiction of Overlapping Waves Model from Siegler (1998)



The findings from the current dissertation support such dynamic models of strategy use. Across both children and adults, I observed evidence for multiple representations being deployed across different conditions. Some participants consistently used the same representation across all tasks, while others flexibly shifted representations depending on whether trials were blocked or intermixed, or whether numbers were presented simultaneously or sequentially. These patterns suggest that representations are not fixed but part of a fluid system that is sensitive to context, development, and cognitive constraints.

Importantly, the data also speak to the question of what factors shape representational choice. While the measures of executive function did not robustly predict representation use, descriptive patterns suggest that certain aspects of executive functioning—particularly cognitive flexibility—may play a supporting role. For example, there was a trend that participants with higher cognitive flexibility were more likely to use integrated representations, suggesting that executive resources might facilitate the coordination or switching between multiple

representational approaches. However, it is likely that these effects are subtle, complex, and may depend on specific combinations of task features and cognitive demands.

Altogether, the evidence from both prior studies and the current dissertation points toward a dynamic model of negative number representation. Rather than following a fixed developmental path or relying on a single representation, people appear to acquire a repertoire of representational tools that they draw on flexibly. This flexibility is shaped by the order of the items, the nature of the comparisons, the way the numbers are presented, and the person's cognitive abilities and experiential background. Such a model moves beyond stage-like accounts and provides a more nuanced view of integer cognition—one in which context, development, and individual difference factors jointly influence how abstract mathematical concepts are represented and used.

The Representational Dynamics in Integer Cognition Framework

Building on the findings from this dissertation and prior research (Blair et al., 2013; Tsang et al., 2015; Varma & Schwartz, 2011; Vest & Alibali, 2024), I propose the Representational Dynamics in Integer Cognition Framework as a theoretical model to account for how people understand and reason about negative numbers (Figure 20). This framework is grounded in three core principles, which I describe next.

First, people possess multiple potential representations of negative numbers. These include at least three distinct types of representations: the components representation, in which sign and magnitude are processed separately relying on rules and natural number magnitude; the extension representation, in which negative numbers are spatially mapped as an extension leftward on the mental number line; and the reflection representation, in which negative numbers are understood as symmetrical opposites of their positive counterparts.

Second, these representations are co-available in the mind. Rather than transitioning from one representation to another in a strict developmental sequence, people appear to develop a repertoire of representations. This repertoire can include both early-emerging rule-based representations and later-developing representations that integrate polarity and magnitude, with different representations being activated in different circumstances.

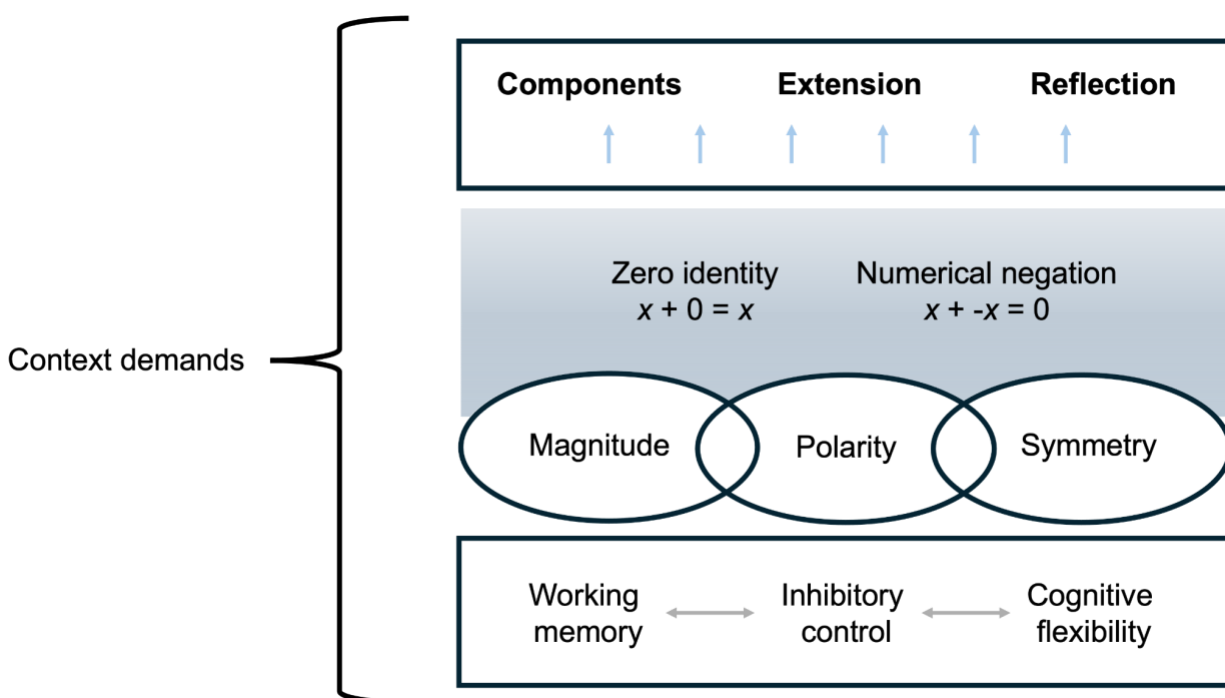
Third, the representation used on any given trial depends on multiple factors, including contextual demands, executive processes, and experience. Representation selection is dynamic and sensitive to factors such as the format of the task, the type of comparison required, prior exposure to mathematical concepts, and individual differences in executive function skills. Critically, representational diversity is not just a transitional or immature feature of children's thinking, but a robust characteristic of mathematical reasoning more broadly.

This framework (see Figure 20 below) can be visualized as a layered structure that rises from low-level cognitive and conceptual foundations to more abstract representational outcomes. At the base of this system are domain-general executive functions—including working memory, inhibitory control, and cognitive flexibility—which act as regulatory mechanisms that influence how conceptual resources are accessed and combined. These executive functions operate like a “dealer of cards,” shaping how people draw upon and integrate core conceptual primitives such as magnitude, polarity, and symmetry. These primitives are likely early-developing or even evolutionarily basic, and they serve as the building blocks for reasoning about integers.

As learners encounter formal instruction and gain experience with symbolic systems, these conceptual foundations support the construction of more abstract mathematical principles, such as zero as the additive identity ($x + 0 = x$) and numerical negation ($x + (-x) = 0$). These principles, in turn, scaffold the emergence of the three major representational approaches:

Figure 20

The Representational Dynamics in Integer Cognition Framework



Note. Executive functions (bottom layer) coordinate the deployment of conceptual primitives (magnitude, polarity, symmetry), which underlie formal number principles (e.g., zero identity, negation) and give rise to distinct mental representations (components, extension, reflection). The representation used is dynamically selected based on contextual demands.

components, extension, and reflection. While these representations differ in the cognitive resources they draw on—components representations emphasize polarity and symbolic rules; extension representations emphasize magnitude and spatial layout; reflection representations emphasize symmetry—they are not mutually exclusive. Instead, they coexist within the mind and may be differentially activated depending on the task and context.

To reflect this dynamic, the framework includes a vertical bracket labeled “Context Demands” that spans the full system, indicating that activation of these representations is shaped

by the surrounding environment and the structure of the task. Features such as whether numbers are presented simultaneously or sequentially, or whether numbers are among similar types or intermixed, can sway the use of one representation or another. Thus, the system is not fixed or hierarchical, but instead it is flexibly configured in response to real-time cognitive and environmental constraints.

The Representational Dynamics in Integer Cognition Framework emphasizes flexibility, variability, and adaptive coordination among multiple representations. It reframes developmental change not as a unidirectional replacement of early representations with mature ones, but as a gradual expansion in the availability, coordination, and strategic deployment of diverse representations. This model echoes Siegler's (1996) overlapping waves theory and extends it by specifying the cognitive structures involved in *integer* reasoning and by shedding light on the conditions under which different representations are selected. It offers a dynamic view of integer cognition that captures the richness, complexity, and adaptability of how children and adults reason about negative numbers.

Theoretical Contributions

Taken together, the findings from this dissertation offer several key theoretical contributions. First, they demonstrate that multiple mental representations of negative numbers coexist and are used flexibly by both children and adults. This challenges traditional developmental models that assume a single "correct" representation gradually emerges with age or instruction. Second, the results show that representation use is sensitive to context, especially for children. Task features such as comparison order and presentation had relatively little impact on which representations adults used, but had a substantial impact on representation use in children. Third, the findings highlight the importance of individual-level analyses for uncovering

cognitive diversity. Group-level data often obscured the rich variability in participants' representations, underscoring the value of examining response patterns at the level of the individual. Fourth, while executive function did not emerge as a robust predictor of representation use, there was preliminary evidence that skills such as cognitive flexibility may support representation use or switching, albeit in subtle or context-dependent ways. Finally, the results (and prior research) support a new theoretical model—the Representational Dynamics in Integer Cognition Framework—that accounts for the observed variability, flexibility, and context-sensitivity in reasoning about negative numbers. This framework moves beyond rigid, stage-like accounts of development and offers a more dynamic, resource-based, and task-sensitive model of integer cognition.

In sum, this work offers a reconceptualization of how people understand negative numbers. Rather than asking which representation is “correct,” we should ask: How do people select and coordinate among multiple representations to solve problems involving negative numbers? This shift in perspective opens new avenues for research and instruction that emphasize flexibility, metacognitive awareness, and adaptive use of multiple representations.

Methodological and Conceptual Limitations

Several methodological and conceptual limitations of this work should be acknowledged. One notable limitation concerns the criteria used to classify individual participants' representations. In this study, I relied on a $+0.25$ and -0.25 SD cutoff around a null distance effect (i.e., 0) to distinguish among components, extension, and reflection representations. While this approach enabled me to identify broad patterns in individual representations, it imposes a somewhat arbitrary threshold on inherently noisy behavioral data. Participants with small but meaningful effects just outside this range may have been misclassified, and the fixed cutoffs may

have obscured more graded or continuous variability in representational use. Future work should explore alternative or more nuanced classification methods that better capture the probabilistic and dynamic nature of representation selection.

A second limitation involves the use of online testing, particularly for adult participants because these participants completed the tasks in an unmoderated context. Although remote testing increases accessibility, it introduces variability in participants' environments, such as differences in device type, screen size, and levels of distraction. These uncontrolled factors may have added noise to the response time data and weakened the reliability of my inferences.

Third, while this study incorporated multiple executive function (EF) measures, the sample sizes and median-split categorization limited the power of these analyses. As a result, the observed relationships between EF and representations should be interpreted cautiously. Larger samples and more sensitive or task-specific EF measures may be needed to clarify the role of executive functioning in representational flexibility.

Additionally, the cross-sectional nature of this research limits my ability to draw conclusions about developmental trajectories. While age group comparisons provide useful snapshots, they cannot capture the within-person changes that occur over time. Longitudinal studies are needed to examine how individuals shift among representations, and whether representational flexibility changes with age, instruction, or experience.

Future Directions

Building on the current findings, several future research directions could deepen our understanding of how negative number representations develop and are used.

The findings suggest that instructional approaches should emphasize not just a single, accurate representation but the *flexible and adaptive use* of multiple representations. Future work

should investigate how metacognitive instruction, like teaching students to reflect on their representation use, can support representational flexibility. In particular, it would be fruitful to test whether explicitly teaching students to compare and contrast different representations (e.g., components vs. reflection) enhances their ability to reason about integers across contexts. This could lead to more effective interventions, particularly for students who struggle with algebraic concepts that rely on negative number understanding.

To better capture developmental change, longitudinal research is needed. Tracking individual students over time would clarify how and when different representations emerge, stabilize, or become coordinated. Such designs could also assess whether executive function skills, particularly cognitive flexibility, predict shifts in representation use as learners gain experience and instruction. Longitudinal data could also help refine the Representational Dynamics in Integer Cognition Framework by clarifying whether certain representational trajectories (e.g., from components to reflection) are more common or stable than others.

The current study relied primarily on distance and semantic congruence effects to infer mental representations. Future research should complement these behavioral measures with additional methods, such as verbal strategy reports, eye-tracking, or neuroimaging, to map participants' cognitive processes. Combining implicit and explicit measures may offer a more comprehensive picture of how people access and coordinate multiple representations.

Another promising avenue involves designing and testing interventions that target specific components of the Representational Dynamics in Integer Cognition framework. For example, activities that explicitly engage symmetry detection (like those in Tsang et al., 2015) or magnitude comparison across zero may encourage students to adopt reflection or extension

representations. Comparing the effectiveness of such targeted instruction across age groups and individual cognitive profiles would help identify which supports are most beneficial for whom.

Finally, future studies could examine whether flexibility with negative number representations predicts success in other areas of mathematics. For example, does helping children with switching between representations support learning about linear functions, algebraic reasoning, or coordinate planes? Investigating such transfer effects could further establish the foundational role of representational dynamics in mathematical development more broadly.

Conclusion

This dissertation challenges the notion that people develop a single, mature representation of negative numbers and instead paints a picture of representational diversity, flexibility, and context sensitivity. Both children and adults were found to use multiple representations, including components, extension, and reflection representations, and they sometimes shifted these representations based on task features. Although children showed more context-driven variability, even young learners demonstrated integrated, magnitude-based reasoning under appropriate conditions. Conversely, some adults relied on earlier-emerging, rule-based strategies, highlighting that development is not linear or uniform.

Critically, group-level analyses often obscured the rich variability observed at the individual level, reinforcing the need for methods that highlight cognitive diversity. While executive function measures were not strong predictors of representation use, exploratory trends suggest that skills like cognitive flexibility may support the coordination and deployment of different representations, warranting further investigation.

To explain these patterns, I proposed the Representational Dynamics in Integer Cognition Framework. This model emphasizes that multiple representations coexist in the mind, that their use is shaped by context and cognitive resources, and that developmental change reflects increasing coordination and adaptive selection among representations. This framework offers a dynamic alternative to stage-based models of numerical development and aligns with broader theories of flexible strategy use in mathematical cognition.

Ultimately, this work reframes how we understand reasoning about negative numbers—not as a progression toward a single “correct” mental model, but as the development of a rich and flexible toolbox of representations. By understanding and supporting this flexibility, educators and researchers alike can foster the development of more robust and adaptable mathematical thinkers.

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Appendix 1

Stimuli for Symbolic Magnitude Comparison Blocks in Left and Right Order

Stim ID	Stim L	Stim R	Comparison	Distance	Size	Correct Ans	Section
1	1	2	Positive	1	Small	right	Section1
3	1	4	Positive	3	Small	right	Section1
5	1	7	Positive	6		right	Section1
7	1	9	Positive	8		right	Section1
9	2	4	Positive	2	Small	right	Section1
11	2	7	Positive	5		right	Section1
13	2	9	Positive	7		right	Section1
15	3	6	Positive	3		right	Section1
17	3	8	Positive	5		right	Section1
19	4	6	Positive	2		right	Section1
21	4	8	Positive	4		right	Section1
23	6	7	Positive	1	Large	right	Section1
25	6	9	Positive	3	Large	right	Section1
27	7	9	Positive	2	Large	right	Section1
29	-1	-2	Negative	1	Large	left	Section1
31	-1	-4	Negative	3	Large	left	Section1

33	-1	-7	Negative	6		left	Section1
35	-1	-9	Negative	8		left	Section1
37	-2	-4	Negative	2	Large	left	Section1
39	-2	-7	Negative	5		left	Section1
41	-2	-9	Negative	7		left	Section1
43	-3	-6	Negative	3		left	Section1
45	-3	-8	Negative	5		left	Section1
47	-4	-6	Negative	2	Small	left	Section1
49	-4	-8	Negative	4	Small	left	Section1
51	-6	-7	Negative	1	Small	left	Section1
53	-6	-9	Negative	3	Small	left	Section1
55	-7	-9	Negative	2	Small	left	Section1
57	-1	2	Mixed	3	Small	right	Section1
59	-1	4	Mixed	5	Small	right	Section1
61	-1	7	Mixed	8		right	Section1
63	-2	3	Mixed	5	Small	right	Section1
65	-2	6	Mixed	8		right	Section1
67	-3	2	Mixed	5	Small	right	Section1
69	-4	1	Mixed	5	Small	right	Section1

71	-4	3	Mixed	7	Small	right	Section1
73	-6	2	Mixed	8		right	Section1
75	2	1	Positive	1	Small	left	Section1
77	4	1	Positive	3	Small	left	Section1
79	7	1	Positive	6		left	Section1
81	9	1	Positive	8		left	Section1
83	4	2	Positive	2	Small	left	Section1
85	7	2	Positive	5		left	Section1
87	9	2	Positive	7		left	Section1
89	6	3	Positive	3		left	Section1
91	8	3	Positive	5		left	Section1
93	6	4	Positive	2		left	Section1
95	8	4	Positive	4		left	Section1
97	7	6	Positive	1	Large	left	Section1
99	9	6	Positive	3	Large	left	Section1
101	9	7	Positive	2	Large	left	Section1
103	-2	-1	Negative	1	Large	right	Section1
105	-4	-1	Negative	3	Large	right	Section1
107	-7	-1	Negative	6		right	Section1

109	-9	-1	Negative	8		right	Section1
111	-4	-2	Negative	2	Large	right	Section1
113	-7	-2	Negative	5		right	Section1
115	-9	-2	Negative	7		right	Section1
117	-6	-3	Negative	3		right	Section1
119	-8	-3	Negative	5		right	Section1
121	-6	-4	Negative	2	Small	right	Section1
123	-8	-4	Negative	4	Small	right	Section1
125	-7	-6	Negative	1	Small	right	Section1
127	-9	-6	Negative	3	Small	right	Section1
129	-9	-7	Negative	2	Small	right	Section1
131	2	-1	Mixed	3	Small	left	Section1
133	4	-1	Mixed	5	Small	left	Section1
135	7	-1	Mixed	8		left	Section1
137	3	-2	Mixed	5	Small	left	Section1
139	6	-2	Mixed	8		left	Section1
141	2	-3	Mixed	5	Small	left	Section1
143	1	-4	Mixed	5	Small	left	Section1
145	3	-4	Mixed	7	Small	left	Section1

147	2	-6	Mixed	8		left	Section1
2	1	3	Positive	2	Small	right	Section2
4	1	6	Positive	5		right	Section2
6	1	8	Positive	7		right	Section2
8	2	3	Positive	1	Small	right	Section2
10	2	6	Positive	4		right	Section2
12	2	8	Positive	6		right	Section2
14	3	4	Positive	1	Small	right	Section2
16	3	7	Positive	4		right	Section2
18	3	9	Positive	6		right	Section2
20	4	7	Positive	3		right	Section2
22	4	9	Positive	5		right	Section2
24	6	8	Positive	2	Large	right	Section2
26	7	8	Positive	1	Large	right	Section2
28	8	9	Positive	1	Large	right	Section2
30	-1	-3	Negative	2	Large	left	Section2
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34	-1	-8	Negative	7		left	Section2
36	-2	-3	Negative	1	Large	left	Section2

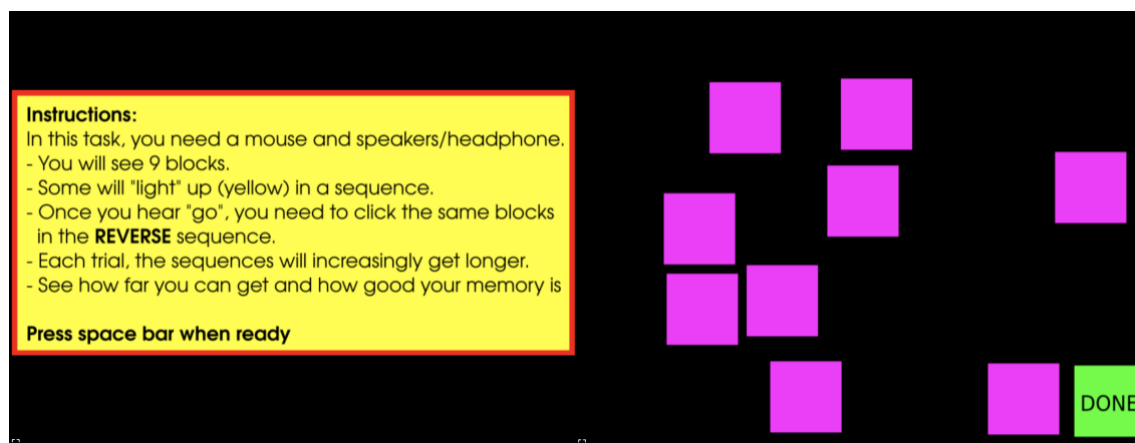
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46	-3	-9	Negative	6		left	Section2
48	-4	-7	Negative	3	Small	left	Section2
50	-4	-9	Negative	5	Small	left	Section2
52	-6	-8	Negative	2	Small	left	Section2
54	-7	-8	Negative	1	Small	left	Section2
56	-8	-9	Negative	1	Small	left	Section2
58	-1	3	Mixed	4	Small	right	Section2
60	-1	6	Mixed	7		right	Section2
62	-2	1	Mixed	3	Small	right	Section2
64	-2	4	Mixed	6	Small	right	Section2
66	-3	1	Mixed	4	Small	right	Section2
68	-3	4	Mixed	7	Small	right	Section2
70	-4	2	Mixed	6	Small	right	Section2
72	-6	1	Mixed	7		right	Section2
74	-7	1	Mixed	8		right	Section2

76	3	1	Positive	2	Small	left	Section2
78	6	1	Positive	5		left	Section2
80	8	1	Positive	7		left	Section2
82	3	2	Positive	1	Small	left	Section2
84	6	2	Positive	4		left	Section2
86	8	2	Positive	6		left	Section2
88	4	3	Positive	1	Small	left	Section2
90	7	3	Positive	4		left	Section2
92	9	3	Positive	6		left	Section2
94	7	4	Positive	3		left	Section2
96	9	4	Positive	5		left	Section2
98	8	6	Positive	2	Large	left	Section2
100	8	7	Positive	1	Large	left	Section2
102	9	8	Positive	1	Large	left	Section2
104	-3	-1	Negative	2	Large	right	Section2
106	-6	-1	Negative	5		right	Section2
108	-8	-1	Negative	7		right	Section2
110	-3	-2	Negative	1	Large	right	Section2
112	-6	-2	Negative	4		right	Section2

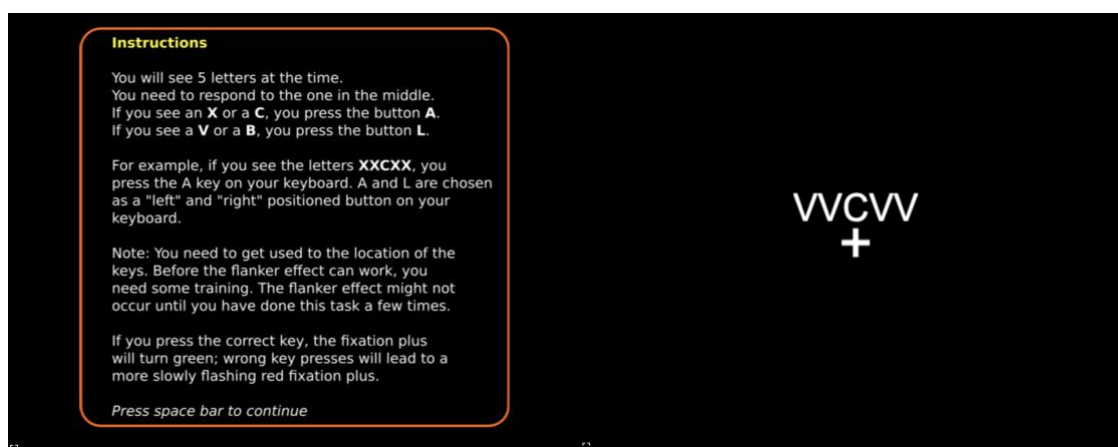
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122	-7	-4	Negative	3	Small	right	Section2
124	-9	-4	Negative	5	Small	right	Section2
126	-8	-6	Negative	2	Small	right	Section2
128	-8	-7	Negative	1	Small	right	Section2
130	-9	-8	Negative	1	Small	right	Section2
132	3	-1	Mixed	4	Small	left	Section2
134	6	-1	Mixed	7		left	Section2
136	1	-2	Mixed	3	Small	left	Section2
138	4	-2	Mixed	6	Small	left	Section2
140	1	-3	Mixed	4	Small	left	Section2
142	4	-3	Mixed	7	Small	left	Section2
144	2	-4	Mixed	6	Small	left	Section2
146	1	-6	Mixed	7		left	Section2
148	1	-7	Mixed	8		left	Section2

Appendix 2

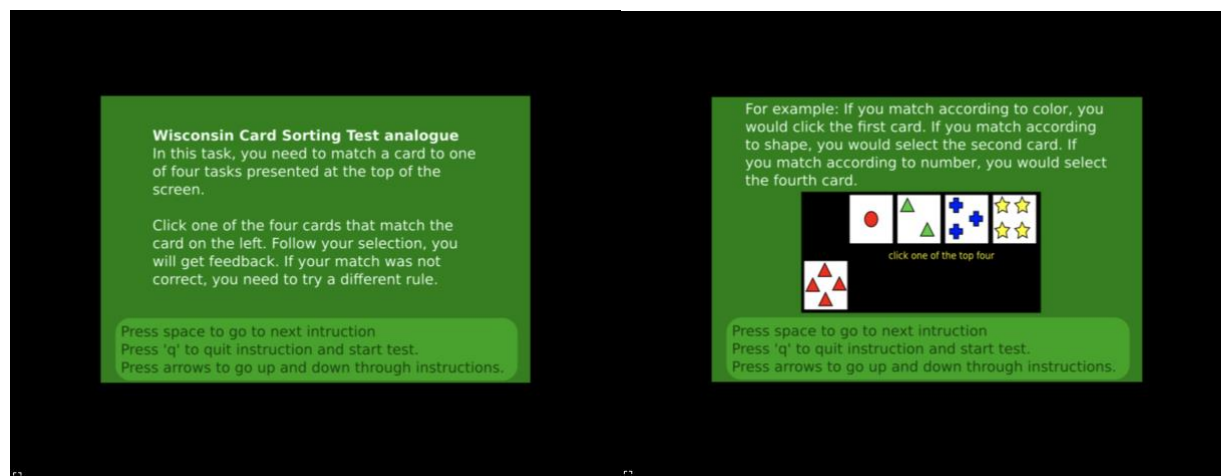
Backward Corsi Block Task Measuring Working Memory



Flanker Task Measuring Inhibitory Control



Wisconsin Card Sorting Task Measuring Cognitive Flexibility



Appendix 3

Model Outputs for Research Questions 1 and 2 in Adults for Group-level Distance Effects in RT

Positive Pairs

Fixed Effects	<i>b</i>	<i>SE</i>	<i>df</i>	<i>t</i> value	<i>p</i>
Intercept	721.14	16.23	51.74	44.36	< .001
Distance	-8.45	2.17	53.18	-3.89	< .001
Order	163.04	18.81	50.25	8.67	< .001
Presentation	472.09	32.12	51.82	14.70	< .001
Predicate	48.39	12.81	52.87	3.78	< .001
Distance*Order	5.56	3.21	220	1.73	.085
Distance*Presentation	-8.80	4.13	53.05	-2.13	.038
Order*Presentation	228.02	36.82	50.63	6.19	< .001
Distance*Order*Presentation	17.87	6.27	401.84	2.85	.005

Negative Pairs

Fixed Effects	<i>b</i>	<i>SE</i>	<i>df</i>	<i>t</i> value	<i>p</i>
Intercept	769.50	18.29	51.85	42.07	< .001
Distance	-12.72	1.77	202.65	-7.18	< .001

Order	175.03	20.06	52.21	8.72	< .001
Presentation	496.48	35.08	51.57	14.15	< .001
Predicate	-63.11	14.05	52	-4.49	< .001
Distance*Order	4.24	3.55	53.27	1.20	.237
Distance*Presentation	-9.36	3.52	201.57	-2.66	.008
Order*Presentation	193.43	37.89	53.21	5.11	< .001
Distance*Order*Presentation	8.73	6.80	53.56	1.43	.158

Mixed Pairs

Fixed Effects	<i>b</i>	<i>SE</i>	<i>df</i>	<i>t</i> value	<i>p</i>
Intercept	711.35	16.33	52.28	43.57	< .001
Distance	-1.82	2.57	6320.52	-0.71	.479
Order	173.43	8.66	6354.16	20.02	< .001
Presentation	409.03	32.27	52.12	12.68	< .001
Predicate	-32.80	13.46	52.14	-2.44	.018
Distance*Order	1.15	5.90	51.75	0.20	.846
Distance*Presentation	-2.26	5.13	6320.51	-0.44	.659

Order*Presentation	369.60	17.33	6353.90	21.33	< .001
Distance*Order*Presentation	-4.19	11.25	50.74	-0.37	.711

Appendix 4

Model Outputs for Research Questions 1 and 2 in Adults for Group-level Semantic Congruence

Effects in RT

Positive Pairs

Fixed Effects	<i>b</i>	<i>SE</i>	<i>df</i>	<i>t</i> value	<i>p</i>
Intercept	737.30	17.73	52.23	41.59	< .001
Size	15.73	9.85	4224.95	1.60	.110
Predicate	70.86	9.90	4233.79	7.12	< .001
Order	161.65	9.91	4235.34	16.310	< .001
Presentation	512.32	35.45	52.23	14.45	< .001
Size*Predicate	22.09	19.67	4224.32	1.12	.262
Size*Order	28.16	19.69	4224.71	1.43	.153
Predicate*Order	91.91	19.75	4229.38	4.65	< .001
Size*Presentation	10.17	19.69	4224.95	0.52	.606
Predicate*Presentation	51.55	19.81	4233.79	2.60	.009
Order*Presentation	223.06	19.82	4235.34	11.25	< .001
Size*Predicate*Order	27.78	39.96	739.82	0.70	.487
Size*Predicate*Presentation	102.77	39.38	4224.32	2.61	.009

Size*Order*Presentation	-42.31	39.38	4224.71	-1.07	.283
Predicate*Order*Presentation	91.29	39.50	4229.38	2.31	.021
Size*Predicate*Order*Presentation	-5.20	79.91	239.82	-0.65	.948

Negative Pairs

Fixed Effects	<i>b</i>	<i>SE</i>	<i>df</i>	<i>t</i> value	<i>p</i>
Intercept	788.48	18.81	52.38	41.91	< .001
Size	9.79	9.79	5539.97	1.00	.318
Predicate	-65.84	9.82	5511.33	-6.70	< .001
Order	168.69	9.85	5514.12	17.13	< .001
Presentation	493.77	37.62	52.38	13.12	< .001
Size*Predicate	39.78	19.59	5537.64	2.03	.042
Size*Order	-4.72	19.60	5536.83	-0.24	.810
Predicate*Order	-3.06	19.68	5511.91	-0.16	.877
Size*Presentation	-25.85	19.59	5538.97	-1.32	.187
Predicate*Presentation	47.74	19.65	5511.33	2.43	.015
Order*Presentation	172.13	19.70	5514.12	8.74	< .001

Size*Predicate*Order	-46.35	42.83	58.42	-1.08	.284
Size*Predicate*Presentation	9.21	29.19	5537.64	0.24	.814
Size*Order*Presentation	-27.67	39.19	5536.83	-0.71	.480
Predicate*Order*Presentation	29.30	39.37	5511.91	0.74	.457
Size*Predicate*Order*Presentation	-55.44	85.67	58.42	-0.65	.520

Appendix 5

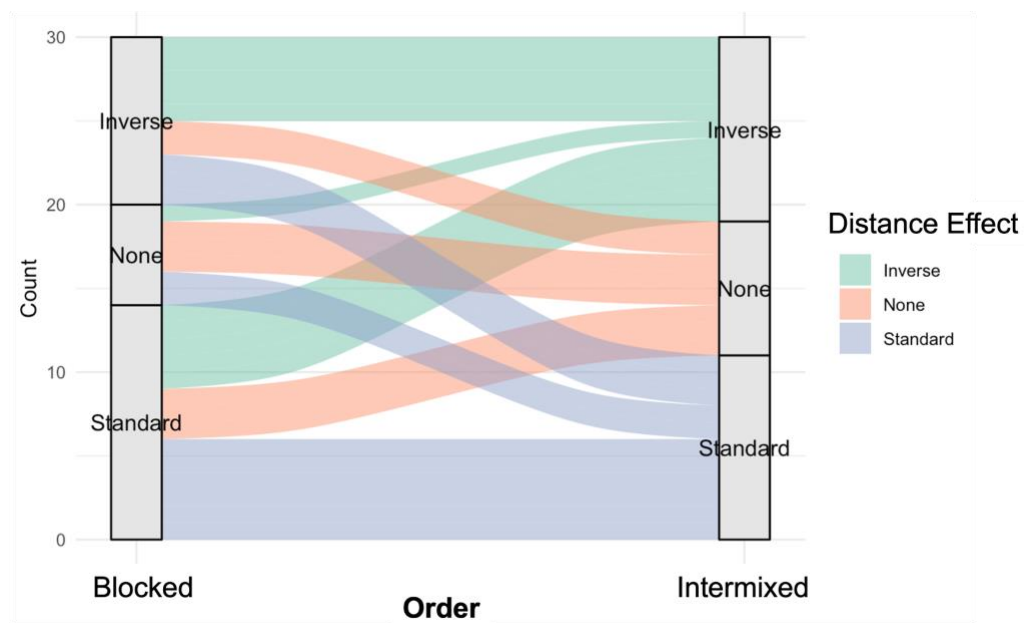
Accuracy during Symbolic Magnitude Comparison for Adults

Comparison	Overall	Near	Far	Lesser	Greater	Small	Large
Mixed	0.90	0.90	0.90	0.88	0.92		
Negative	0.89	0.88	0.90	0.86	0.92	0.88	0.88
Positive	0.90	0.90	0.90	0.87	0.92	0.89	0.89

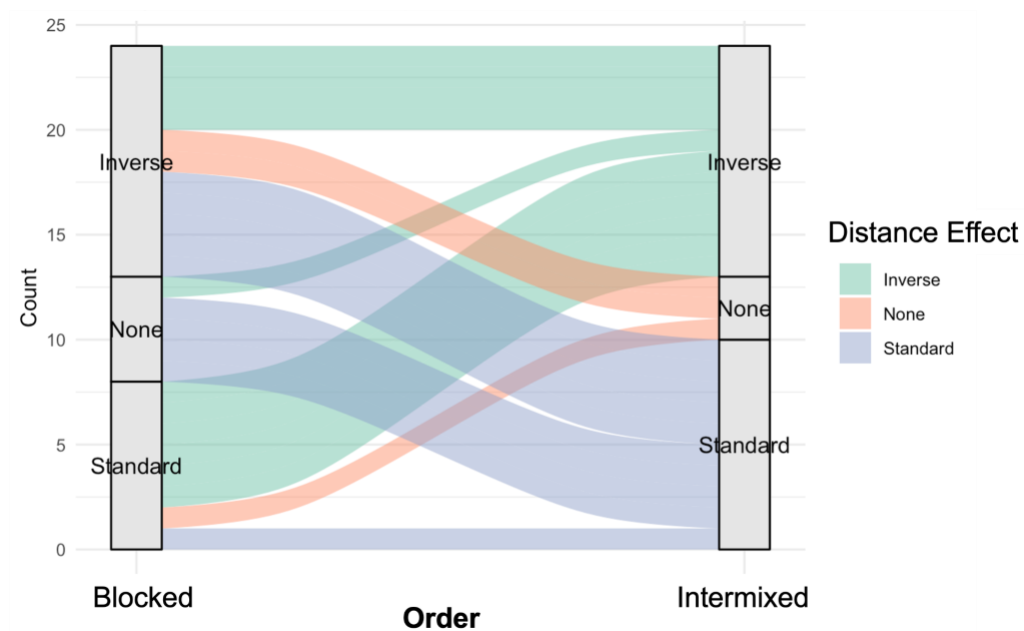
Appendix 6

Sankey Diagram for Adults Depicting Representation Change (via Distance Effects) Across Order Conditions Between Presentation Conditions

Simultaneous



Sequential



Appendix 7

Model Outputs for Research Questions 1 and 2 in Children for Group-level Distance Effects in RT

Positive Pairs

Fixed Effects	<i>b</i>	<i>SE</i>	<i>df</i>	<i>t</i> value	<i>p</i>
Intercept	940.29	25.06	51.11	37.52	< .001
Distance	-10.70	2.71	5148.73	-3.94	< .001
Order	209.21	11.37	5155.51	18.45	< .001
Presentation	726.14	49	50.92	14.82	< .001
Predicate	105.35	49.72	51.07	2.12	.039
Distance*Order	5.49	5.49	882.45	1.00	.318
Distance*Presentation	-16.75	5.43	5148.68	-3.09	.002
Order*Presentation	231.80	22.73	5155.52	10.20	< .001
Distance*Order*Presentation	13.17	10.98	883.21	1.20	.231

Negative Pairs

Fixed Effects	<i>b</i>	<i>SE</i>	<i>df</i>	<i>t</i> value	<i>p</i>
Intercept	967.97	29.60	50.75	32.71	< .001

Distance	-13.35	2.92	5089.21	-4.57	< .001
Order	239.18	12.29	5092.73	19.46	< .001
Presentation	718.01	57.87	50.69	12.41	< .001
Predicate	-54.30	59.11	50.71	-0.92	.363
Distance*Order	-0.38	5.86	1964.73	0.06	.949
Distance*Presentation	-12.54	5.84	5089.23	-2.15	.032
Order*Presentation	171.73	24.58	5092.73	6.99	< .001
Distance*Order*Presentation	11.93	11.73	1954.75	1.02	.309

Mixed Pairs

Fixed Effects	<i>b</i>	<i>SE</i>	<i>df</i>	<i>t</i> value	<i>p</i>
Intercept	920.08	30.63	50.29	30.04	< .001
Distance	6.03	4.55	3274.50	1.33	.185
Order	282.31	15.12	3243.79	18.68	< .001
Presentation	598.64	59.89	50.25	10.00	< .001
Predicate	-17.12	61.28	50.26	-0.28	.781
Distance*Order	16.57	10.18	48.96	1.63	.110

Distance*Presentation	-16.25	9.10	3274.49	-1.79	.074
Order*Presentation	266.46	30.24	3243.82	8.81	< .001
Distance*Order*Presentation	-32.44	20.34	48.96	-1.60	.117

Appendix 8

Model Outputs for Research Questions 1 and 2 in Children for Group-level Semantic

Congruence Effects in RT

Positive Pairs

Fixed Effects	<i>b</i>	<i>SE</i>	<i>df</i>	<i>t</i> value	<i>p</i>
Intercept	962.95	25.30	50.22	38.07	< .001
Size	13.06	18.15	2154.51	0.72	.472
Predicate	88.50	50.60	50.22	1.75	.086
Order	228.03	18.23	2161.14	12.51	< .001
Presentation	765.85	50.60	50.22	15.14	< .001
Size*Predicate	61.92	26.30	2156.51	1.71	.088
Size*Order	6.31	36.44	1198.11	0.17	.863
Predicate*Order	148.98	36.45	2161.14	4.09	< .001
Size*Presentation	30.63	36.30	2154.51	0.84	.399
Predicate*Presentation	121.57	101.19	50.22	1.20	.235
Order*Presentation	246.08	36.45	2161.14	6.75	< .001
Size*Predicate*Order	96.90	72.88	1198.11	1.33	.184
Size*Predicate*Presentation	8.68	72.59	2154.51	0.12	.905

Size*Order*Presentation	50.88	72.88	1198.11	0.70	.485
Predicate*Order*Presentation	187.96	72.90	2161.14	2.58	.010
Size*Predicate*Order*Presentation	-57.47	145.76	1198.11	-0.40	.693

Negative Pairs

Fixed Effects	<i>b</i>	<i>SE</i>	<i>df</i>	<i>t</i> value	<i>p</i>
Intercept	986.16	30.34	50.05	32.50	< .001
Size	-22.06	17.80	2843.96	-1.24	.215
Predicate	-59.92	60.68	50.05	-0.99	.328
Order	236.11	17.89	2819.93	13.20	< .001
Presentation	740.64	60.68	50.05	12.21	< .001
Size*Predicate	61.90	35.61	2843.96	1.74	.082
Size*Order	59.16	40.77	54.67	1.45	.152
Predicate*Order	31.89	35.78	2819.93	0.89	.373
Size*Presentation	-34.37	35.61	2843.96	-0.97	.335
Predicate*Presentation	52.78	121.36	50.05	0.44	.667
Order*Presentation	166.97	35.78	2819.93	4.67	< .001

Size*Predicate*Order	78.09	81.53	54.67	0.96	.342
Size*Predicate*Presentation	-54.64	71.22	2843.96	-0.77	.443
Size*Order*Presentation	21.89	81.53	54.67	0.27	.789
Predicate*Order*Presentation	54.87	71.56	2819.93	0.77	.443
Size*Predicate*Order*Presentation	-246.24	163.07	54.67	-1.51	.137

Appendix 9

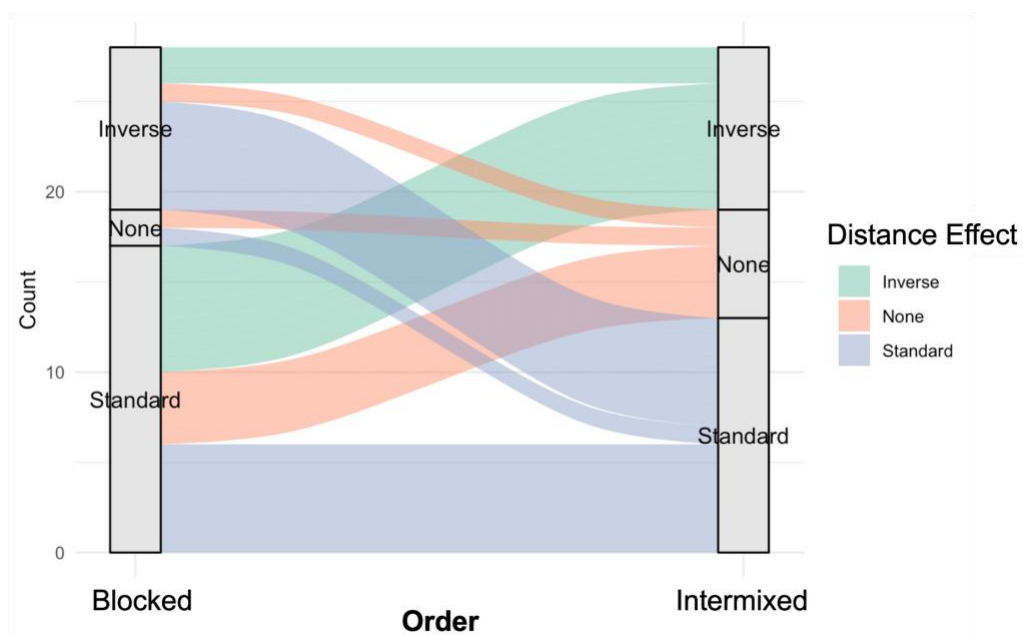
Accuracy during Symbolic Magnitude Comparison for Children

Comparison	Overall	Near	Far	Lesser	Greater	Small	Large
Mixed	0.93	0.92	0.93	0.90	0.94		
Negative	0.92	0.91	0.92	0.89	0.93	0.91	0.91
Positive	0.92	0.92	0.92	0.89	0.95	0.92	0.92

Appendix 10

Sankey Diagram for Children Depicting Representation Change (via Distance Effects) Across Order Conditions Between Presentation Conditions

Simultaneous



Sequential

